

RECORD OF THE REGULAR MEETING OF THE BROOKLINE SCHOOL COMMITTEE HELD ON **THURSDAY, APRIL 27, 2017** AT 6:00 PM IN THE WALSH SCHOOL COMMITTEE ROOM, 5TH FLOOR, TOWN HALL. STATUTORY NOTICE OF THIS MEETING WAS FILED WITH THE TOWN CLERK.

School Committee Members Present: Ms. Ditkoff (Chairman), Mr. Pollak (Vice Chairman), Mr. Chang, Ms. Charlupski, Mr. Glover, Dr. Jackson, Ms. Schreiner-Oldham, Ms. Scotto, and Ms. Stram. Also present: Mr. Bott, Ms. Dunn, Ms. Gittens, Mr. Zimmerman, Mr. Lummis, Ms. Jennings, and Ms. Coyne.

Others Present: Sandra Brown, Blue Wave Solar and Pip Lewis, HMFH Architects.

1. ADMINISTRATIVE BUSINESS

a. Calendar

Ms. Ditkoff called the meeting to order at 6:00 PM. She reported that current and former School Committee members and other honored guests convened earlier this evening to honor Dr. Lisa Jackson and thank her for her service to the Public Schools of Brookline and the community. The School Committee is grateful for her dedication and leadership on tough issues and wishes her well. Ms. Ditkoff then noted upcoming events on the calendar.

b. Consent Agenda **ACTION 17-41**

On a motion of Mr. Glover and seconded by Ms. Stram, the School Committee VOTED with the following abstentions: October 20, 2016 Workshop (Dr. Jackson because she was not at the workshop and Ms. Schreiner-Oldham because she was not on the School Committee); March 2, 2017 Meeting (Ms. Charlupski and Dr. Jackson because they were not at the meeting); March 30, 2017 (Mr. Pollak and Ms. Scotto because they were not at the meeting); and April 6, 2017 Meeting (Ms. Ditkoff and Mr. Chang because they were not at the meeting), to approve the items included in the Consent Agenda.

- i. Past Record: October 20, 2016 School Committee Workshop
- ii. Past Record: March 2, 2017 School Committee Meeting
- iii. Past Record: March 30, 2017 School Committee Meeting
- iv. Past Record: April 6, 2017 School Committee Meeting
- v. Out-of-State Student Trip: Upper Devotion 8th Grade to Camp Cody; Freedom, New Hampshire; May 18, 2017-May 19, 2017 (Attachment A)
- vi. Out-of-Country Student Trip: Brookline High School Angers, France Exchange; February 15, 2018-March 3, 2018 (Attachment B)

2. SUBCOMMITTEE REPORTS

a. Finance

Ms. Stram reported on a recent meeting with the Brookline Town Meeting Members Association Board. Board members appreciated the increase in transparency of the school budget document and process. Discussion focused on the need for an override.

An operating override will be needed and the School Committee looks forward to working with the Board of Selectmen on this matter. The next meeting of the Finance Subcommittee will take place on May 22, 2017. The agenda will include review of the FY 2017 3rd Quarter Financial Report.

b. Capital Improvements

Mr. Pollak reported that the Capital Improvements Subcommittee met earlier this evening to hear a presentation on a proposed gift of a new education building for the Brookline Early Education Program. Counsel for the donors and the architects will be presenting to the full School Committee on May 4, 2017. Mr. Pollak noted various meetings related to proposed funding for schematic design for the 9th Elementary School at Baldwin and the Brookline High School Expansion, including this evening's Advisory Committee meetings.

c. Curriculum

Ms. Scotto reported that School Committee members had the opportunity to meet with some Driscoll parents and staff at this morning's School Site Council meeting. Discussion focused on 2016-2017 benchmarks, meeting the needs of diverse learners, and use of technology. Also this morning, the Curriculum Subcommittee met to discuss the plan for data reporting and the K-8 Literacy/English Language Arts Program. The next meeting of the Curriculum Subcommittee will take place on May 17, 2017. The agenda will include student data reporting.

d. Government Relations

Ms. Charlupski reported that the House Budget was released today. The Senate budget is expected to be released in mid-May. There will be a Bridge for Resilient Youth in Transition (BRYT)/Bridge Program Legislative Briefing on May 4, 2017.

e. Policy Review

Dr. Jackson reported that the next meeting of the Policy Review Subcommittee will take place on May 10, 2017. The agenda will include continued review of the proposed Public Schools of Brookline (PSB) Authorized Signature Policy and Wellness Policy.

f. Additional Liaisons and Updates

Mr. Chang reported that he attended the Brookline Climate Action Committee meeting on April 24, 2017. Discussion focused on Town Meeting Warrant Article 21: Resolution regarding a Net Zero Energy Ninth Elementary School and the Expansion of Brookline High School, which will be discussed later this evening. He encouraged support for Article 21, as amended by the Advisory Committee. Dr. Jackson reported that her son has been selected as the first student from Brookline High School to participate in the Japan Society Student Leaders Program.

3. PUBLIC COMMENT

Elizabeth Myska, Co-Chairman of the Brookline Special Education Parent Advisory Council (SEPAC), spoke to clarify that SEPAC does not endorse political candidates. Rebecca Mautner, Town Meeting Member Precinct 11, spoke in support of Town Meeting Warrant Article 22: Resolution to Support the Town of Brookline's Participation in the Government Alliance on Race and Equity.

4. STUDENT REPORT

Ms. Jennings reported that students in the Brookline High School Environmental Action Club filed a Town Meeting Warrant Article asking Town Meeting to pass a resolution that would allow a gas tax to fund transportation infrastructure improvements around town. The Student Council supports the proposal. Ms. Jennings also reported on steps taken to prevent rodent issues in the school, school renovation, a recent ceremony to honor three enslaved men who marched from Brookline to the Battle of Lexington, and Student Legislature review of government structure.

5. SUPERINTENDENT'S REPORT

Mr. Bott provided the following report:

Brookline Education Foundation (BEF) Caverly Award Winners

Each year since 1984, the Brookline Education Foundation has presented the Ernest R. Caverly Award to Brookline teachers in recognition of their professional achievements and contributions to the Public Schools of Brookline. Two awards are given annually: one to a high school educator and one to an elementary school educator. The award is named for Dr. Caverly who served as a distinguished public school leader, educator and Brookline Schools Superintendent (1931-1964).

On April 12, I had the pleasure of joining BEF Executive Director Liz Ascoli, Board President Ian Roffman and Associate Director Molly Schmitt in surprise visits to the recipients of this year's Caverly Awards.

This year's elementary school recipient is Dominique Ferdinand, a kindergarten teacher at Lawrence. Early Education Principal Vicki Milstein joined us as we surprised Dominique, who was deep in grade-level planning work with her colleagues. This year's high school recipient is mathematics teacher Betty Strong. Headmaster Anthony Meyer, Dean of Students Lisa Redding, and High School math coordinator Josh Paris joined us to surprise Betty just as her Trigonometry class was about to begin. Dominique and Betty are incredibly deserving of this recognition. We look forward to joining educators and administrators from across the district to publicly honor them at the BEF Celebrating Teachers Reception at 3:30 p.m. on May 16 at Newbury College. The BEF will also announce the recipients of 20 teacher grants and 11 collaborative grants at the Celebrating Teachers Reception.

Brookline Education Foundation (BEF) Grants for Administrative Leadership and Educational Initiatives

We are grateful to the Brookline Education Foundation for funding several important system-wide grants in FY18 totaling \$59,000.

Leadership Team Retreats

The District Leadership Team (Superintendent, Deputy Superintendents, Special Assistant, and school principals/headmaster) continues to include a high number of administrators in their first or second year in Brookline. This grant will support District Leadership team building and leadership development work by funding team retreats – one in summer 2017 and one during the school year.

New and Interim Principal Mentoring Program

The challenges for new principals are many. Beyond the support of their Leadership Team colleagues, experience shows that new leaders benefit from additional mentoring, both individually and as a group. This grant provides for a mentored support program for new building leaders during this period of transition. (This is year two of this BEF grant.)

Superintendent Professional Development Grant

This grant enables the Superintendent to attend state and national conferences and learn how other school districts address problems similar to those Brookline faces, keeping Brookline schools in the forefront of educational innovation.

Making Informed and Intentional Instructional Decisions

During 2017-18, this grant was used to send a team of district leaders to the Harvard Graduate School of Education's Data Wise Institute. This group will embark on data inquiry work with teams during the summer and school year.

This grant will also fund a second cohort of district leaders to attend the January Data Wise Institute.

District-wide Educational Equity

For the second year the District will hold a full day of professional learning, focused on the core values of Educational Equity and Respect for Human Differences, to provide a common experience for all educators to reinvigorate ongoing work to eliminate racially predictable achievement gaps. BEF funding will support District hiring of high-profile experts in the area of cultural proficiency to work with educators on this professional development day.

The support of the BEF is fundamental to our district-wide efforts to ensure that we achieve equitable outcomes for all students. We are thankful for their continued support and we look forward to reporting on how these funds are successfully utilized next year to further our work.

Lawrence Principal Search Update

I communicated with Lawrence School families and staff earlier this week to provide them with an update on the search process for a new principal. The 14-member principal search committee completed first round interviews on April 13. The committee identified candidates to move forward into the next phase of the search process, which is already

underway. Finalists are participating in interviews with Senior Staff this week after which we will conduct in-depth reference checks. We have tentatively scheduled site visits next week to each of the finalist's current schools. A subset of the search committee will join me in these site visits. My goal remains to name a preferred candidate by May 11.

I want to publicly thank the members of the search committee who volunteered so much of their time for this work. The committee reviewed over 35 applications, developed interview questions, interviewed candidates over the course of three days, and identified finalists for the next phase of the search. The insight, expertise and thoughtful collaboration of the search committee has been an invaluable part of this important process.

Three Hidden Veterans

I had the honor of attending the Three Hidden Veterans ceremony on Tuesday, April 25, in front of the High School. The ceremony honored three enslaved Brookline men - Adam, Peter and Prince - who marched to the Battle of Lexington, April 19, 1775. Since 2015, the Town of Brookline has held flag-raising ceremonies memorializing Brookline veterans who served honorably in World War II, Korea and Vietnam. But the service of these three Brookline veterans of the Revolutionary War had been hiding in plain sight.

Adam, Peter and Prince marched to Lexington with the Brookline Militia, engaged with the British, and helped create a new nation. They were memorialized on Tuesday as some of Brookline's first veterans with a flag-raising and very meaningful ceremony.

BHS educator and Brookline resident Malcolm Cawthorne led the ceremony, which was sponsored by the Brookline Board of Selectmen, the Brookline Veterans Services Department, the Brookline Department of Diversity, Inclusion and Community Relations, The Hidden Brookline Committee: bringing to light the hidden history of slavery & freedom in our town, the Brookline American Legion Post 11 and the Brookline Veterans of Foreign Wars, Stephen F. Rutledge Post.

2017 U.S. Presidential Scholars

Please join me in congratulating Brookline High School senior Amir Siraj who was selected as one of the 723 semifinalists to advance to the final round of the 2017 U.S. Presidential Scholars competition. Of 3.5 million graduating seniors, only 5,100 students were identified as candidates in the program nationwide. The 723 semifinalists have been selected from across the country. The program, established in 1964, seeks to honor some of the nation's most distinguished graduating high school seniors and represents the highest honor for a high school student. The White House Commission on Presidential Scholars will select up to 161 U.S. Presidential Scholars in May. We wish Amir the best of luck!

13 Reasons Why

You may have heard news reports of a new television series airing on Netflix, 13 Reasons Why. The series is based on a young adult novel and focuses on a high school student who dies by suicide. The popularity of the show among young people has caused concern among suicide prevention professionals.

6. BROOKLINE HIGH SCHOOL (BHS) EXPANSION DESIGN ALTERNATIVES

a. Presentation of BHS Expansion Design Alternatives

Pip Lewis, HMFH Architects, presented BHS Expansion Design Alternatives (Attachment C), including Option 1 (enrollment accommodation), Option 2 (Massachusetts State Standards with Competition Gym at Tappan Gym), Option 3 (BHS Education Plan and with Field House Addition to Tappan Gym), Option 4A (New Cypress Building with BHS-Greenough; Renovate Existing Science and New 5-Classroom Addition); Option 4B (New Cypress Building with BHS-Greenough; New Science at Roberts Wing and Renovate 3rd Floor as Core Academics); Option 4C (New Cypress Building over the MBTA with BHS-Greenough; Renovate Existing Science and New 5-Classroom Addition); and Option 4D (New Cypress Building over the MBTA with BHS-Greenough; New Science at Roberts Wing and Renovate 3rd Floor as core Academics). Mr. Lewis provided a summary showing how the options address Program Priorities, as well as the estimated square footage and project cost. The estimated project costs range from \$105,342,903 (Option 4A; note does not include acquisition costs) to \$348,352,647 (Option 3). Mr. Lewis also provided cost estimates for optional projects (Tappan Gym Upgrades, Cypress Park Field and Playground, and Underground Parking).

b. Public Hearing on BHS Expansion Design Alternatives

Several residents, including Nina Brown (50 Gorham Avenue), Alan Price (12 White Place), Naomi Sweitzer (13 Griggs Terrace), and Elizabeth Schlosberg (275 Beverly Road) spoke regarding the proposed BHS Expansion Design Alternatives. Comments/questions included the following: See many benefits to Option 4; suggest that we investigate the possibility of placing some parking below the classrooms at 111 Cypress Street (instead of using parkland) and some parking in the neighborhood; Option 4 would seem to be the least disruptive to students (many of whom experienced elementary school renovations), take the least amount of time, and be the least expensive; like the notion of a 9th grade academy; need to be mindful of costs and would be unlikely to support a field house; asked how long the options might take (estimated at 18 to 30 months, depending on the option); would need to address student pedestrian safety with Option 4; asked how much of the existing campus would be taken down under Option 4 (would probably be a partial tear-down, with a significant portion getting reused; need to carefully consider the implications of taking parkland (was noted that the park is under the jurisdiction of the Park and Recreation Commission; the Commission is represented on the School Building Committee); be mindful of the potential impact of Option 4 on traffic flow; as enrollment grows, there should be a proportionate increase in arts space (Option 4 would include new space for art, science and music); asked whether 111 Cypress could accommodate an auditorium that would fit all students (no, because of

space and financial constraints); asked about next steps (the Building Committee for BHS Expansion is expected to vote on a preferred option on May 3, 2017); noted that the timeline is fast and that it is important that residents understand the process. There being no more people wishing to speak, Ms. Ditkoff closed the public hearing.

School Committee comments/questions included the following: strong preference for Option 4, noting that it best improves our science facilities, which may, in turn, provide opportunities to use existing space to address other deficiencies, that it seems to be least disruptive, least expensive, is best pedagogically of the options; concern that the process to seek authorization to build over the tracks would significantly delay the project; asked if the School Committee needs to vote to include 111 Cypress (no, Feasibility can explore that question); supports the concept of a 9th grade academy; strong preference for Options 4B and 4D because they improve art and science space and free up space for other improvements; would like to retain the courtyard and preserve the character of the main building; building over the tracks could provide additional needed space. The consensus of the School Committee was strong support for Option D as the preferred alternative.

7. Annual Town Meeting Warrant Articles

a. Article 16 – Authorize a 20-Year Lease of the Runkle School Rooftop to Host a Rooftop Solar Photovoltaic Installation

i. Presentation

Sandra Brown, Director of BlueWave Solar, explained regional solar initiative work done to date in Brookline - site assessments, Letter of Intent execution for short- and long-term projects, rate calculations, and finalization of system designs (Attachment D). Ms. Brown described next steps (finalize net metering credit purchase agreement and lease, obtain approvals, and construction/operations. Assuming Town Meeting approval, Advanced Solar will commence construction once all necessary permits and approvals are secured, likely in July 2017. The system is anticipated to begin operating during the fall of 2017.

ii. Public Hearing

George Cole, Brookline resident and member of the Building Commission, spoke in support of Warrant Article 16, but cautioned that we ensure that the Town/Schools are not responsible for repairing any roof damage that occurs during installation, nor for the cost of moving the panels when there are future roof repairs; be mindful of aesthetics; and keep the neighborhood informed. Ms. Brown responded that the contract will provide protections for the Town/Schools. When decommissioned, the roof will need to be returned to the condition appropriate for its age. The Runkle roof is the newest school roof and was designed for solar. Their crews will do regular maintenance. There being no more members of the public wishing to speak, Ms. Ditkoff closed the public hearing.

iii. Potential Vote

School Committee asked for further clarification on the process for reviewing the contract. It was noted that Town Counsel and the Building Department will review the

contract, which will be signed by the Board of Selectmen. The Metropolitan Area Planning Council (MAPC) will be available to provide economic and technical advising. Ms. Brown stated that a neighborhood meeting was held, but was not well attended. Ms. Brown stated that she will send the School Committee last year's presentation, which more clearly shows massing.

The School Committee agreed to convey to the Board of Selectmen the School Committee's understanding that in-house legal and building staff (with outside assistance, if appropriate) will thoroughly review the terms of the proposed contract.

ACTION 17-42

On a motion of Mr. Chang and seconded by Ms. Stram, the School Committee VOTED UNANIMOUSLY (8-0, Mr. Pollak was not present for the vote) to recommend that Town Meeting vote favorable action on Town Meeting Warrant Article 16 – Authorize a 20-Year Lease of the Runkle School Rooftop to Host a Rooftop Solar Photovoltaic Installation

b. Article 21 – Resolution regarding a Net Zero Energy Ninth Elementary School and the Expansion of Brookline High School

i. Presentation

Petitioners Werner Lohe, Alan Christ, and Kathleen Scanlon presented an explanation of Town Meeting Warrant Article 21: Resolution regarding a Net Zero Energy Ninth Elementary School and the Expansion of Brookline High School (Attachment E). Their explanation, in part, states the following:

Brookline was one of the first communities in Massachusetts to address climate change, adopting its first Climate Action Plan in 2002. In 2012, it accepted Massachusetts' target for greenhouse gas emissions—the reduction in emissions to 80% below 1990 levels by 2050. In part in response to this statement of our community's environmental values, recent school construction (i.e., the Runkle School and the Devotion School) has achieved high standards of energy efficiency. Similarly, in the Educational Plan for the Ninth Elementary School at Baldwin, the School Committee has stated its strong commitment to a state-of-the-art school. Nevertheless, overall town-wide progress toward reduced emissions has been slow, not yet approaching the rate needed to reach our goal.

New construction of any sort inevitably leads to a slight increases in emissions. Therefore, communities throughout Massachusetts and New England have begun to address that reality by designing "Net Zero Energy" (NZE) schools, that is, schools that minimize on-site energy use as much as possible, and offset that energy use with renewable energy generated on site, with the goal of equalizing, on an annual basis, energy consumed and renewable energy generated on site. (Examples of such schools are the Martin Luther King School, Cambridge, MA, the King Open School, Cambridge, MA, and the Pell Elementary School, Newport, RI.) Schools such as this not only address climate change, but typically save money by reducing energy costs. Brookline now has an opportunity to design its own NZE school.

About ten years ago, Brookline improved its design process for municipal buildings by including consideration of “environmental and sustainability goals and objectives, includ[ing] design and construction practices that explicitly consider Green technologies.” Bylaws, § 7.3.2(a). Because of the Town’s sound design practices and the Massachusetts School Building Authority’s (MSBA) standards, the Runkle and Devotion projects have good energy performance, though they are not NZE. The Ninth Elementary School at Baldwin, however, is not subject to MSBA requirements, and no specific standards for energy performance have been set. This resolution provides direction to the design architects, under the supervision of the School Committee, the Building Commission, and the Board of Selectmen, by establishing Net Zero Energy as the community’s goal for its new school. This goal encourages energy efficient building design, places a major focus on energy, and encourages building users to reduce their energy needs without compromising building programs or mission.

The most widely used and accepted rating system in the United States and in the world for green buildings is LEED v4 (Leadership in Energy and Environmental Design, version 4). Building performance is measured by awarding up to a total of 110 checklist points in eight categories and 57 sub-categories. The single sub-category in which the most points are available—with 16 points—is Energy and Atmosphere: Optimize Energy Performance. Consultation with architects and energy efficiency consultants familiar with schools, NZE principles, and LEED indicates that the realistic goal for achieving NZE for the Ninth Elementary School at Baldwin is 16 Optimize-Energy-Performance points with certification at the Platinum rating level with Energy Use Intensity (EUI) of 25 kBtu/sq.ft.

NZE schools, in addition to addressing climate change, typically save money by reducing energy costs. Further, the Town and its architect are committed to “integrated design,” which is the most effective process for ensuring not only energy efficiency, but also cost savings. Integrated design relies on careful planning and goal setting early in the design process. Project costs and savings are evaluated during the schematic design phase of the project. Thus, the savings (or costs) that may be realized by designing an NZE school cannot be accurately assessed at present since schematic design has not yet been authorized by Town Meeting. But, consistent with best practices, an important provision in the Town’s bylaw requires that “the consultant [(the architect) prepare] a cost estimate for the project (including life-cycle costs) [and] consider the investigation, cost-benefit analysis, and recommendation of appropriate options that address the environmental and sustainability goals and objectives....” Bylaw, § 3.7.2(b).

The bylaw does not specify the methodology to be used in the cost-benefit analysis. The Town has frequently used “simple payback period” methodology. But, because of the complexity of the design of a large project such as the Ninth Elementary School at Baldwin and the desire to achieve the goal of NZE, financial analysis of energy-related features such as geothermal HVAC, solar PV panels, triple-glazed windows, and the like should include a Discounted Cash Flow (DCF) analysis, considering Internal Rate of Return (IRR) or Net Present Value (NPV) or both, based upon the life of the features, and should also consider the Social Cost of Carbon.

The petitioners stated that they understand the concerns that have been raised with regard to budget constraints. The Advisory Committee has recommended changes, which include the following: changing “Platinum” to “Silver” for the LEED minimum standard, changing the minimum energy optimization points from 16 to 13, while keeping a goal of 16; changing the EUI standard, and deleting “at Baldwin” throughout so that the ninth school is not specific to any site for the purposes of the article (Attachment F).

ii. Public Hearing

Claire Stampfer, Town Meeting Member Precinct 5, spoke in favor of the warrant article. Pip Lewis noted that the Brookline Devotion School will be more energy efficient than the Pell School in Newport, Rhode Island. There being no more members of the public wishing to speak, Ms. Ditkoff closed the public hearing.

iii. Potential Vote

School Committee members expressed support for the warrant article, as amended by the Advisory Committee. Ms. Stram moved to recommend that the Advisory Committee’s proposed language be amended by adding back “at Baldwin.” The motion was seconded by Ms. Schreiner-Oldham. The School Committee unanimously supported this amendment.

ACTION 17-43

On a motion of Mr. Chang and seconded by Dr. Jackson, the School Committee VOTED UNANIMOUSLY to recommend that Town Meeting approve the Advisory Committee’s proposed motion on Article 21, as amended by adding back “at Baldwin.”

8. NEW BUSINESS

There was no new business to discuss.

9. PROPOSED EXECUTIVE SESSION

By unanimous roll call vote at 9:15 PM, the School Committee entered into Executive Session for Purpose 3, to discuss anticipated litigation strategy pertaining to the proposed 9th elementary school at Baldwin because an open meeting may have a detrimental effect on the School Committee’s litigating position and for Purpose 2, to conduct strategy sessions in preparation for negotiations with nonunion personnel or to conduct collective bargaining sessions or contract negotiations with nonunion personnel regarding architectural services pertaining to the proposed Brookline High School Expansion.

Note: Ms. Ditkoff made the following statement at the May 4, 2017 School Committee meeting: “At last week’s School Committee meeting (April 27, 2017), the School Committee went into executive session for two purposes, one of which was Purpose 2 (Proposed executive session for the purpose of conducting strategy sessions in preparation for negotiations with nonunion personnel or to conduct collective bargaining

sessions or contract negotiations with nonunion personnel regarding architectural services pertaining to the proposed Brookline High School Expansion). We have since been told that discussion of vendor contracts is not a permissible reason under this purpose. Minutes of the entire discussion will be posted with the open session records on the Town Website.” Here are the minutes of the discussion.

The School Committee discussed whether to request that the Building Commission issue a Request for Quotations for architectural services for the schematic design phase of the Brookline High School Expansion project. Co-Chairmen of the Building Committee for 9th School Expansion Nancy Heller and George Cole were also present. The School Committee discussed how this might impact the schedule, noting the time involved in soliciting requests and for potentially bringing on a new architect who is not familiar with the project. There was general agreement that we want the best possible design for the school, even if this would result in a delay.

10. ADJOURNMENT

Ms. Ditkoff adjourned the meeting at 10:55 PM.

Respectfully Submitted,

Robin E. Coyne, Executive Assistant
Brookline School Committee

PUBLIC SCHOOLS OF BROOKLINE

APPLICATION FOR ALL OUT-OF-STATE, OVERNIGHT FIELD TRIPS

TRIP LOGISTICS:

1. Name of Field Trip: **Upper Devotion 8th Grade Field Trip to New Hampshire**
2. Educator Requesting Field Trip Approval: **Monica Crowley, Principal**
3. School: **Upper Devotion School** Grade Level: **8th Grade**
4. Have you reviewed the *School Committee Policy I 3 a. for School Sponsored Field Trips*? **Yes**
5. Have you reviewed the document "*Administrative Procedures for School Sponsored Field Trips*"? **Yes**
6. What is your destination? **Camp Cody, 9 Cody Rd, Freedom, NH 03836**
7. What is the date and time you are leaving school? **Thursday, May 18, 2017-3:15pm**
8. What is the date and time you are returning to school? **Friday, May 19, 2017-8pm**
9. Do the dates of the trip conflict with any religious holidays or observances? **No**
10. How many days will students miss from school? **1**
11. How are students being transported (school bus, chartered bus, plane, rail, etc)? **Chartered bus.**
12. How many students will be participating in the field trip? **Approximately 72**
13. What members of the student body are eligible for the trip? **All eighth graders are eligible.**
14. How are students selected to participate in this field trip? **All are included.**
15. Where will students be staying?

Camp Cody, 9 Cody Rd, Freedom, NH 03836

16. What are the names and cell phone numbers of the primary staff chaperones on the trip?

Monica Crowley, Vice Principal, 1-617-947-2614

17. Other than those listed in #16 above, what are the names and roles (teacher, parent, etc) of other adult chaperones who will be on the trip?

Parents Going up on Thursday and Staying Overnight: Cindy Tungate, Abby Cox, Christina Kovach, David Levy, Roberto Sanchez-Dahl, Mark O'Neil, Ed Dumas, Len Watkins, Mark Gigliette

EDUCATIONAL RELEVANCE

18. What is the purpose of the trip and how does it relate to Brookline's Learning Expectations (K-8) or BHS Course Syllabi?

The purpose of this trip is to provide an educational culminating experience for the students before they graduate from Upper Devotion School.

19. Describe activities planned before the trip to prepare students:

N/A

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20. Describe activities planned after the trip for students to wrap-up/reflect:

Students will debrief in class and complete written reflections of their experiences.

ACCESSIBILITY AND STUDENT SAFETY

21. To what extent does the field trip group reflect the diversity of the school population? What efforts will be undertaken to ensure that, to the greatest extent possible, the participants in this field trip reflect the diversity of the school population?

All eighth grade students will participate in the field trip and all of the related activities.

22. What measures are planned to ensure student safety on the trip, including chaperone coverage, student behavior contracts, etc.

The only staff member attending will be principal, Monica Crowley, with nine parents. Camp Cody will provide one camp staff member for every ten students, so 7-8 in total to run activities while we are at the camp.

23. What is the name and location of the medical facility closest to your destination?

Memorial Hospital, 3073 White Mountain Hwy, North Conway, NH, 03860: 1-603-356-5461

24. Will students be swimming? **YES**

(please note: if swimming is planned, parents must give explicit written permission for their child to swim; students who do not have this express written permission will not be allowed to swim).

25. If travelling outside of the United States, please attach the appropriate Department of State Advisory Review and Notification and a copy of Consular statement. **N/A**

FUNDING

26. What is the total cost of the trip?

The cost of the trip is only \$63/student. This very affordable cost is due to the fact the Devotion PTO is generously funding the cost of the busses which totals about \$3,500.

27. How will the field trip be funded?

Families will be asked to pay as much as they can afford toward the cost of the trip. The PTO makes a donation to the trip scholarship fund, and students and families sponsor fundraising efforts to raise money for the trip scholarship fund.

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28. What accommodations are made for students who cannot afford the expenses of this trip? Are partial and/or full scholarships available?

Yes. Partial and full scholarships are available. No student will be denied a chance to participate in the trip because of an inability to pay.

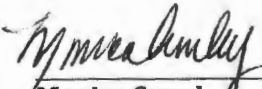
29. If fundraisers are planned to help lower the cost of the trip for all students, please describe those plans here:

Bake sales during many school wide events

Family donations

Signatures:

Principal:


Monica Crowley

3/18/17
Date: 3/18/17

Brookline High School Field Trip Application Brookline-Angers Exchange 2017-2018

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Additional Documents:

Forms to be completed by parents/students

Prepared by:

Andrew Kimball, French Teacher, BHS, Lead Chaperone
Jen Martin, History Teacher (SWS), BHS, Lead Chaperone
Laura Gurry, French Teacher, BHS, Lead Chaperone
Scott Butchart, French Teacher, Dean of Students, Chaperone

Agnès Albérola, World Language Curriculum Coordinator

STATEMENT OF PURPOSE

French-American Exchange between Angers and Brookline
October 7-18, 2017 (French students in Brookline)
February 15, 2017 – March 3, 2018 (BHS students in France)

Brookline High School (BHS) students and teachers are continuing an exchange program with a French *lycée* in Angers, France. This bi-annual exchange was new to Brookline in 2010 when we completed an incredibly successful exchange, and it has been run three times total (again in 2012, 2014 and 2016).

The French-Angers Exchange directly responds to the third goal of the Strategic Plan of the Public Schools of Brookline, which commits us to equipping students with the intellectual and social skills necessary to thrive in an increasingly complex and diverse global society. In accordance with Strategy 3.1, we will “develop opportunities to enhance the global awareness of every student.”

The exchange will include approximately 27 BHS students. For the first part of the exchange, Brookline High School families will host the French students and their English teachers in October of 2017. For the second part of the exchange, Brookline High School students will be hosted by the Angers high school students’ families. They will also spend five days in Paris, and then three days discovering Normandy. The eligible students study French, and are interested in French culture, art, architecture, or history. BHS French teachers Andrew Kimball and Laura Gurry, as well as BHS History (SWS) teacher Jen Martin; Dean of Students and French teacher Scott Butchart will help chaperone.

The goals of this trip are to experience life as a member of a French family, to speak and interact in French, to visit Paris and Angers, to eat food typical of these regions, and to expand students’ knowledge of French culture.

Highlights of the itinerary will include:

Angers and its surroundings:

- Life in the host family
- Classes taken at the French *lycée*
- Visits of the Loire valley and the Atlantic coast

Paris and its surroundings (Following list not exhaustive)

- Notre Dame Cathedral
- The Arc de Triomphe
- The Eiffel Tower
- The Louvre, Orsay, and Quai Branly (non-Western art) Museums
- The Seine River, by boat

Normandy

- The abbey at Mt. St. Michel
- The Museum of Fine Arts, Rouen
- The Bayeux Tapestry
- A visit of the D-Day beaches and the Battle of Normandy museum in Caen

It is worth noting that this year for the first time, we will be planning our trip with the help of ACIS (a Boston-based student travel company). We find this to be necessary in an increasingly complex political climate that may affect international travel abroad. For instance, they will provide Brookline families with additional resources in the event of an emergency, optional supplements on trip insurance, additional financial aid and organizational help to the teacher chaperones.

APPLICATION FOR ALL OUT-OF-STATE, OVERNIGHT TRIPS

TRIP LOGISTICS:

1. **Name of Field Trip:** French-American Exchange between Angers and Brookline
2. **Educators Requesting Field Trip Approval:**
Andrew Kimball, Jen Martin, Laura Gurry
3. **School:** Brookline High School **Grade Level:** 11th and 12th
4. **Have you reviewed the new School Committee Policy I 3 a. for Field Trips?** Yes
5. **Have you reviewed the document, *Administrative Procedures for School Sponsored Field Trips*?** Yes
6. **What is your destination?** Angers, Caen and Paris, France
7. **What is the date you are leaving Brookline?** Thursday, Feb. 15, 2018
(anticipated evening departure)
8. **What is the date you are returning to Brookline?** Saturday, March 3, 2018
(anticipated afternoon return)
9. **Do the dates of the trip conflict with any religious holidays or observances?** No
10. **How many days will students miss from school?** Six
11. **How are students being transported?**
 - a. from Boston to Paris: airplane
 - b. in Paris: foot and public transportation (subway)
 - c. from Paris to Angers: coach bus
 - d. Excursions from Angers: coach bus
 - e. Angers to CDG Airport: coach bus
 - f. in and around Normandy: coach bus
 - h. Paris-Boston: airplane
12. **How many students will be participating in the field trip?** +/- 27 students
13. **What members of the student body are eligible for the trip?** All 11th and 12th grade students who study French and in French 2 or higher are invited to participate.
14. **How are students selected to participate in this field trip?** Students will be selected based on the following criteria:
 - Strong academic record (good study habits, participation, attendance)
 - Interest in learning the French language, history and culture
 - Ability to serve as an ambassador from Brookline and the United States
 - Sensitivity to other people and cultures
 - Maturity level
 - Successful completion of the application
 - Perceived potential for personal growth
15. **Where will students be staying?**

while in Angers, students will be staying with host families in the small city of Angers, France. In Normandy, we will be in a hostel run by the national hostelling association. Finally, while in Paris, students and chaperones will be staying in MIJE, International Student Residence in 4-bed dormitories which are centrally located in the Marais. Afterwards,
16. **What are the names and cell phone numbers of the primary staff chaperones on the trip?** All of our chaperones have native or native-like fluency in French, have experience with international travel, and have led student groups and/or exchange trips in the past. They will be supervising the students and available to them on a 24-hour basis.
 - a. Andrew Kimball, BHS French Teacher, 617-595-6711
 - b. Jen Martin, BHS Soc. St. Teacher, 617-642-2416
 - c. Laura Gurry, BHS French Teacher, 617-721-2238

While in France, the teacher-chaperones will be using their American cell phones for easy parent contact.

17. Other than those listed above, what are the names of other adult chaperones on the trip?

Scott Butchart, BHS Dean of Students, French Teacher

We also have three contacts who are available to us throughout the trip:

Beatrice Ernot
beatrice.ernot@laposte.net
011.33.2.41.68.16.26

Sophie Daudin
sophiethomas@free.fr

Christine Eychenne

EDUCATIONAL RELEVANCE:

18. What is the purpose of the trip and how does it relate to the Brookline High School Course Syllabi?

The purpose of this trip is to create an opportunity for BHS students to further build their language skills developed in the classroom, in addition to learning the skill of adapting to a new culture, essentially executing their role as a new global citizen. A secondary goal is to build a reciprocal relationship between Brookline and Angers, and well as among the community of French students at BHS. The French students will spend two weeks in October in host families. Our students will join them as guests for ten days in February. Goals include increasing the linguistic proficiency and cultural understanding of our students of French. The exchange will also foster deep and lasting connections between our Brookline students and students of Angers High School in France. Students will live with host French families, interact with French people, attend a French High School, and explore Angers and the surrounding Anjou region. In addition, five days will be spent in Paris visiting cultural sites. Throughout the trip, the students will have a rich and unique immersive experience. They will be asked to keep a journal, make presentations to French classes about their own culture, and share their experience upon return. During the exchange in Angers, the students will shadow their peer host through their morning classes at the high school. Most afternoons will be dedicated to exploring the city, and participating in local excursions to better understand this particular region of France. (see complete description attached) While in Paris, students will discover the major sites, including guided tours of museums. Students will also get a more personal feel of the city through spontaneous interactions and walking tours. This exchange directly complements our French program at Brookline High School. Our goals for students of French at BHS are that they develop as fully as possible their communicative competence and familiarize themselves with another culture.

19. Describe activities planned before the trip to prepare students:

Students will meet regularly during the first semester (typically during X-Block) to prepare the students for the hosting and travel experience. Here they will become familiar with the ground rules and expectations, the trip itinerary, and their responsibilities as hosts and guests. They will each research and present one of the sites we will visit. Correspondence between exchange students will be ongoing. This will encourage real ownership of this trip. Students should also be prepared to make a presentation to their host school concerning the town of Brookline, Boston, New England or another area of interest to them.

We also plan to have a few team-building activities in September in advance of the French students' arrival. In the past, this has taken place at the Brookline Teen Center.

Students participating in the exchange will be responsible for completing academic assignments while participating in the exchange and will be responsible for presenting those assignments to their teachers within two weeks after their return. While in Angers, we have a wing of their library reserved for us, which the teachers will run as a sort of Tutorial in which we monitor and help kids stay focused on keeping current with their other courses, perhaps even facilitating communication between student and BHS teachers.

20. Describe activities planned after the trip to wrap-up/reflect:

Upon return, students will share their experience at a French-American Exchange Celebration (pot-luck dinner, slide show). Families, BHS educators, school administrators and School Committee members will be invited. Students will be encouraged to pursue connections with their French peer in the following months.

By the spring of 2016, this forum may have evolved to resemble a sort of conference where all student trips share with one another to parents, other students and the Brookline community.

ACCESSIBILITY AND STUDENT SAFETY:

21. To what extent does the field trip group reflect the diversity of the school population?

What efforts will be undertaken to ensure that, to the greatest extent possible, the participants in this field trip are representative of the school population?

This trip is open to all 11th and 12th grade students of French at BHS, taking French 2 or higher. The exchange will be widely advertised in class by teachers of French, as well as through programs such as Steps to Success and African-American/Latino Scholars. In announcing the trip, we will be sure that students know about available financial aid (see #28) so that they understand that the cost of the trip should not deter them from applying to participate. Historically, each trip has had at least three STS students accepted who received financial aid.

In 2010, on our first exchange, we had 22 students participate. Among the participants, 15 were White (68.2%), 6 were Hispanic/Latino (27.3%), and 1 was Asian (4.5%). Of the 22 students who participated, 10 had a home language other than English. The 2012, 2014 and 2016 groups represented a similar demographic. There is no reason to believe our 2018 group would be any different.

Once the School Committee has approved the trip, we will make a concerted effort to encourage participation from a broad spectrum of BHS students so that the group reflects the makeup of the larger student body. We will make every effort to accommodate any physical and emotional disabilities that students may have. Most of the museums and tourist sites that we plan to visit are wheelchair accessible and offer audio tours, and the transportation we plan to utilize (plane, tour bus and public transportation) are generally handicap accessible.

22. What measures are planned to ensure student safety on the trip, including chaperone coverage, student behavior contracts, etc?

At all times three faculty members will be on duty as chaperones. In all cases, at least one chaperone will have served as a past trip leader. The adults know the students well based on classroom contact and/or previous group-building exercises. These chaperones will be with our student group 24-hours a day in Paris and Normandy, and available by telephone while in homestays. We will meet with parents during the planning portion of the exchange to outline our behavior expectations, and all students who participate will sign behavior contracts. France is the most visited country in the world and generally exceedingly safe for visitors. We will follow any State Department advisories in the unlikely event any are issued.

23. What is the name and location of the medical facility closest to your destination?

Angers: Hôpital Local SAINT-NICOLAS
14, rue de l'Abbaye 49016 ANGERS

Paris: Hôpital St. Vincent de Paul
82, avenue Denfert-Rochereau 75014 01.40.48.81.11
Bayeux: Centre Hospitalier Universitaire de Caen
53 Boulevard de la Charité 14000 Caen, France 02 31 06 31 06

24. Will students be swimming? No

25. If travelling outside of the United States, please attach the appropriate Department of State notices and advisories. There are currently no State Department travel alerts or travel warnings for France. We will continually monitor the State Department website in the unlikely event of changing conditions. The United States Embassy is located at: 2 Avenue Gabriel, Paris 8 (tel. [33] (1) 4312-2222).

FUNDING:

26. What is the cost of the trip?

The projected cost is \$3,100 per student. This cost includes: round trip airfare on scheduled airlines, tour buses / airport transfers, and miscellaneous administrative expenses as well as the following:

In Angers:

- accommodations in French homes
- breakfast, lunch (provided at school) and dinner daily
- excursions in the surrounding areas

In Paris:

- accommodations in MIJE in the Marais
- breakfast **and** lunch or dinner at MIJE
- museum entries
- transportation (metro tickets)

In Normandy

- entry fees
- transportation
- accommodation
- breakfast, lunch and dinner at the youth hostel

For a detailed budget, please refer to the budget document.

27. How will the field trip be funded?

The trip will be funded by the families of participants. As described in #28 and #29 below, we will actively engage in fundraising efforts to raise money for financial aid, and students will be encouraged to assist their family by contributing to the cost through their own work.

28. What accommodations are made for students who cannot afford the expenses of the trip? Are partial and/or full financial aid available? Yes, fin will be available. As we did for the last exchange trips, we will advertise throughout the school: at Advisory Meetings, in French classes, during "InfoTime" and to students in BHS programs such as Steps to Success, METCO, and African American/Latino Scholars. In all of our outreach efforts we will include language that makes it clear that all students are encouraged to apply regardless of financial ability. Furthermore, we will let students know that financial aid are available and that all financial aid requests will remain confidential. Students may complete a separate financial aid application that

will in no way restrict their selection to participate. The equivalent of two full tuitions will be divided among students in need. If additional funds are needed, additional fundraisers will be planned to supplement them. Scott Butchart, Chair of the Brookline High School Scholarship Committee, has agreed to help us review financial aid applications, determine financial needs, and explore additional possibilities for funding. Acceptance to the exchange will be determined separately from financial aid awards so as not to influence selection process.

In the past, we have not had to deny anyone admission based on financial need.

29. If fundraisers are planned to help lower the cost of the trip for all students, please describe those plans here.

A number of fundraising opportunities will take place, including bake sales, France-themed t-shirt sales and other events (movie nights, etc). Families will have the option of contributing financially to a travel financial aid fund. We will also solicit corporate/private donations from local French businesses and US companies. A committee of students and parents will be formed to assist in fundraising efforts. All students involved are expected to participate actively in all fundraising activities.

Other relevant information for the Brookline-Angers Exchange:

Emergency Numbers in Angers:

- SAMU : 15
- Police secours : 17
- Pompiers : 18
- Pharmacie de garde : La nuit, jusqu'à 22h, appelez le commissariat central au 02 41 57 52 00.
- Après 22h se munir d'une pièce d'identité et d'une ordonnance
- Grande Pharmacie : 3 Bd Gaston Ramon Carrefour St Serge - 49100 Angers - 02 41 43 50 01
- Médecin de garde : 02 41 33 16 33
- Maison médicale de garde : 02 41 32 54 49
- Urgences : 02 41 35 37 12
- Clinique de la main : 02 41 86 86 41
- Centre Antipoison : 02 41 48 21 21
- Urgence dentiste (le week-end) : 02 41 87 22 53

Centre Hospitalier Universitaire d'Angers: <http://www.chu-angers.fr/>

- Hôpital Local SAINT-NICOLAS
- 14, rue de l'Abbaye 49016 ANGERS
- Tel : 02 41 73 41 00 Fax : 02 41 73 43 00
- hospital-local@hlmr-stnicolas-angers.fr

Emergency numbers in Paris:

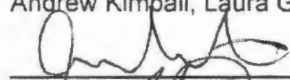
The clinic of the American Hospital of Paris provides a medical and surgical emergency service 24 hours a day, 7 days a week. Patients are looked after by a bilingual team (French-English). These practitioners can call on specialists, accredited by the hospital, who are on call in more than twenty medical and surgical fields. There is always a cardiologist and intensive care doctor on duty at the American Hospital.

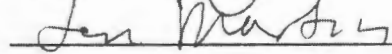
- American Hospital of Paris - 63 boulevard Victor Hugo - 92200 Neuilly-sur-Seine
Telephone: 33 (0)1 46 41 25 25 - Fax: 33 (0)1 46 24 49 38
<https://www.american-hospital.org/index.php?L=3>
- Hôpital Saint-Vincent de Paul
82 avenue Denfert Rochereau
75014 Paris
Phone: 011.33.01.40.48.81.11

Submitted by:

Educators: Andrew Kimball, Laura Gurry and Jen Martin, BHS

Signatures :





Date: 4/13/17

Date: 4/13/17

Date: 4/13/17

World Language Curriculum Coordinator: Agnes Alberola



Date: 4/13/17

Headmaster: Anthony Meyer



Signature :

Date: 4/13/17

PROPOSED TIMELINE

APPLICATION & RECRUITMENT TIMELINE 2017-2018

Sept.-Oct.	Teacher chaperones meet to prepare documents
Nov.-Feb.	Work with representative at ACIS to plan budget and itinerary
March	Teachers meet to prepare relevant documents
mid-April	X-block meeting of interested students: Q and A (panel of past participants) Applications available
May 5, 2017	Student Application Deadline Review applications, interview as needed
May 19,	Notify students/families of acceptance Accepted student bonding activity at BTC
May 31, 2017	Parent Meeting #1 Deposit due
June	Pairing of exchange students: Brookline-Angers
2x monthly throughout 1st semester	X Block Meetings
fall	families pay ACIS, in installments as necessary
X Blocks	Organizational Meetings: fund-raising, trip information, planning French students' visit
Oct. 7 th -18 th	French Students Visit
early December	Full payment due (remaining \$850)
Back-to-school in January	All documents due: passports, proof of medical insurance, etc.
mid-January	Final Parent/Guardian & Student meeting
Feb 15 th – March 3 rd	Trip to France

Student Guidelines and Contracts

Students participating in the 2012 Brookline-Angers Exchange Trip to Angers and Paris, France will abide by Brookline High School rules and regulations as stated in the Handbook, as well as those enumerated in this document.

1. Students and parents/guardians will agree to and sign the rules explained in the STUDENT BEHAVIOR CONTRACT (see attached).
2. Students and parents/guardians will agree to and sign the information and rules explained in the HEALTH INFORMATION SHEET FOR FIELD TRIPS (see attached). Students on medication must be able to self-medicate or give early notification if a parent/guardian must accompany the trip.
3. Parents/guardians must agree to and sign the information explained in the PERMISSION AND RELEASE FORM FOR PARIS TRIP (see attached).
4. Students must be in good academic standing to participate in the program.
5. Students must be aware of and follow airport and terminal procedures.
6. Students are responsible for their own spending money. The chaperones will be responsible for the group expenses.
7. Students are responsible for acquiring or renewing their own passports. They must make five copies of their passport and give them to Ms. Albérola, Ms. Gurry or Mr. Kimball.
8. Students must honor all deadlines and attend all student and parent/student meetings. If this is not possible, they must see Ms. Albérola, Ms. Gurry or Mr. Kimball in advance to receive the information.

In addition, the student and his/her parent are advised that School Committee and the Superintendent may cancel a trip at any time, including up to the day of departure, if that it deemed necessary for student safety.

I, (print student name) _____, have read the packet and agree to the information listed above.

Student's Signature: _____ Date: _____

I, (print parent name) _____, have read the packet agree to the information above.

Parent's/Guardian's Signature: _____ Date: _____

STUDENT BEHAVIOR CONTRACT

CODE OF CONDUCT:

1. I agree that while I am traveling with the Brookline High School Exchange Trip to Angers and Paris, France, I will not use or possess non-prescription drugs, tobacco or alcohol, nor will I be present while someone else is using or possessing non-prescription drugs, tobacco, or alcohol.
2. I agree to and will abide by the curfew. At the appointed time, I will go to my room and observe quiet. Bed checks will occur after curfew to ensure that the students are where they are supposed to be.
3. I agree to uphold all laws and regulations as dictated by local, State and Federal governments (i.e. under age drinking, assault, shoplifting). I agree to respect the property of others and to pay for any losses and/or damages for which I am found responsible.
4. I agree never to leave the hotel without permission from my chaperone. Should I be granted permission to leave, I will be accompanied by other members of my group.
5. I understand the responsibilities of traveling in a group and will respect meeting times and places.
6. I understand that I am under the direct responsibility and authority of the assigned chaperones, and I will follow and abide by any rules and/or decisions made by these persons.

I am signing this Contract, and hereby giving my word of honor that I will follow the above Code of Conduct while traveling in France with Brookline High School from February 18th – March 2nd, 2012. As a representative of Brookline High School, I understand that my behavior is critical to the success of the trip and as a precedent for future exchanges. **Any violation of these rules will result in disciplinary action (i.e. being sent back to the United States at my parent's expense, or being suspended from a performance or activity).**

Student's Printed Name: _____ Date: _____

Student's Signature: _____ Date: _____

I have read the above contract signed by my child.

Parent/Guardian Signature: _____ Date: _____

APPENDIX I.1 : PARENT TESTIMONIALS AND FEEDBACK

A majority (65%) of parents surveyed were in favor of extending the trip's length as well as its cost.

A majority (73%) of parents would support their Senior missing two weeks of school.

A majority (92%) felt that the 2013-2014 price (dollars) per trip length (days) ratio was fair. In 2015-2016 we're proposing an even more favorable ratio. The cost break-downs are \$160/day (2014) compared to \$119/day (2016).

When asked about the task of catching up on missed work, parent responses included ...

"Not a big deal." (x2)

"It was fine, neither 'not a big deal' nor 'overwhelming'. I'd say manageable. It was unfortunate that SAT's were the first weekend back.

"Not overwhelming.....but being a senior made a big difference in the degree of stress."

"Pretty manageable."

"Seemed OK."

"The benefits of going to France far outweighed any academic challenges. Teachers have been helpful in allowing extra time to catch up."

APPENDIX I.2 : STUDENT TESTIMONIALS AND FEEDBACK

A majority (100%) of Seniors and a majority (88%) of Juniors would have liked to spend three weeks in France instead of two, even considering the added concerns of more work to make up.

"Although one would miss more school, it would be totally worth it for the experience. I don't think we were there long enough."

"21 is a huge number. I would not be concerned about missing 8 class [sessions per course]."

"Every second it was so worth it and I would have liked staying in Angers for the extra week."

"Knowing what I know now I certainly would have jumped at the opportunity, I'm not sure how you'll sell it to future classes though."

A majority (80%) of students report that Paris would have been a more enjoyable experience with more time so that students could visit sites more leisurely, see more sites, and have more time to rest. Students self-reported moderate or great linguistic and cultural improvement in the following areas:

- fluency/pace of speech – 87%
- pronunciation/accent – 88%
- vocabulary building – 70%
- grammar – 43%
- cultural awareness – 93%
- confidence – 87%

Students self-reported that a third week in France would have helped increase their French skills in the following areas.

- fluency/pace of speech – 68%
- pronunciation/accent – 62%
- vocabulary building – 81%
- grammar – 37%
- cultural awareness – 50%
- confidence – 75%

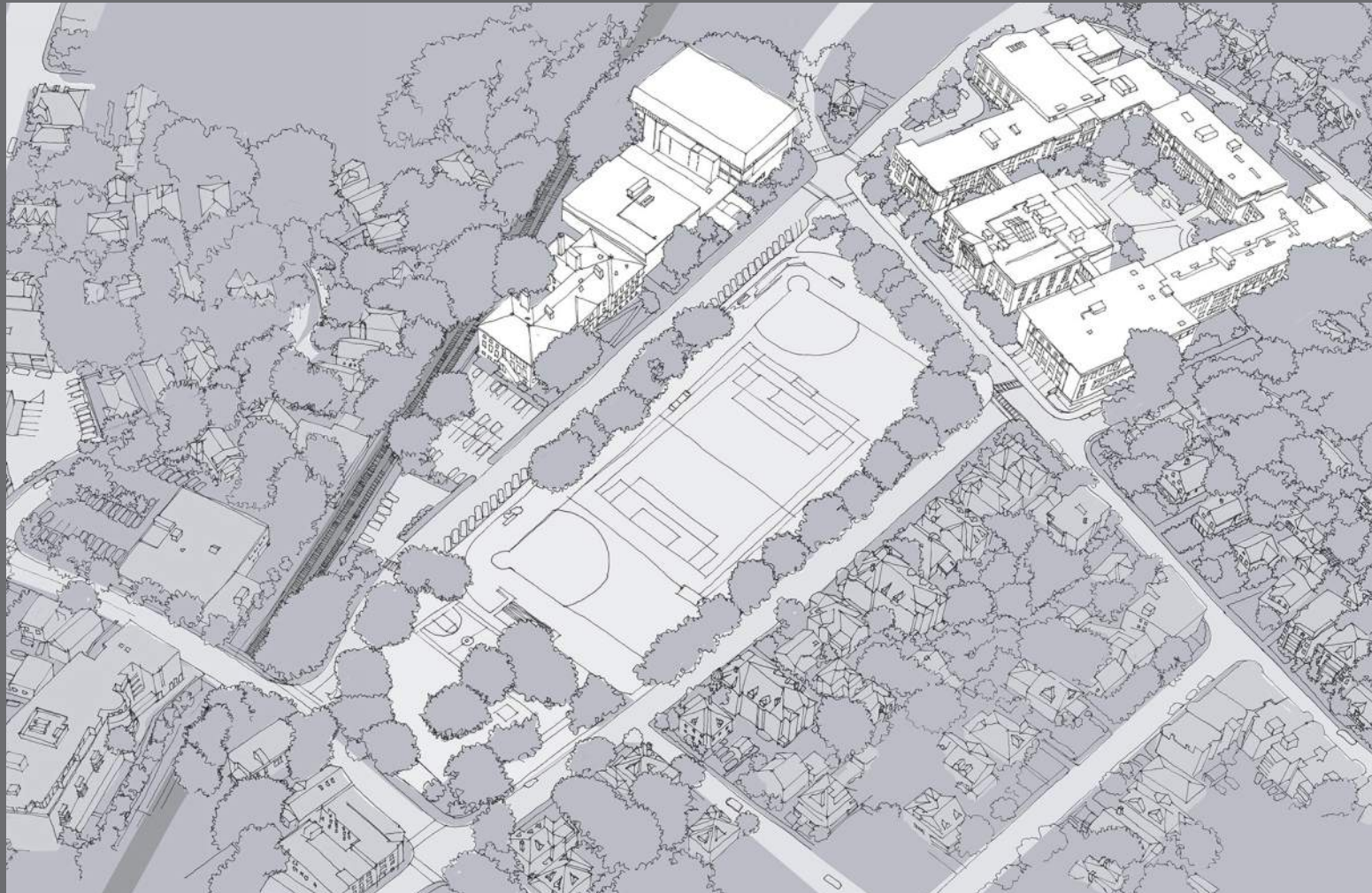


Brookline
School Committee

Brookline High School
Campus Expansion Update

Feasibility Study

APRIL 27, 2017



HMFH ARCHITECTS

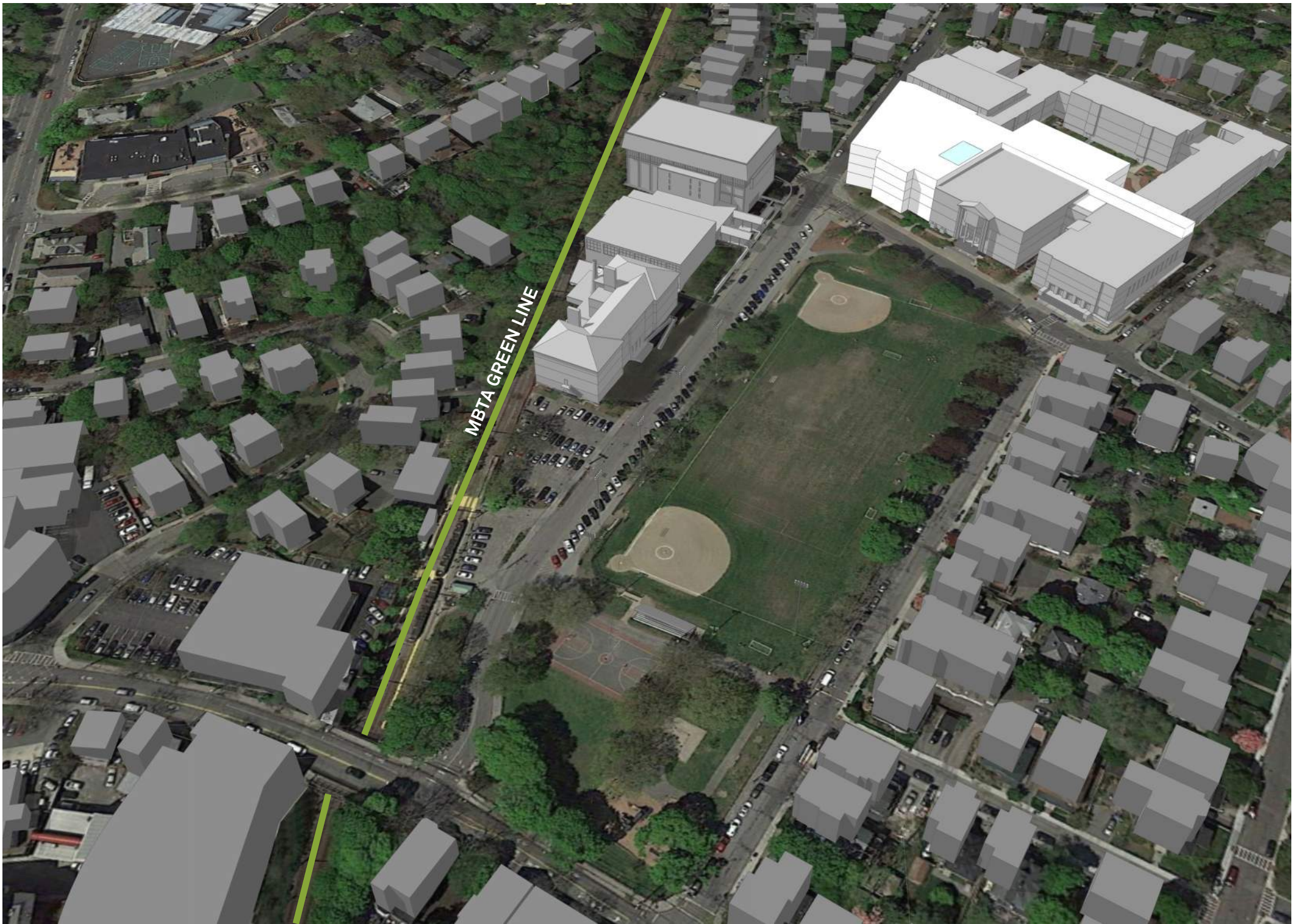
130 Bishop Allen Drive, Cambridge, MA 02139 617 492 2200 hmfh.com

Existing



Massing

OPTION 1
Enrollment Accommodation

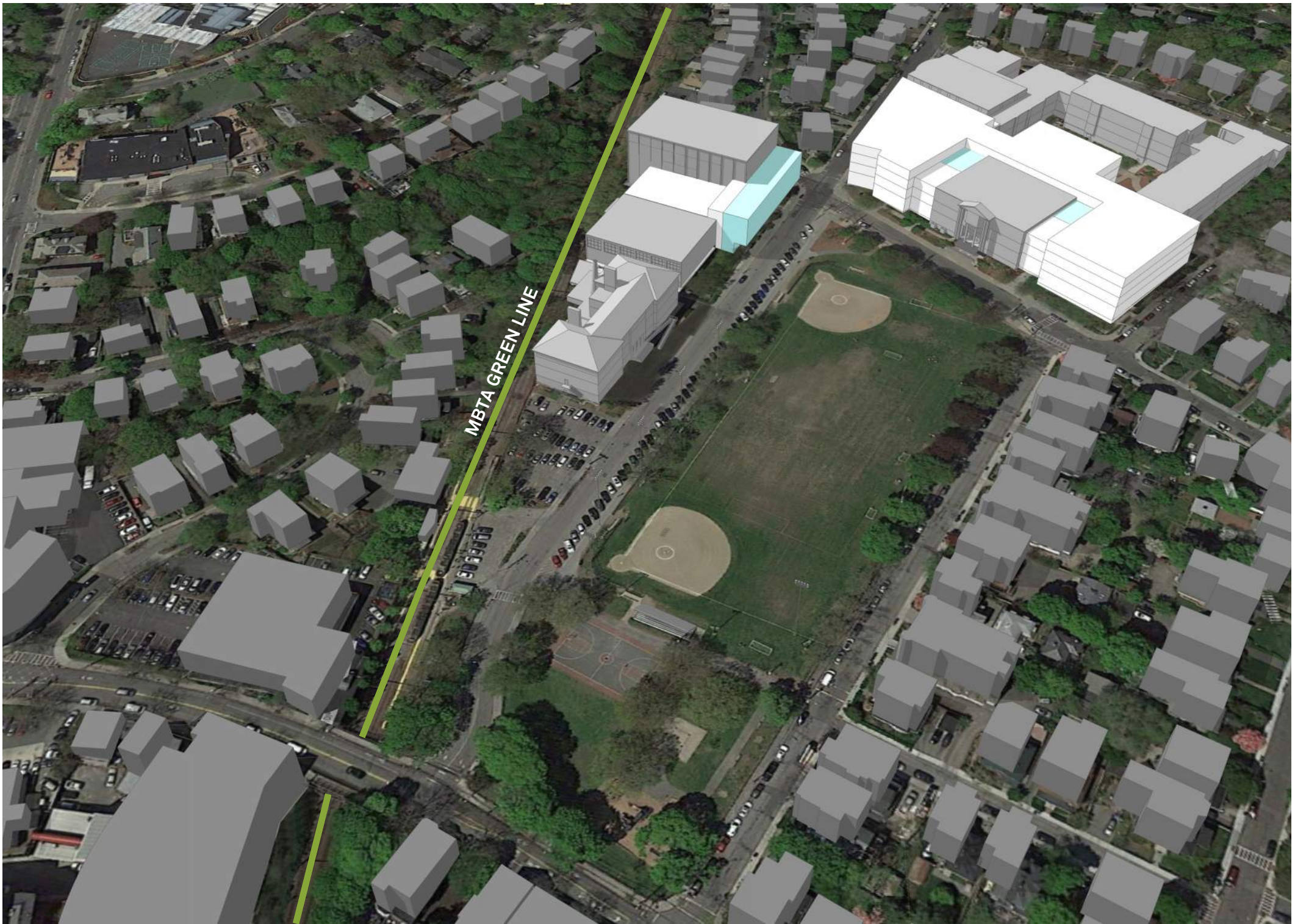


BHS Existing to Remain
294,310 sf

New Construction at BHS
165,000 sf

Massing

OPTION 2
MA State Standard with Competition Gym at Tappan Gym



BHS Existing to Remain
242,280 sf

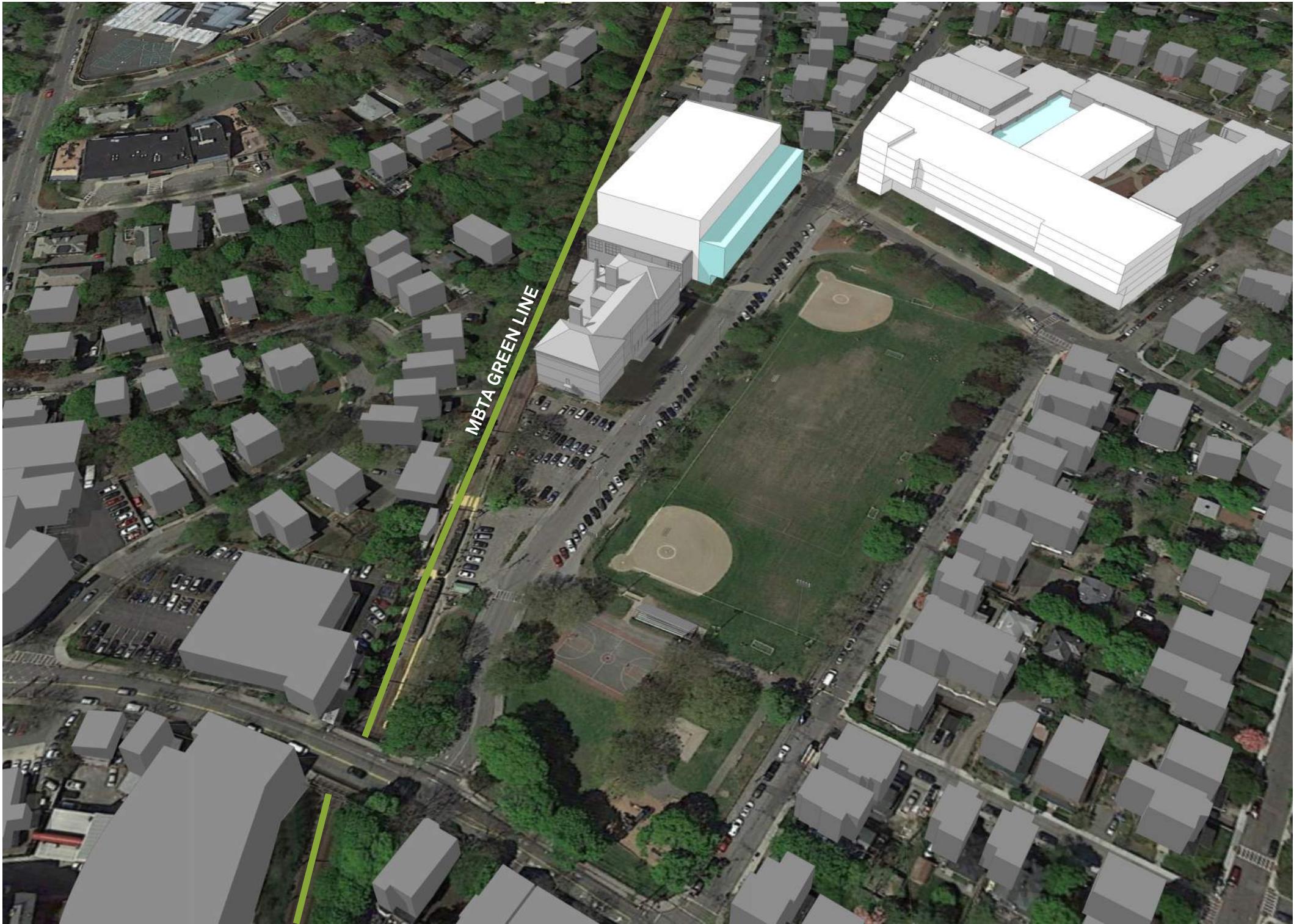
New Construction at BHS
228,690 sf

plus
2. Tappan Gym Moderate
Renovation with
Competition Gym and
Practice Gym Addition

Massing

OPTION 3

BHS Educational Plan and with Field House Addition to Tappan Gym



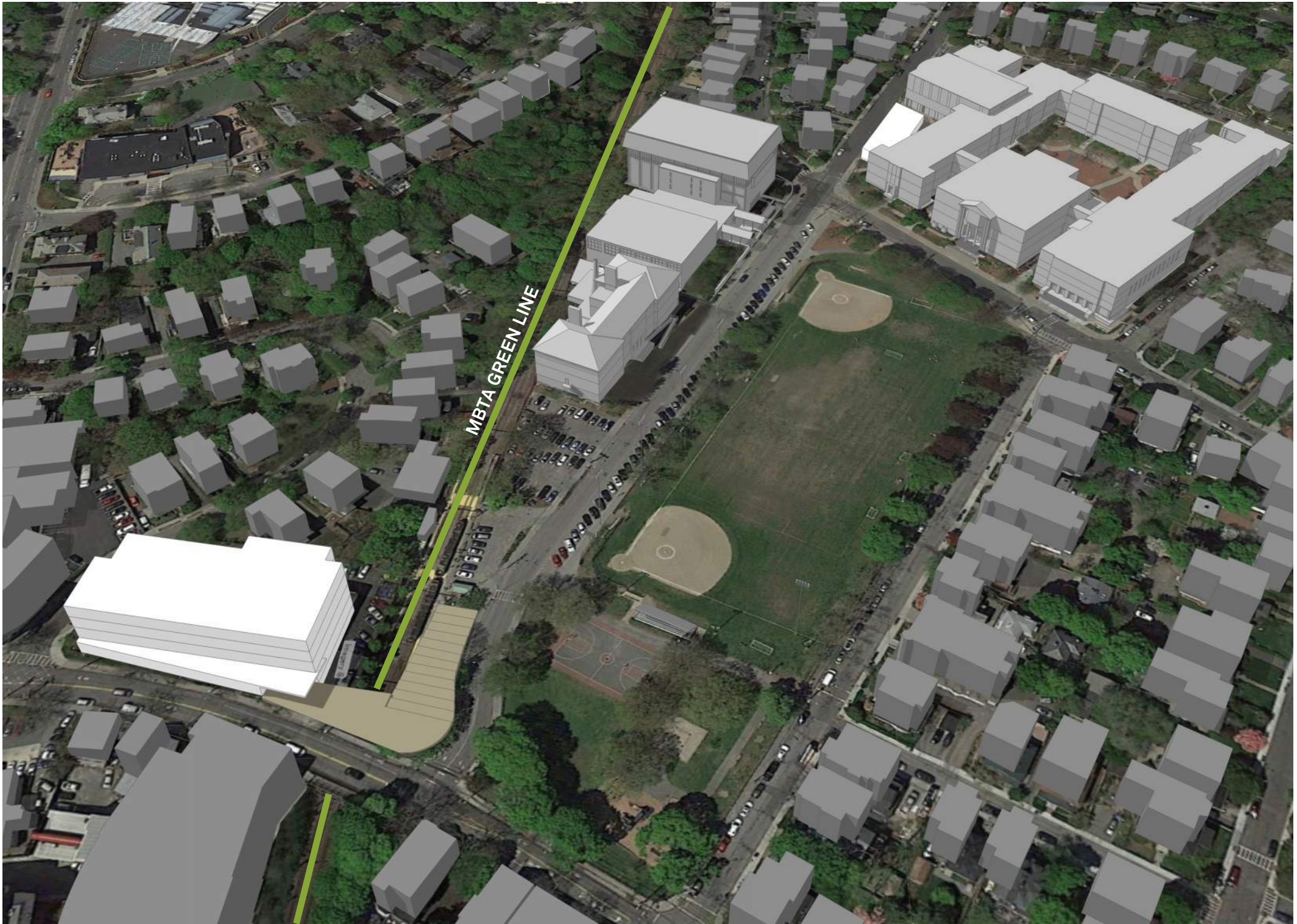
BHS Existing to Remain
181,050 sf

New Construction at BHS
298,130 sf

plus
3. Tappan Gym
Field House Addition

Massing

OPTION 4A
New Cypress Building with BHS-Greenough
Renovate Existing Science & New 5-Classroom Addition



Renovate Existing BHS
Science
35,500 sf

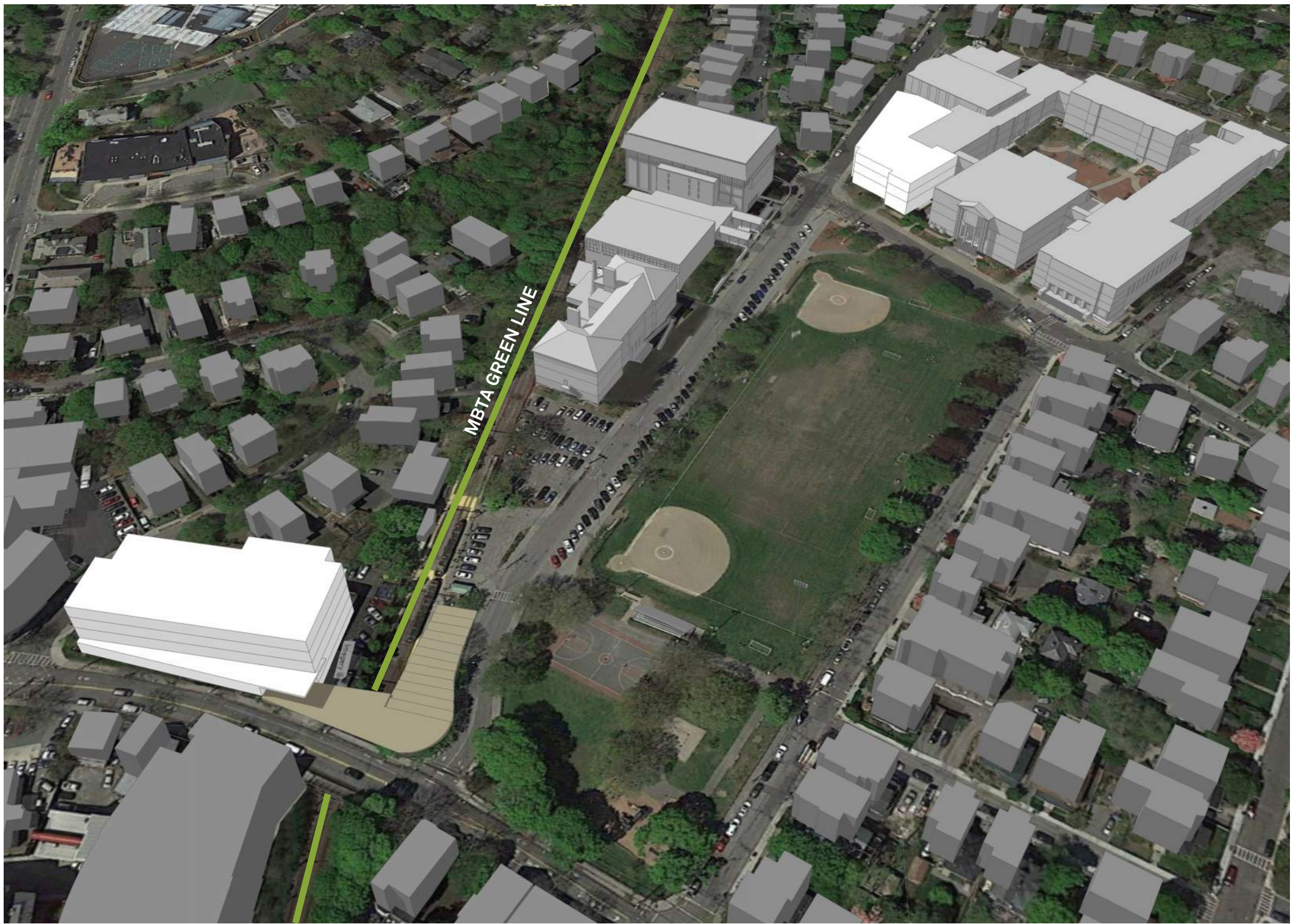
New Construction at BHS
7,500 sf

New Construction at
Cypress Building
109,810 sf

Massing

OPTION 4B

New Cypress Building with BHS-Greenough
New Science at Roberts Wing & Renovate 3rd Floor as Core Academics



Renovate Existing BHS
Science into Classrooms
36,500 sf

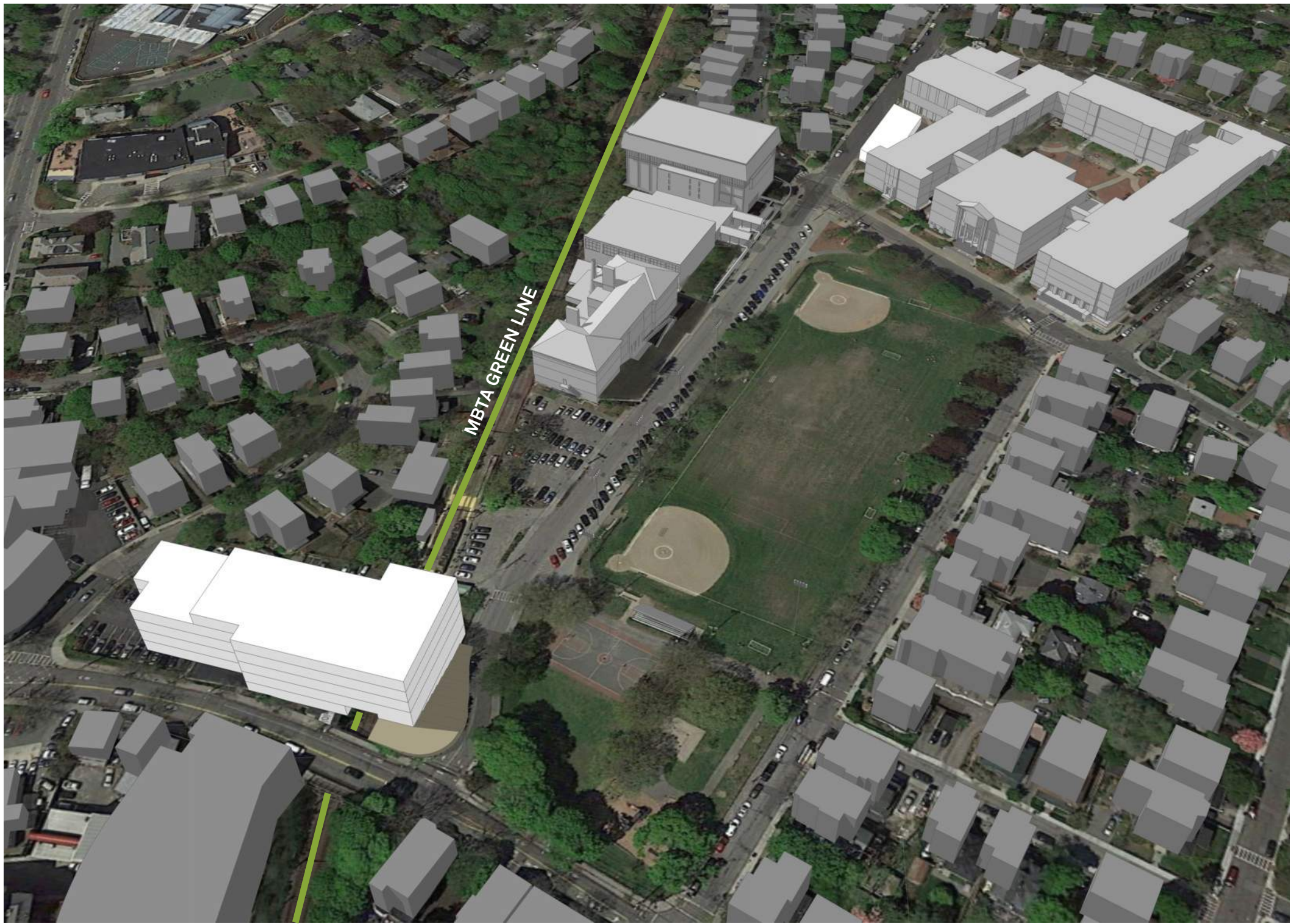
New Construction at BHS
44,100 sf

New Construction at
Cypress Building
109,810 sf

Massing

OPTION 4C

New Cypress Building over the MBTA with BHS-Greenough
Renovate Existing Science & New 5-Classroom Addition

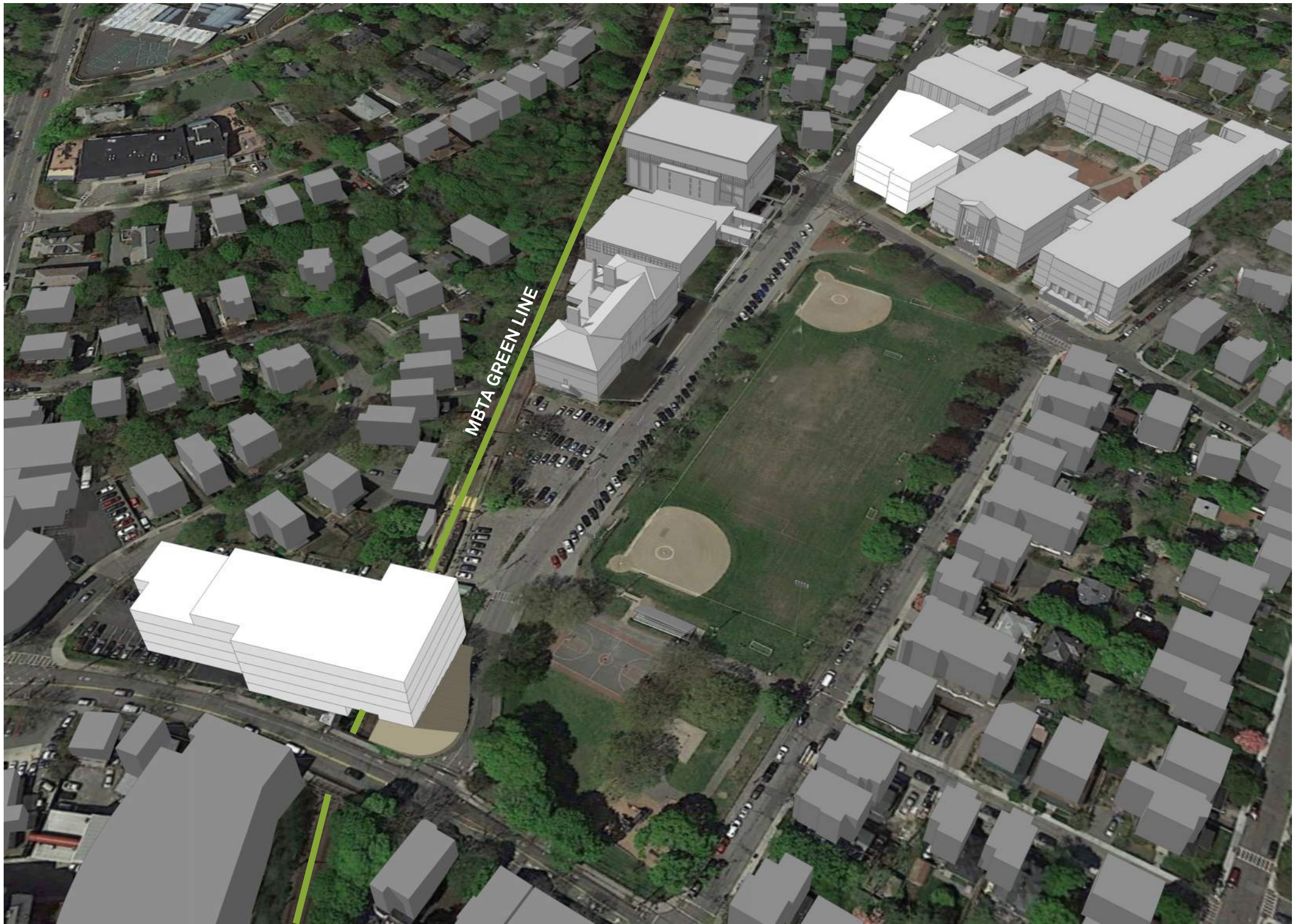


- Renovate Existing BHS Science
35,500 sf
- New Construction at BHS
7,500 sf
- New Construction at Cypress Building
112,700 sf

Massing

OPTION 4D

New Cypress Building over the MBTA with BHS-Greenough
New Science at Roberts Wing & Renovate 3rd Floor as Core Academics



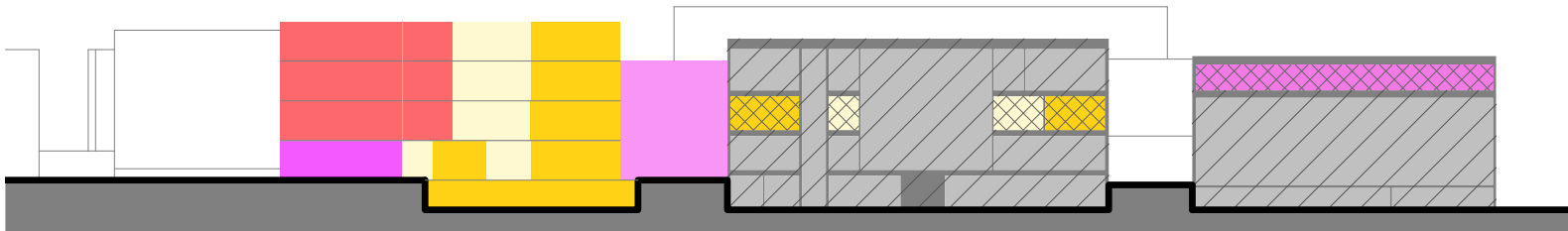
- Renovate Existing Science into Classrooms
36,500 sf
- New Construction at BHS
50,400 sf
- New Construction at Cypress Building
112,700 sf

Massing

OPTION 1
Enrollment Accommodation



Massing



Section

CIRCULATION

CORE ACADEMICS

SCIENCE

COLLABORATIVE

LIBRARY

CULINARY ARTS

DINING COMMONS

PERFORMANCE

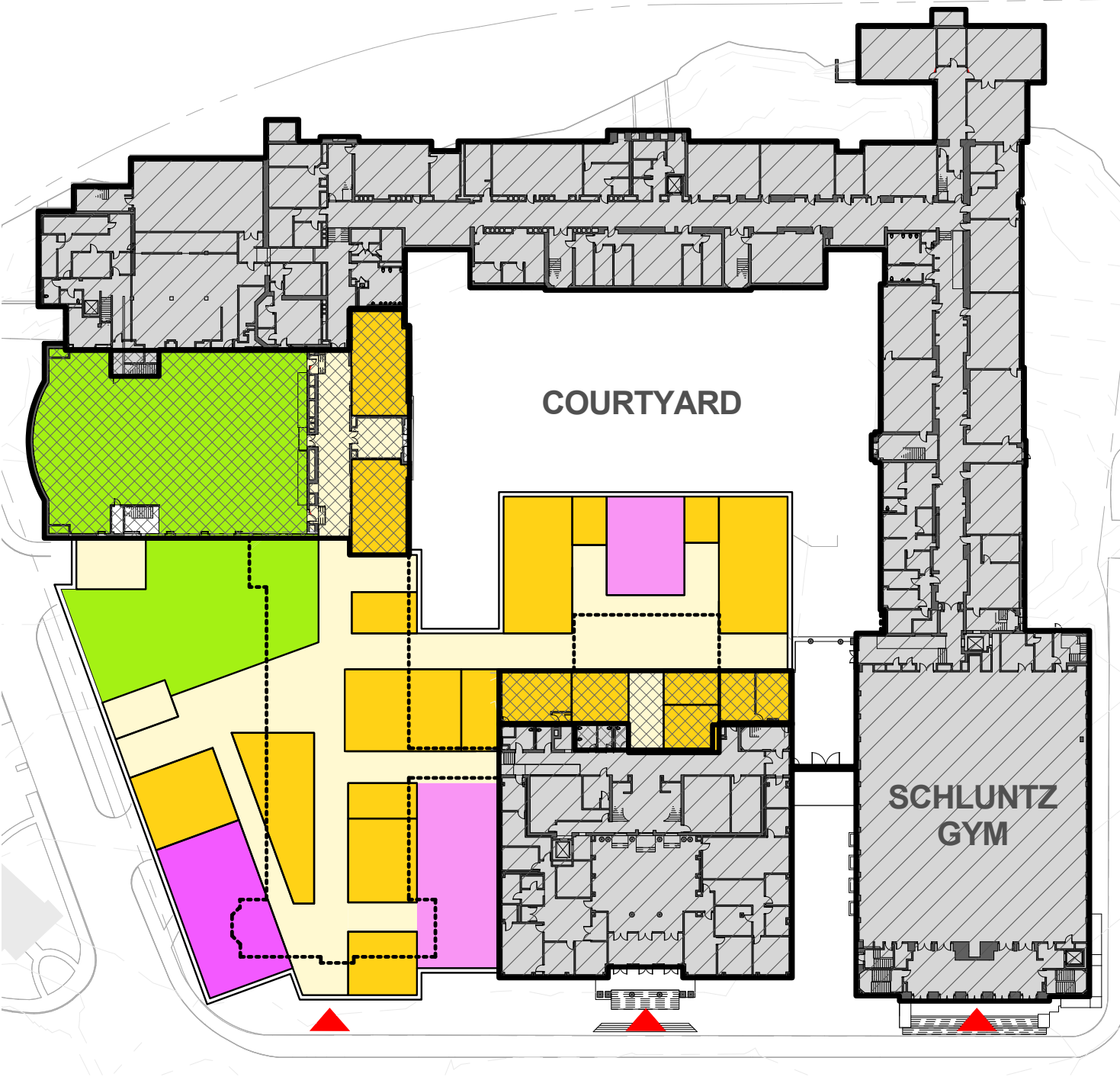
..... EXTENT OF DEMOLITION

EXISTING - NO WORK

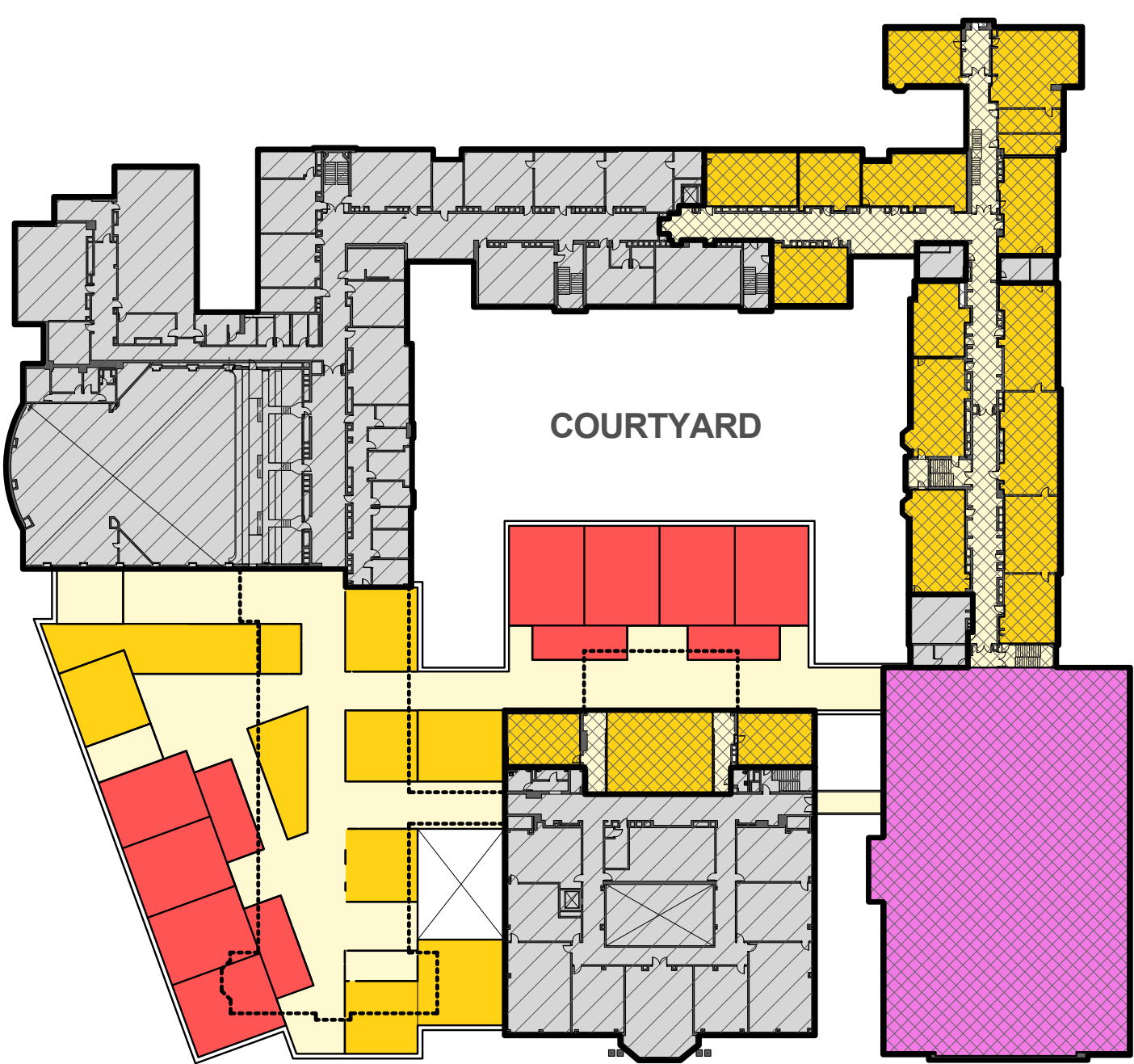
RENOVATION

OPTION 1

Enrollment Accommodation



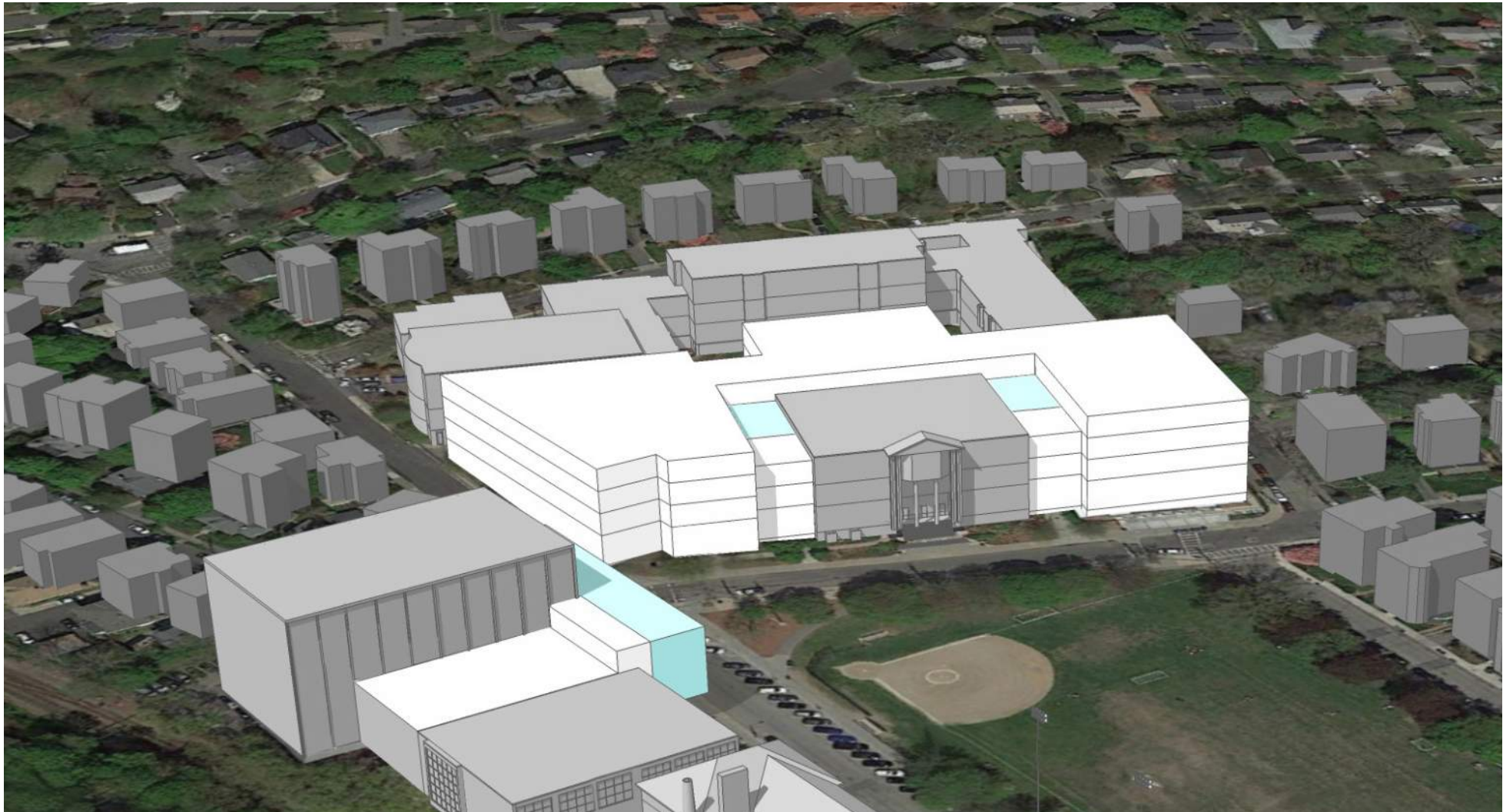
1st Floor Plan



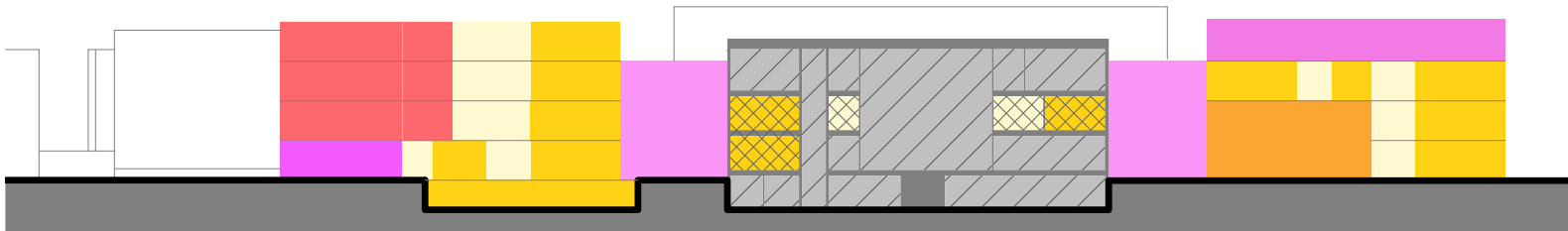
3rd Floor Plan

CIRCULATION CORE ACADEMICS SCIENCE COLLABORATIVE LIBRARY CULINARY ARTS DINING COMMONS PERFORMANCE EXTENT OF DEMOLITION EXISTING - NO WORK RENOVATION

OPTION 2
MA State High School Standards



Massing



Section

- CIRCULATION

CORE ACADEMICS

SCIENCE

COLLABORATIVE

LIBRARY

CULINARY ARTS

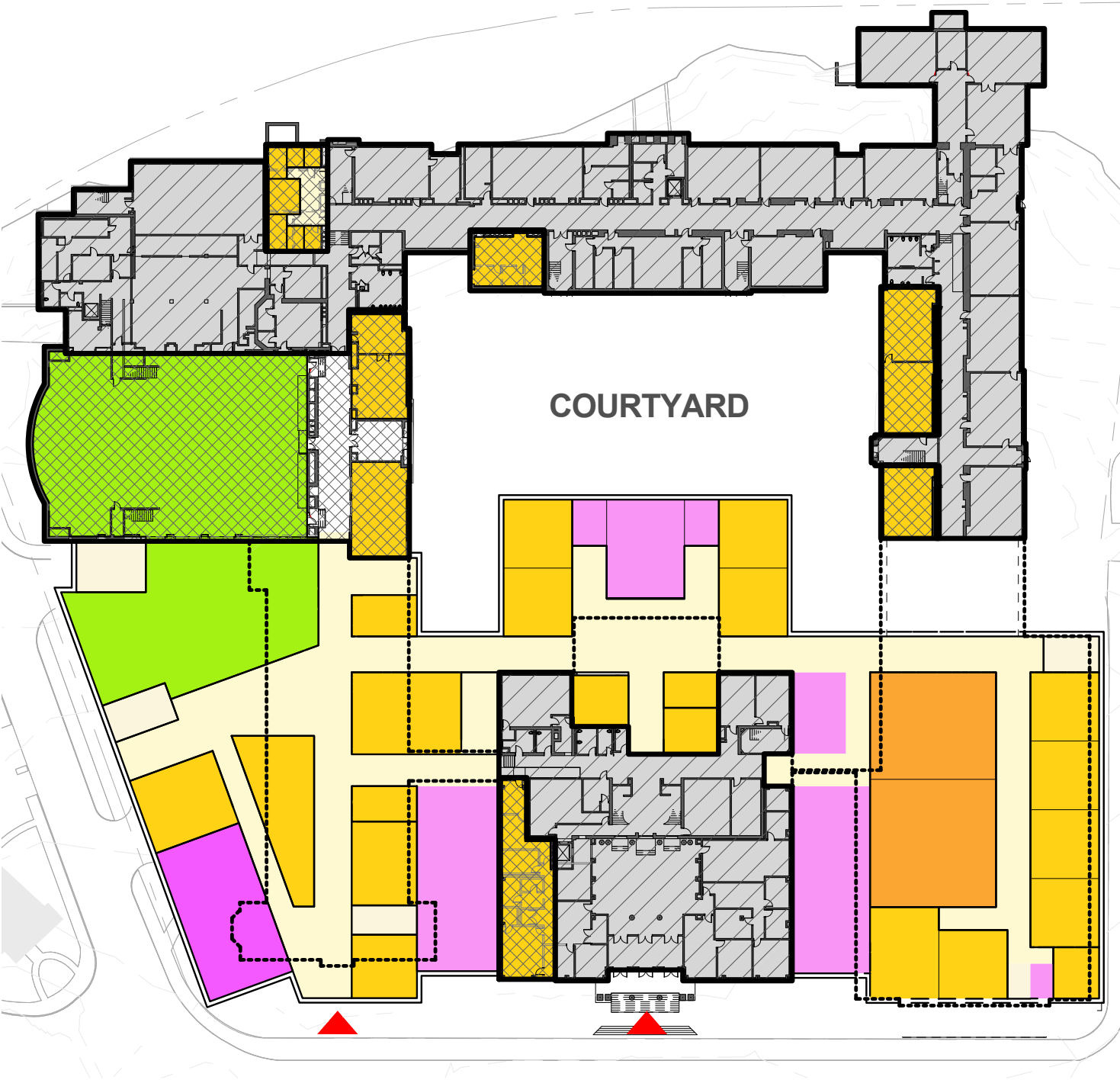
DINING COMMONS

PERFORMANCE
- EXTENT OF DEMOLITION

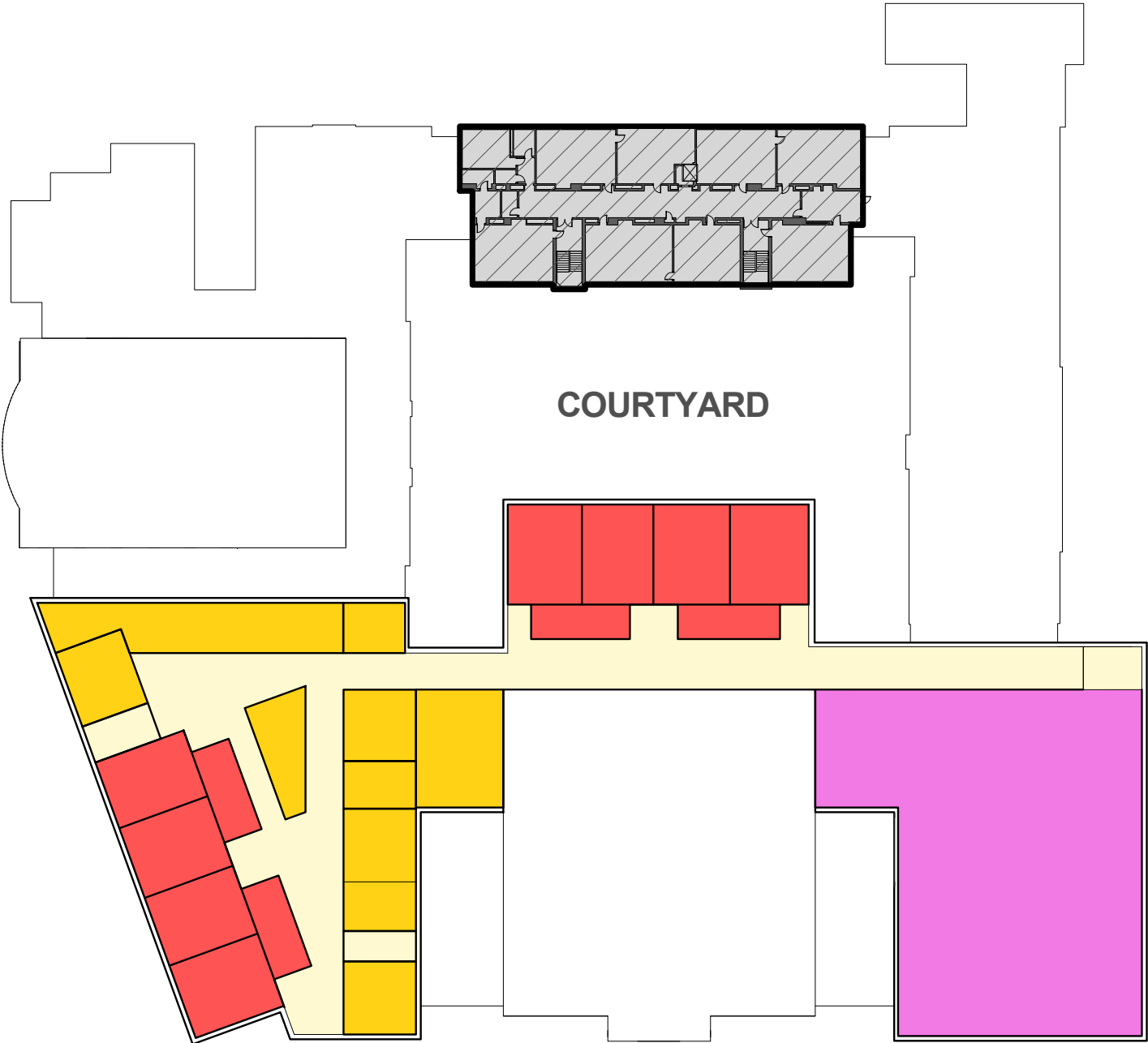
EXISTING - NO WORK

RENOVATION

OPTION 2 MA State High School Standards



1st Floor Plan



4th Floor Plan

CIRCULATION

CORE ACADEMICS

SCIENCE

COLLABORATIVE

LIBRARY

CULINARY ARTS

DINING COMMONS

PERFORMANCE

..... EXTENT OF DEMOLITION

EXISTING - NO WORK

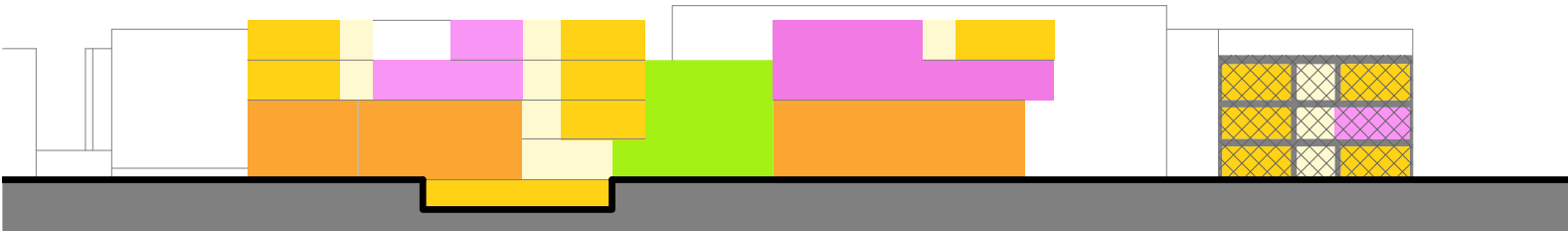
RENOVATION

OPTION 3

Brookline High School Education Plan



Massing 3 With Tappan Gym



Section 3

CIRCULATION

CORE ACADEMICS

SCIENCE

COLLABORATIVE

LIBRARY

CULINARY ARTS

DINING COMMONS

PERFORMANCE

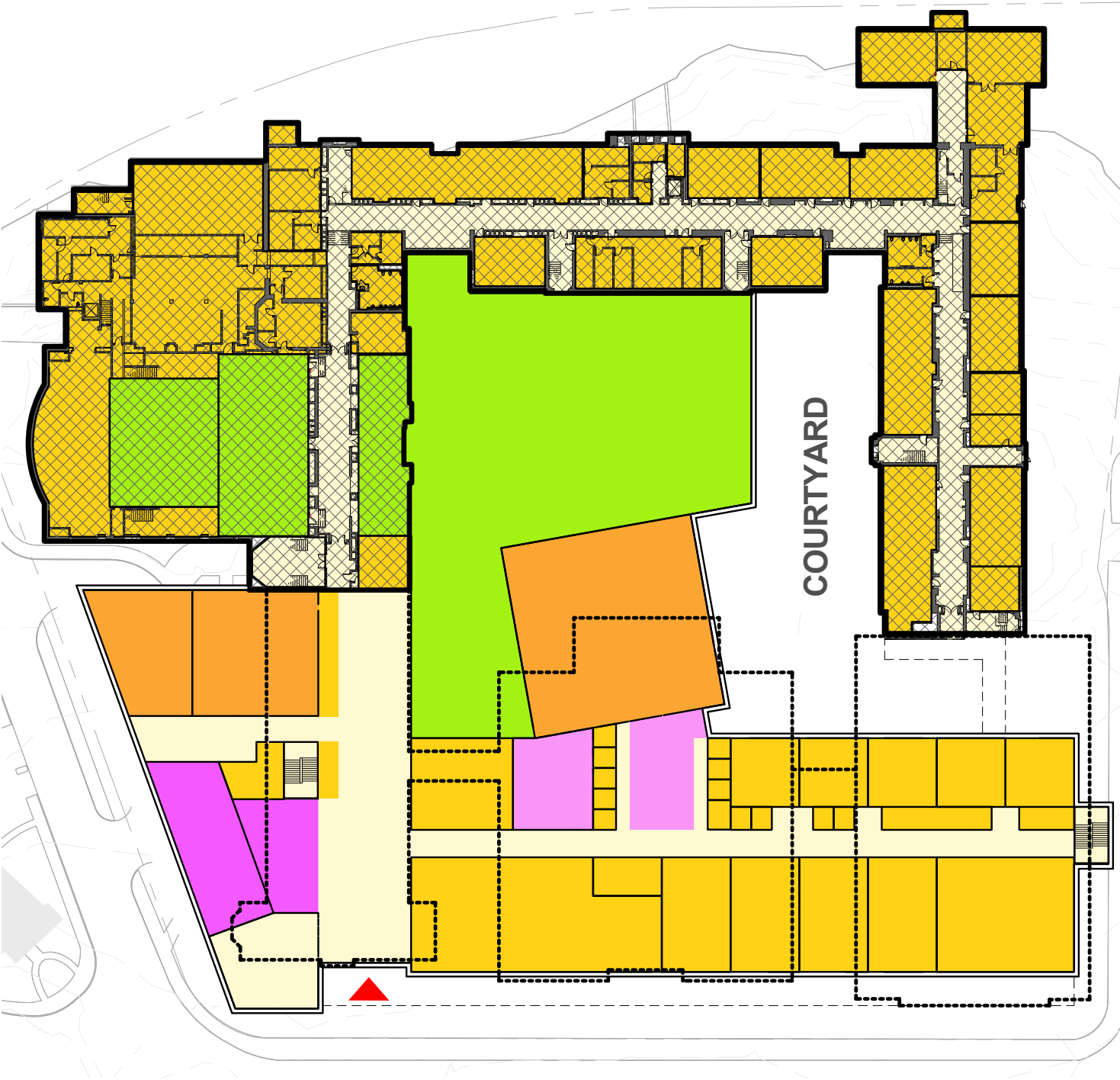
..... EXTENT OF DEMOLITION

EXISTING - NO WORK

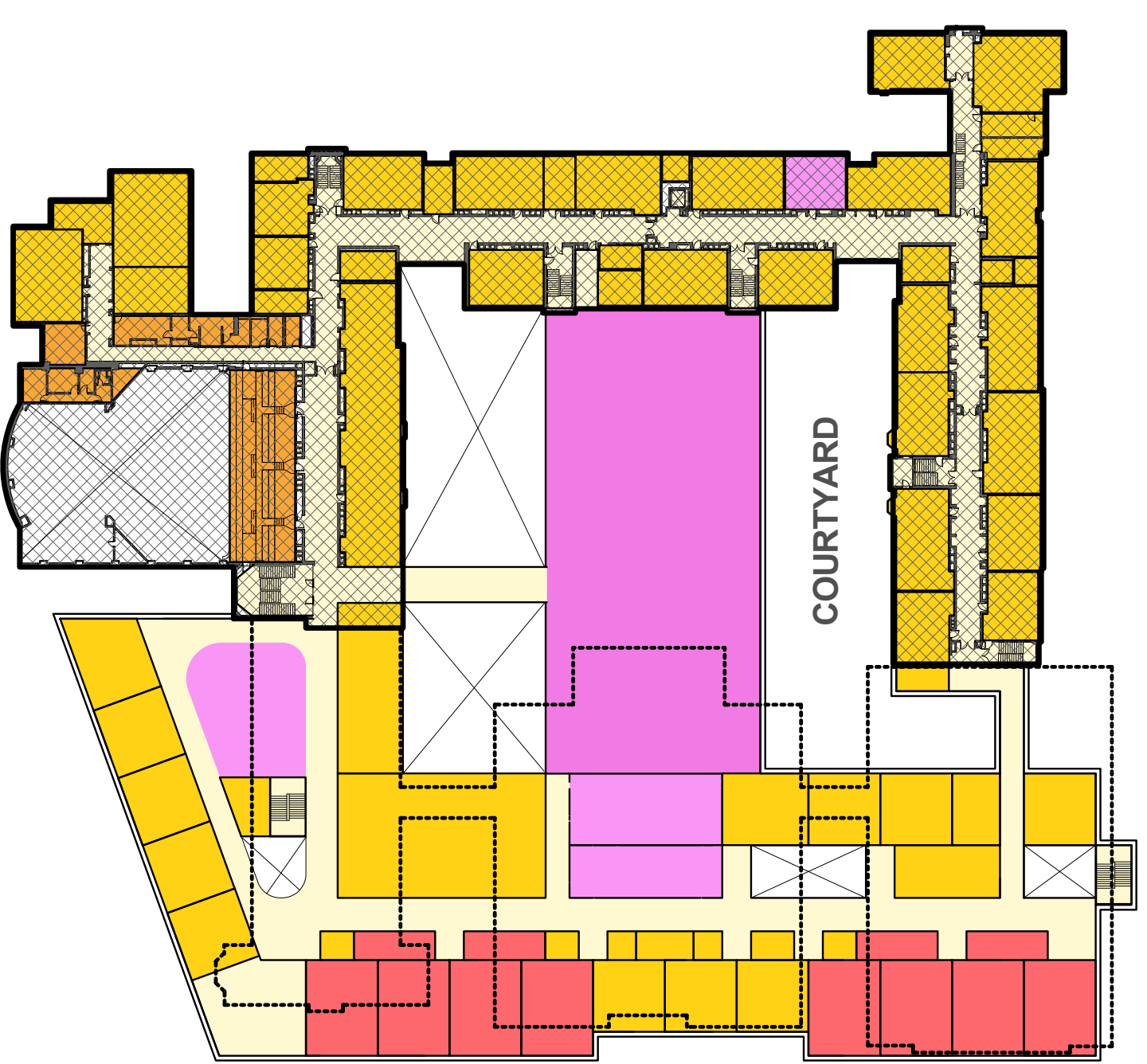
RENOVATION

OPTION 3

Brookline High School Education Plan



1st Floor Plan



3rd Floor Plan

CIRCULATION

CORE ACADEMICS

SCIENCE

COLLABORATIVE

LIBRARY

CULINARY ARTS

DINING COMMONS

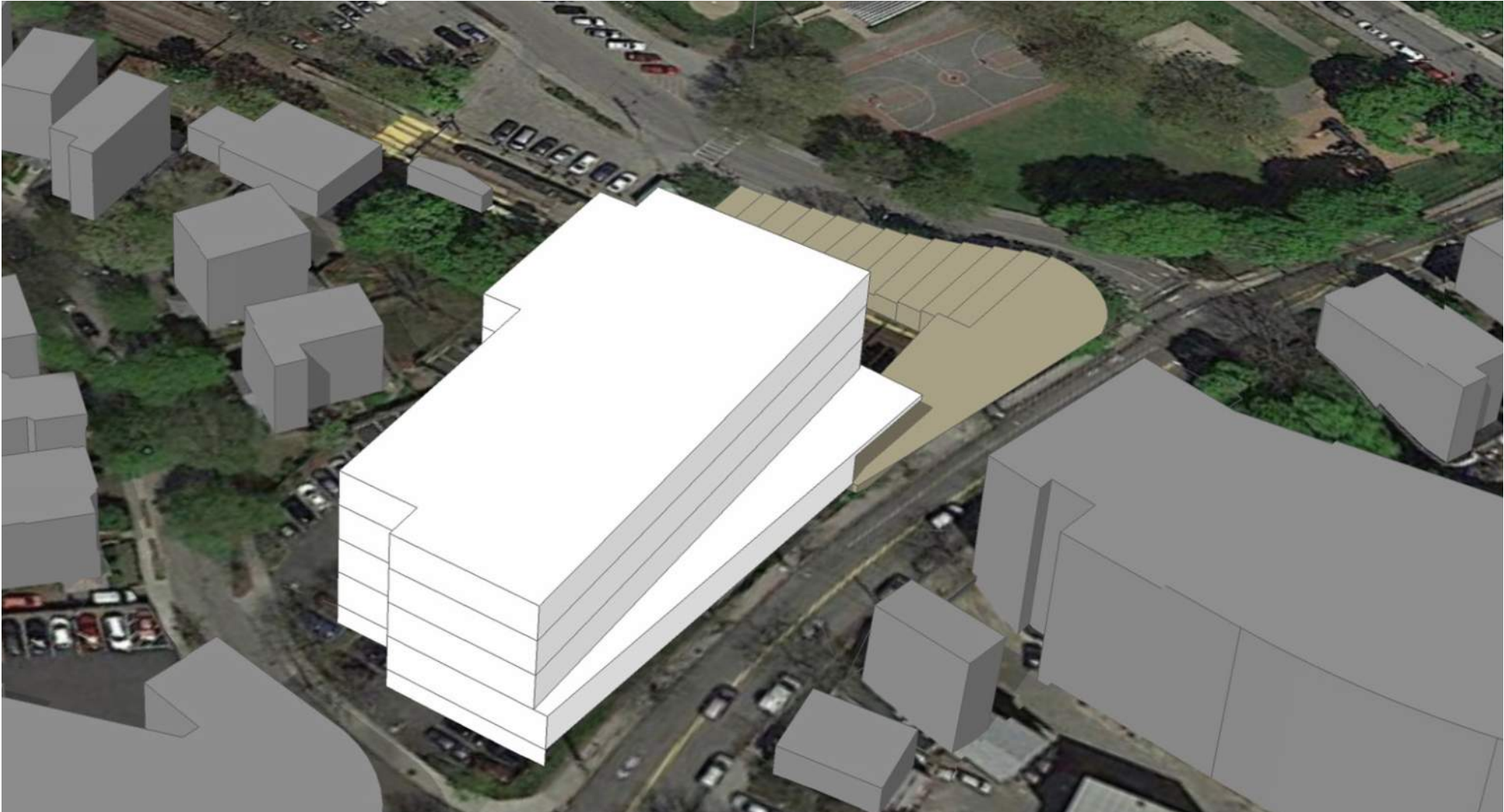
PERFORMANCE

EXTENT OF DEMOLITION

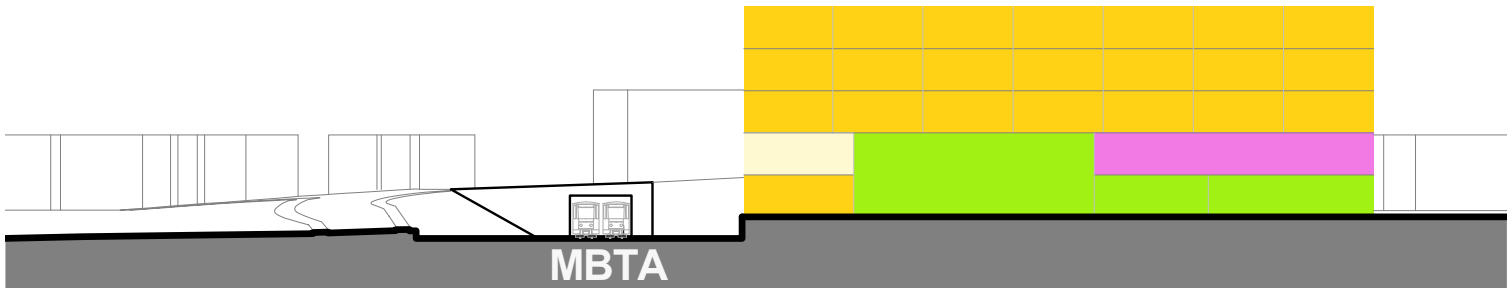
EXISTING - NO WORK

RENOVATION

OPTION 4A & 4B New Cypress Building



Massing 4A & 4B



Section 4A & 4B

CIRCULATION

CORE ACADEMICS

SCIENCE

COLLABORATIVE

LIBRARY

CULINARY ARTS

DINING COMMONS

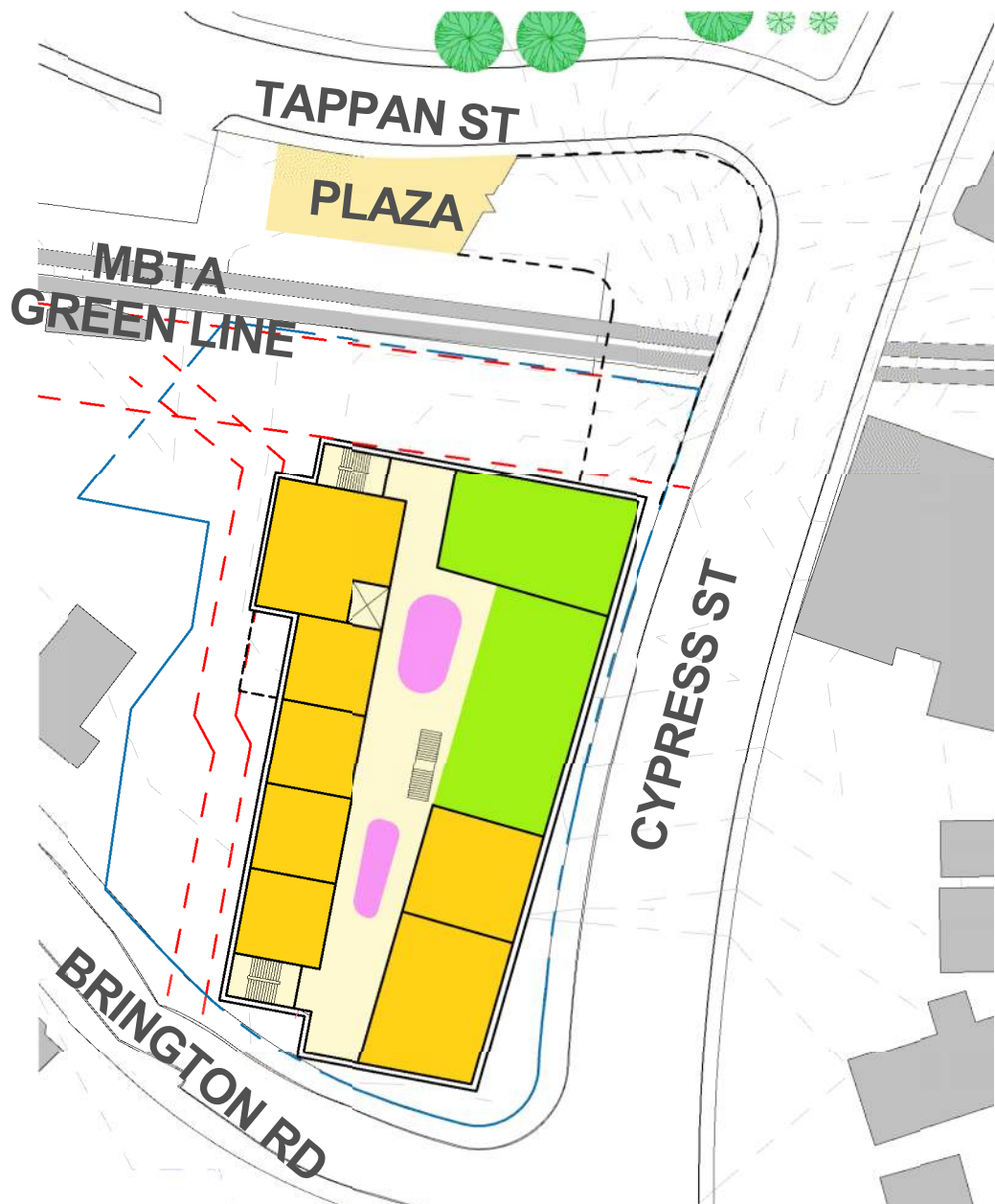
PERFORMANCE

..... EXTENT OF DEMOLITION

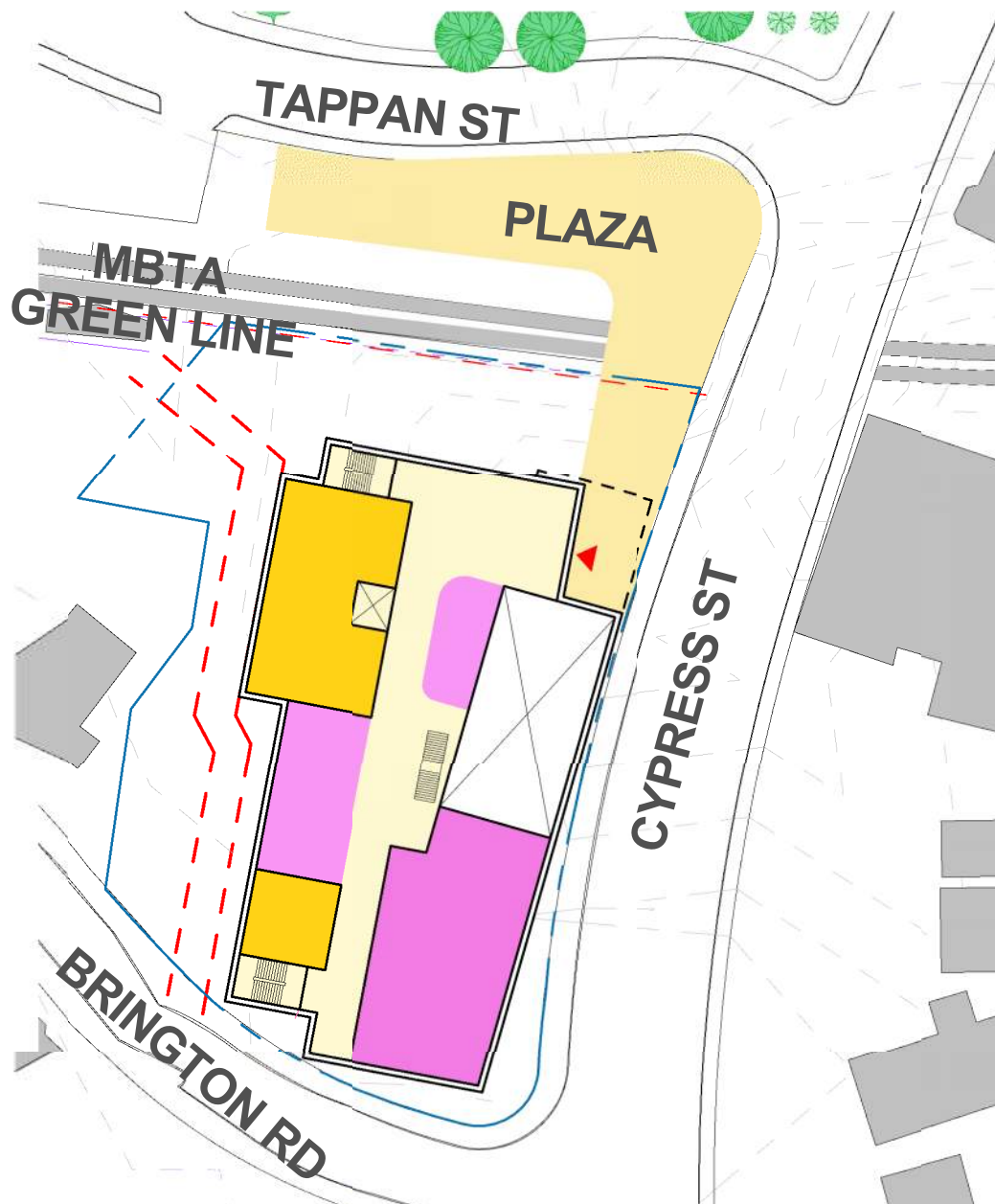
EXISTING - NO WORK

RENOVATION

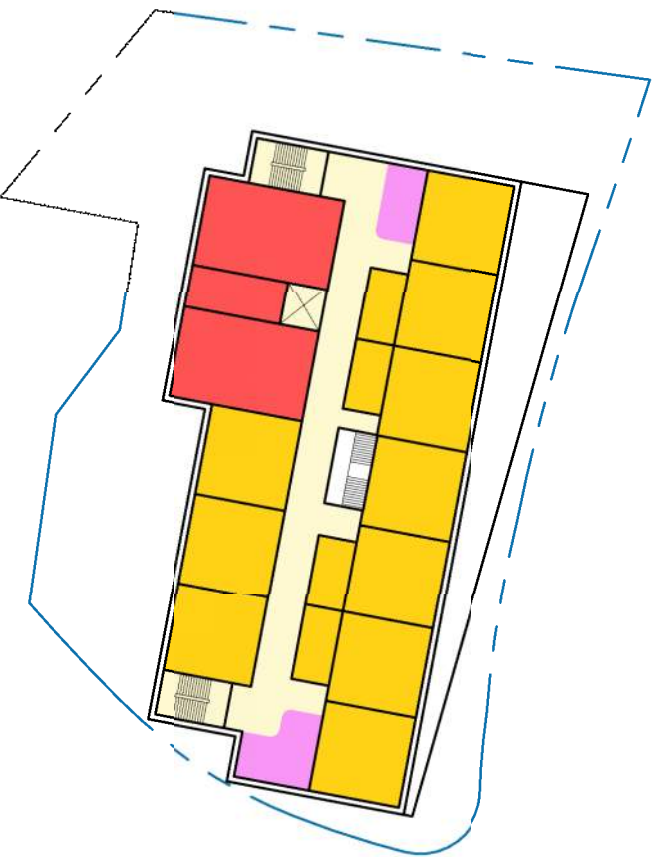
OPTION 4A & 4B New Cypress Building



Ground Level Plan



1st Floor Plan



2nd - 4th Floor Plan

CIRCULATION

CORE ACADEMICS

SCIENCE

COLLABORATIVE

LIBRARY

CULINARY ARTS

DINING COMMONS

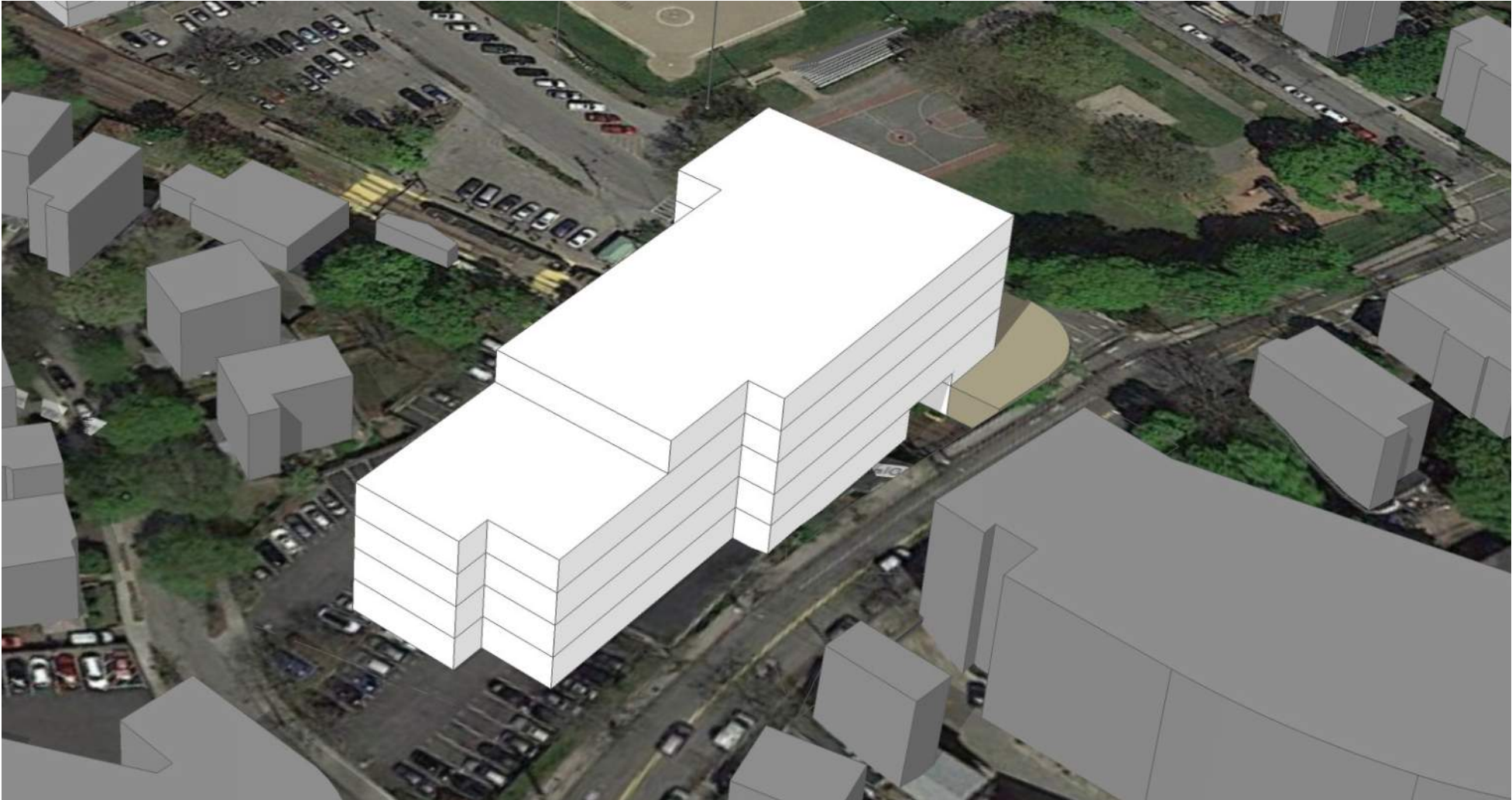
PERFORMANCE

..... EXTENT OF DEMOLITION

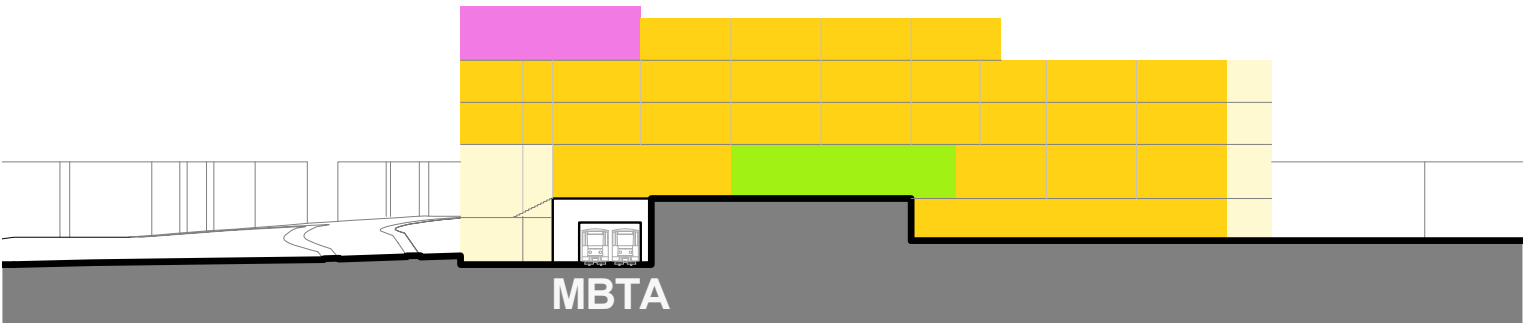
EXISTING - NO WORK

RENOVATION

OPTION 4C & 4D: New Cypress Building over the MBTA



Massing 4C & 4D



Section 4C & 4D

- CIRCULATION

CORE ACADEMICS

SCIENCE

COLLABORATIVE

LIBRARY

CULINARY ARTS

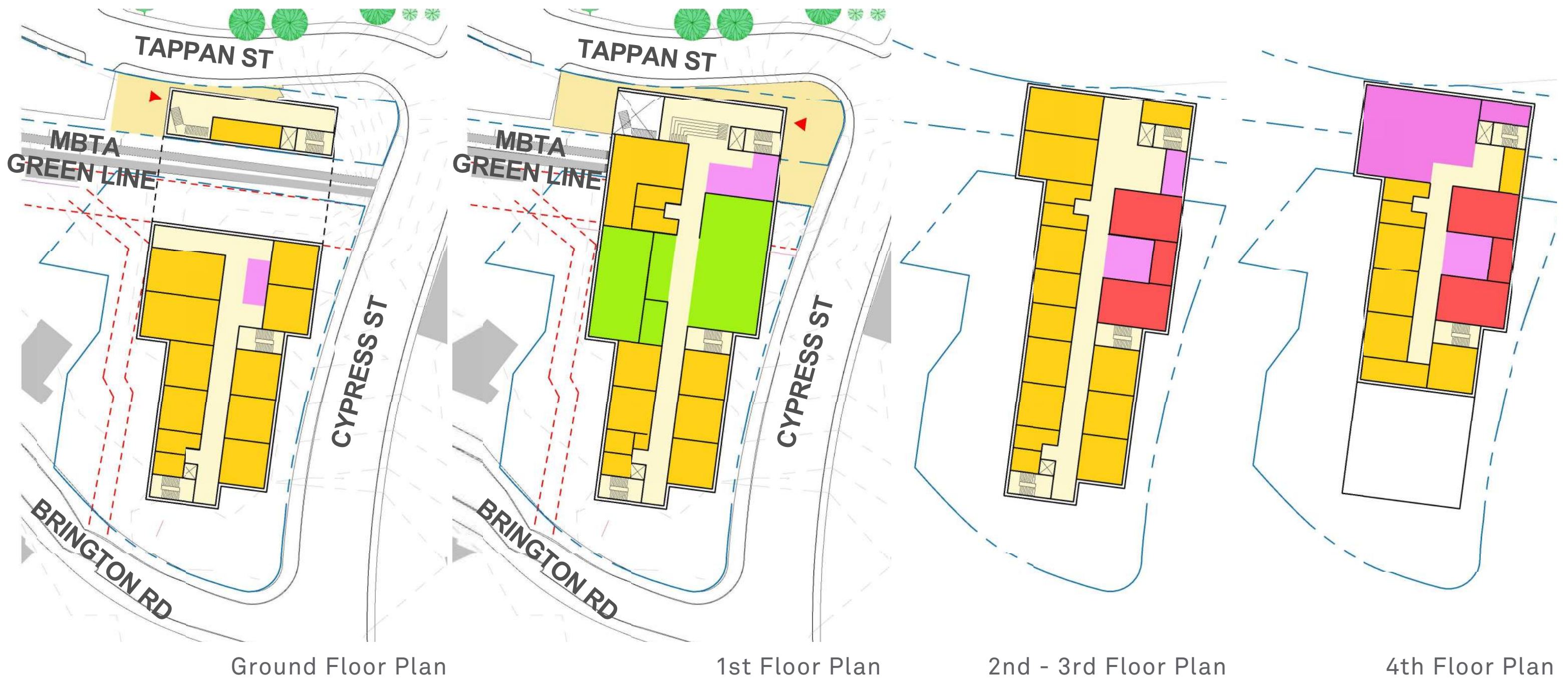
DINING COMMONS

PERFORMANCE
- EXTENT OF DEMOLITION

EXISTING - NO WORK

RENOVATION

OPTION 4C & 4D: New Cypress Building over the MBTA

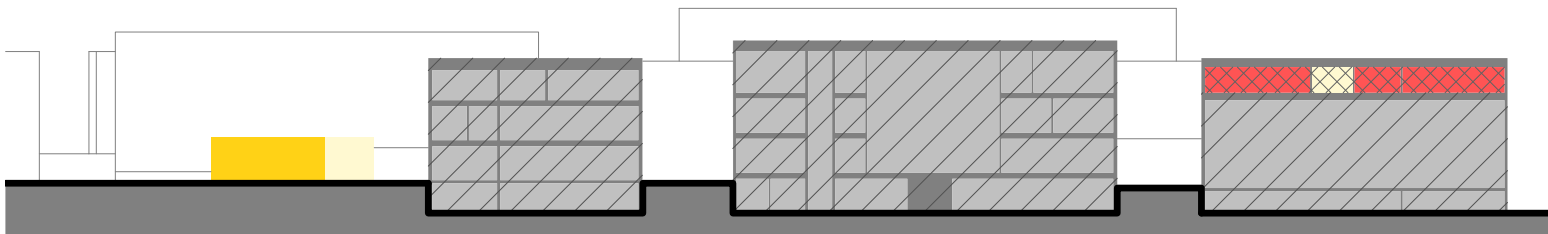


OPTION 4A & 4C

Brookline High School New 5-Classroom Addition & Renovate Existing Science



Massing 4A & 4C

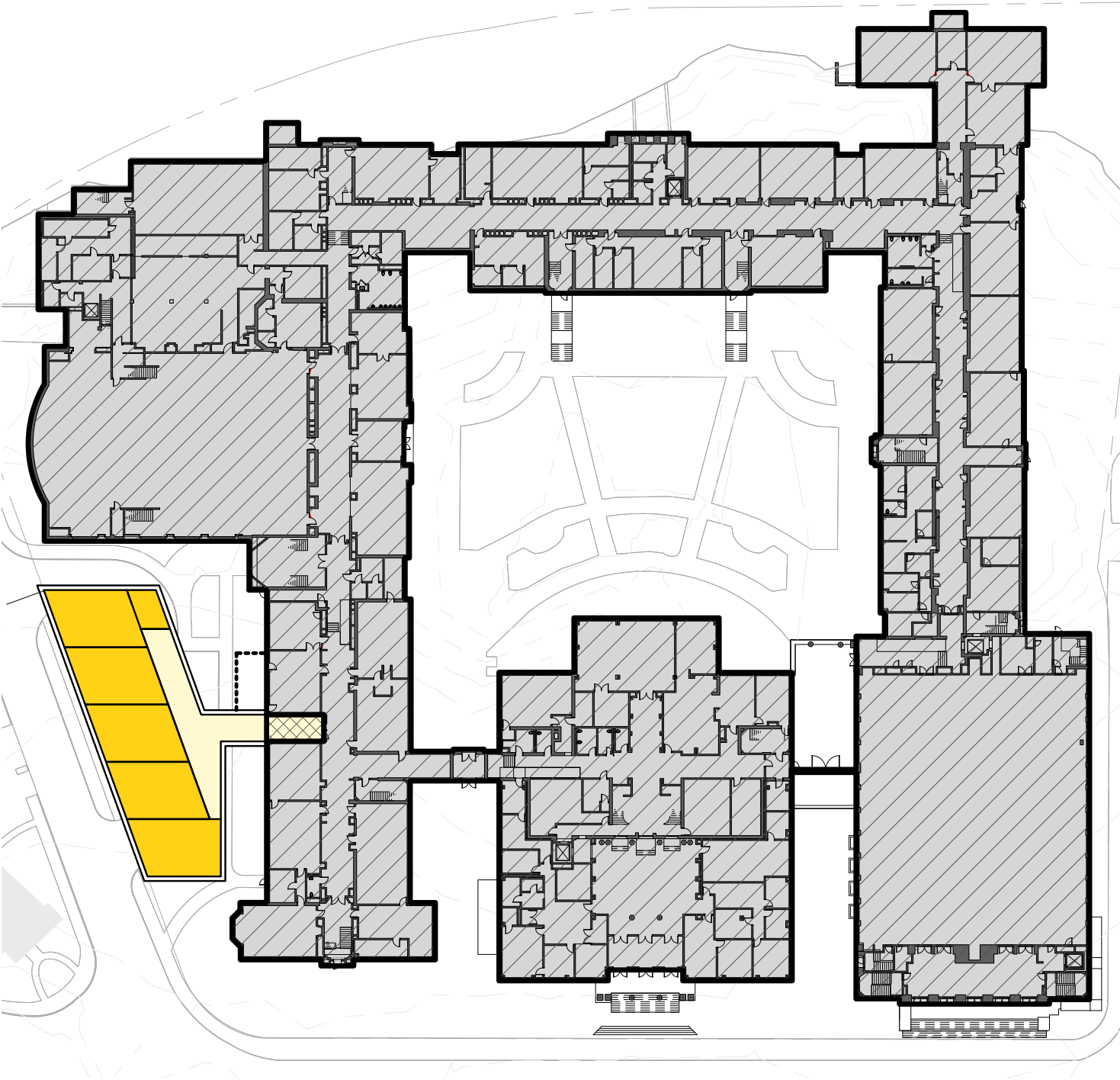


Section 4A & 4C

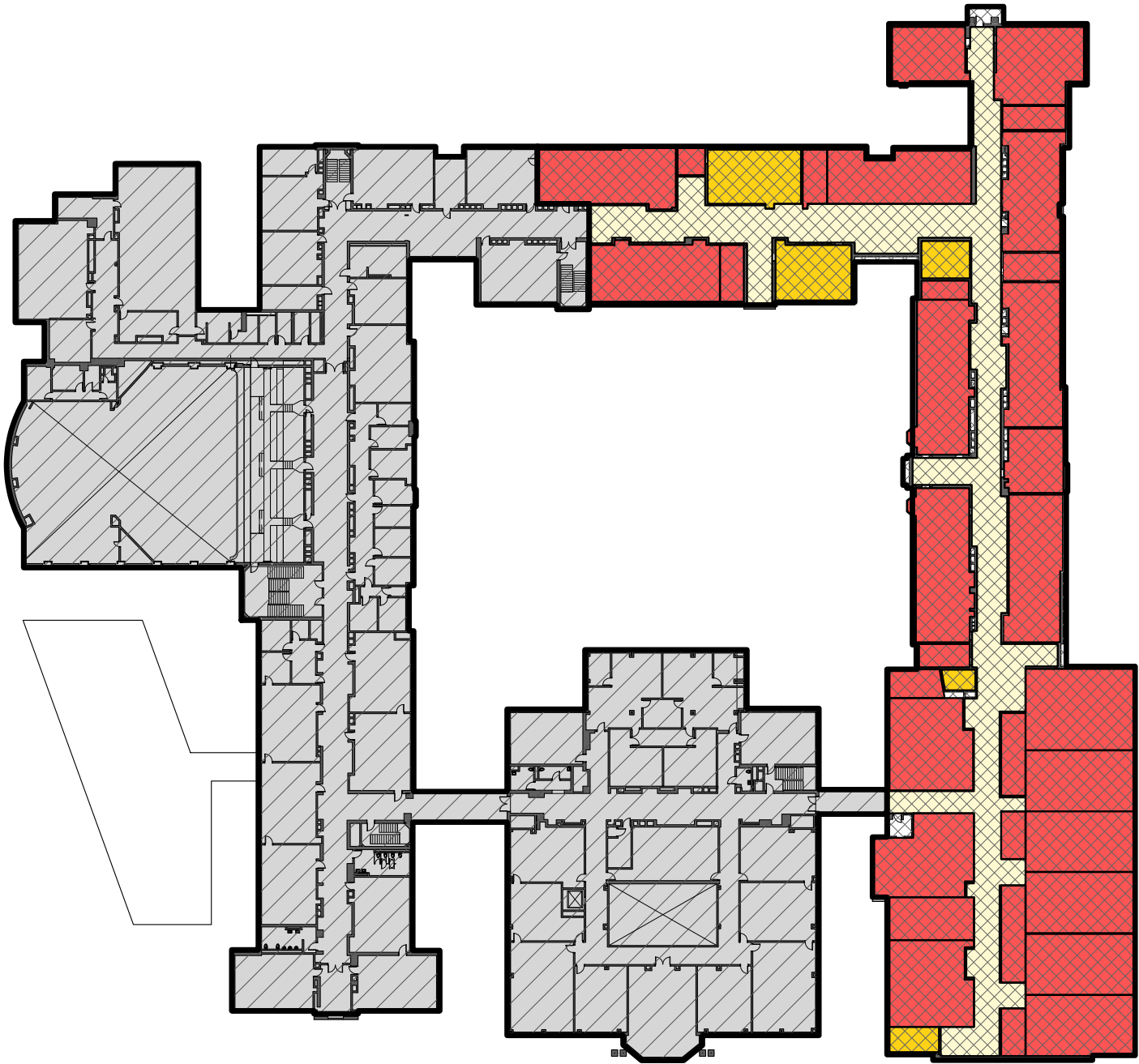
- CIRCULATION
- CORE ACADEMICS
- SCIENCE
- COLLABORATIVE
- LIBRARY
- CULINARY ARTS
- DINING COMMONS
- PERFORMANCE
- EXTENT OF DEMOLITION
- EXISTING - NO WORK
- RENOVATION

OPTION 4A & 4C

Brookline High School New 5-Classroom Addition & Renovate Existing Science



1st Floor Plan



3rd Floor Plan

CIRCULATION

CORE ACADEMICS

SCIENCE

COLLABORATIVE

LIBRARY

CULINARY ARTS

DINING COMMONS

PERFORMANCE

EXTENT OF DEMOLITION

EXISTING - NO WORK

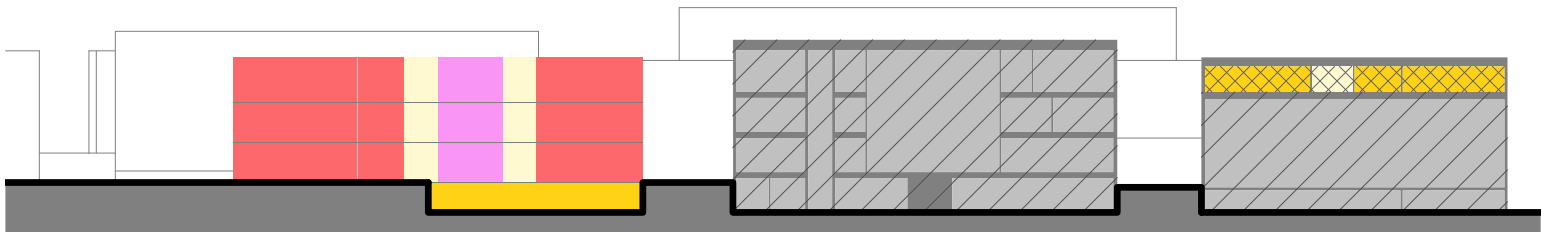
RENOVATION

OPTION 4B & 4D

Brookline High School New Science at Roberts Wing & Renovate 3rd Floor as Core Academics



Massing 4B & 4D



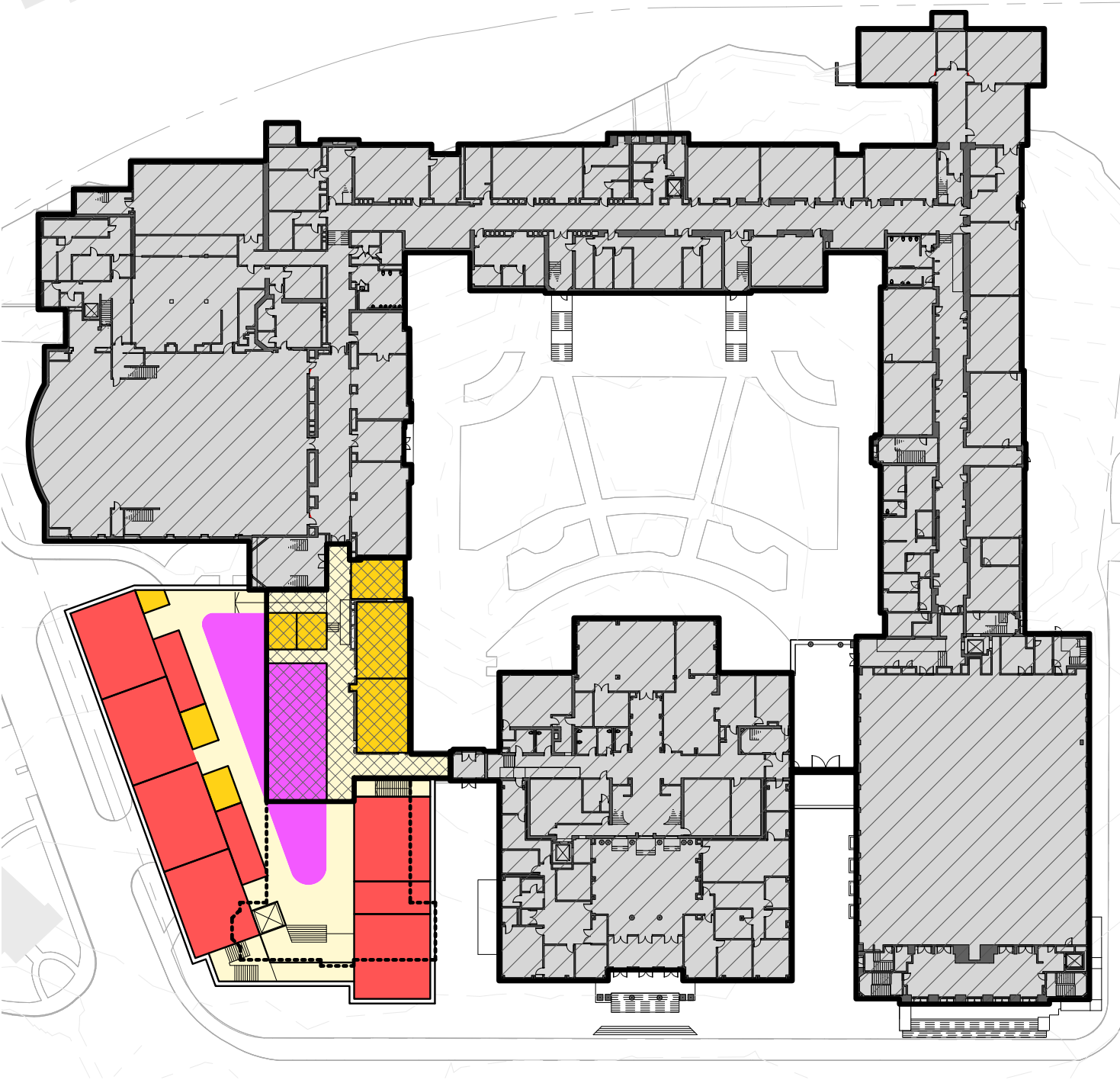
Section 4B & 4D

CIRCULATION CORE ACADEMICS SCIENCE COLLABORATIVE LIBRARY CULINARY ARTS DINING COMMONS PERFORMANCE

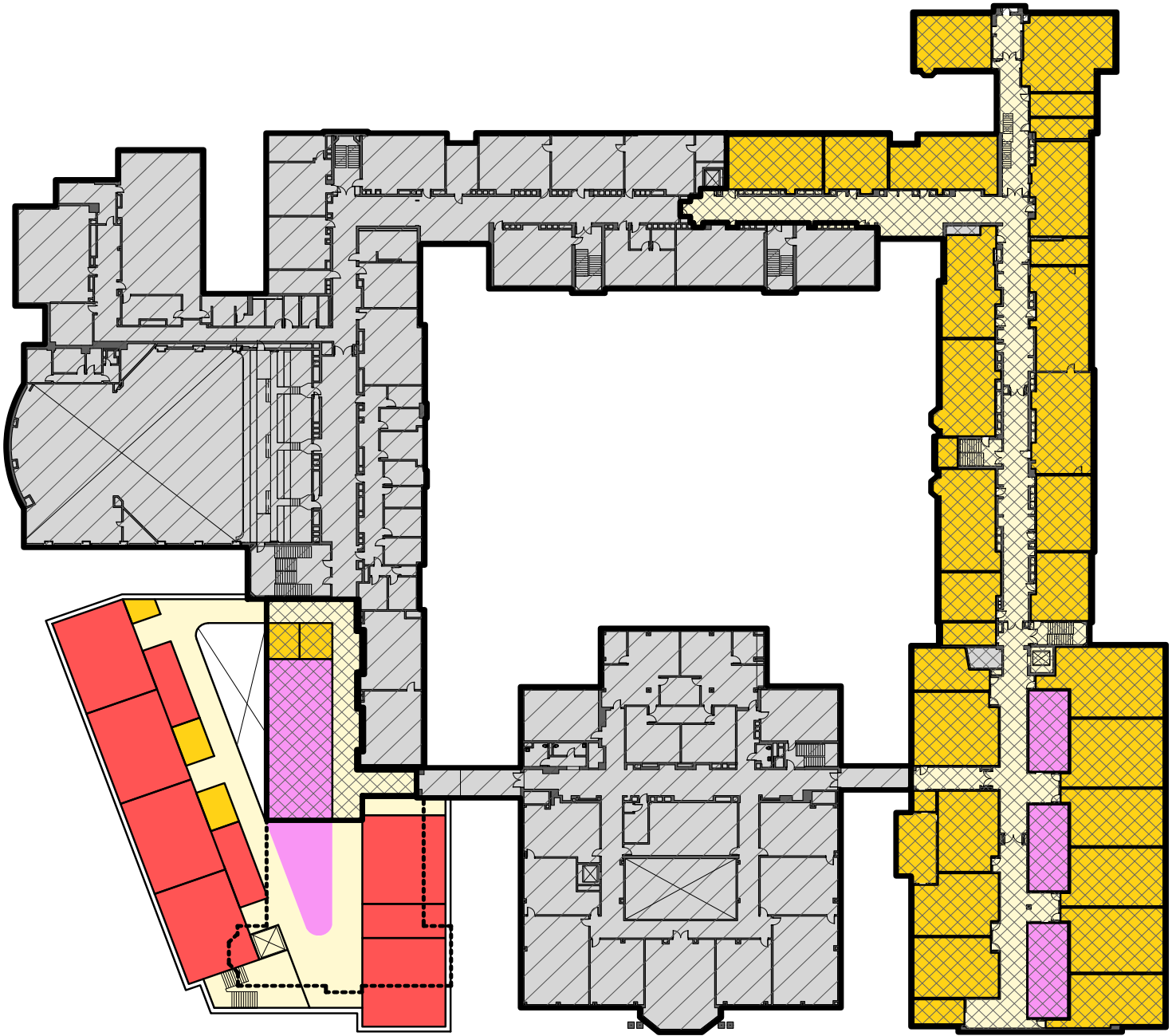
..... EXTENT OF DEMOLITION EXISTING - NO WORK RENOVATION

OPTION 4B & 4D

Brookline High School New Science at Roberts Wing & Renovate 3rd Floor as Core Academics



1st Floor Plan



3rd Floor Plan

CIRCULATION

CORE ACADEMICS

SCIENCE

COLLABORATIVE

LIBRARY

CULINARY ARTS

DINING COMMONS

PERFORMANCE

..... EXTENT OF DEMOLITION

EXISTING - NO WORK

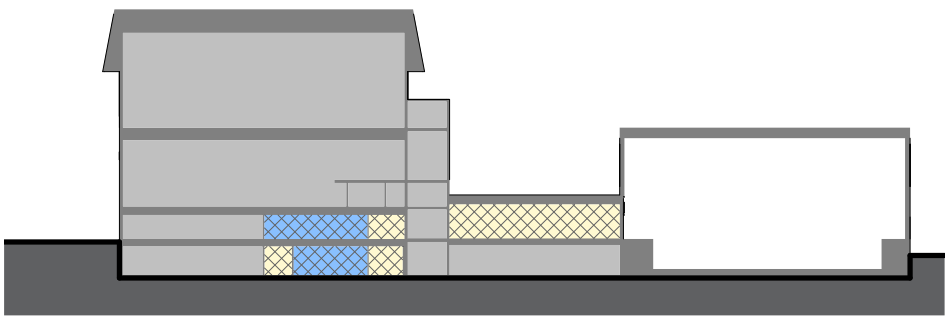
RENOVATION

OPTIONAL PROJECT - with Option 1 or Option 4

1. Tappan Gym - Minimal Renovations



Tappan Gym Massing



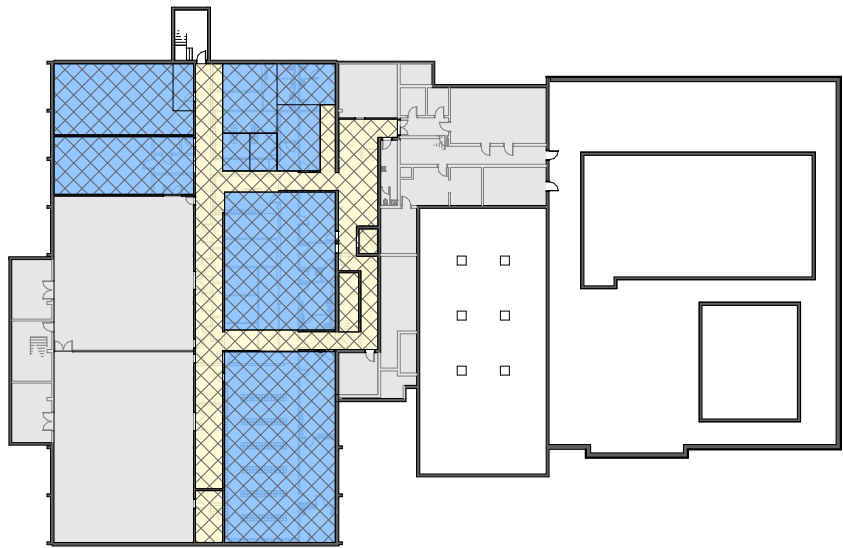
Tappan Gym Section

CIRCULATION HEALTH AND PHYSICAL EDUCATION

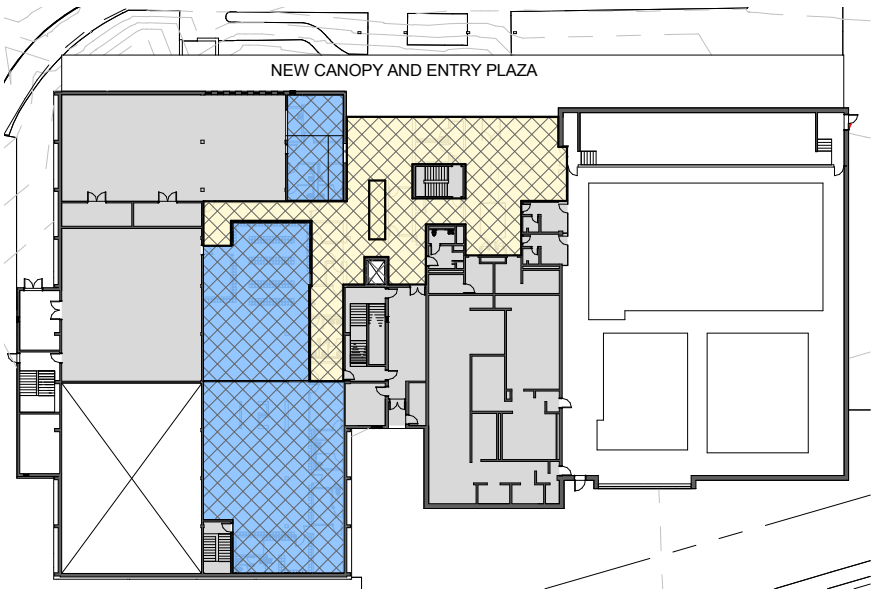
EXTENT OF DEMOLITION EXISTING - NO WORK RENOVATION

OPTIONAL PROJECT - with Option 1 or Option 4

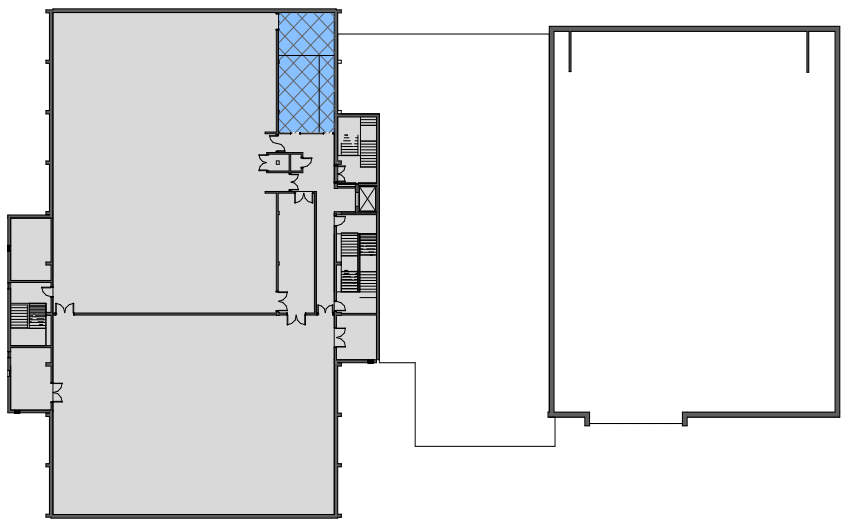
1. Tappan Gym - Minimal Renovations



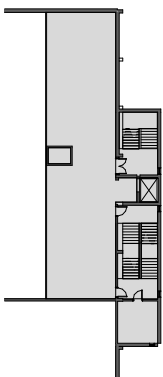
Basement



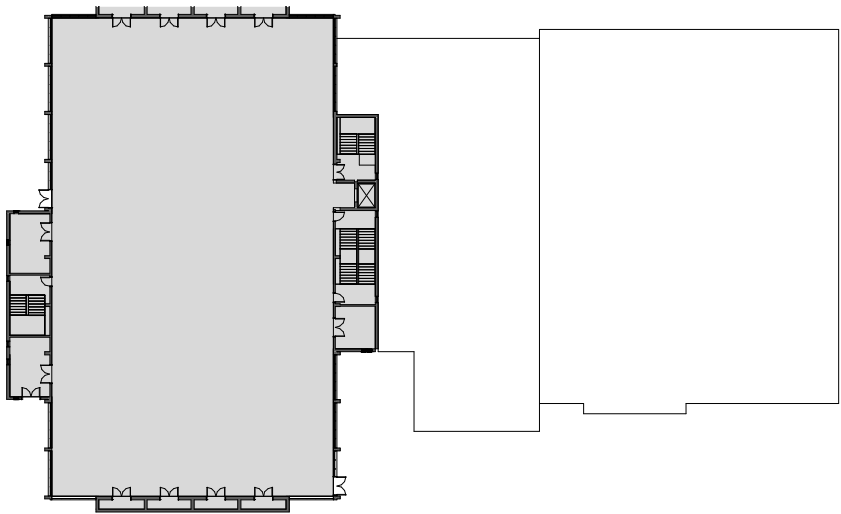
First Floor



Second Floor



Mezzanine



Third Floor

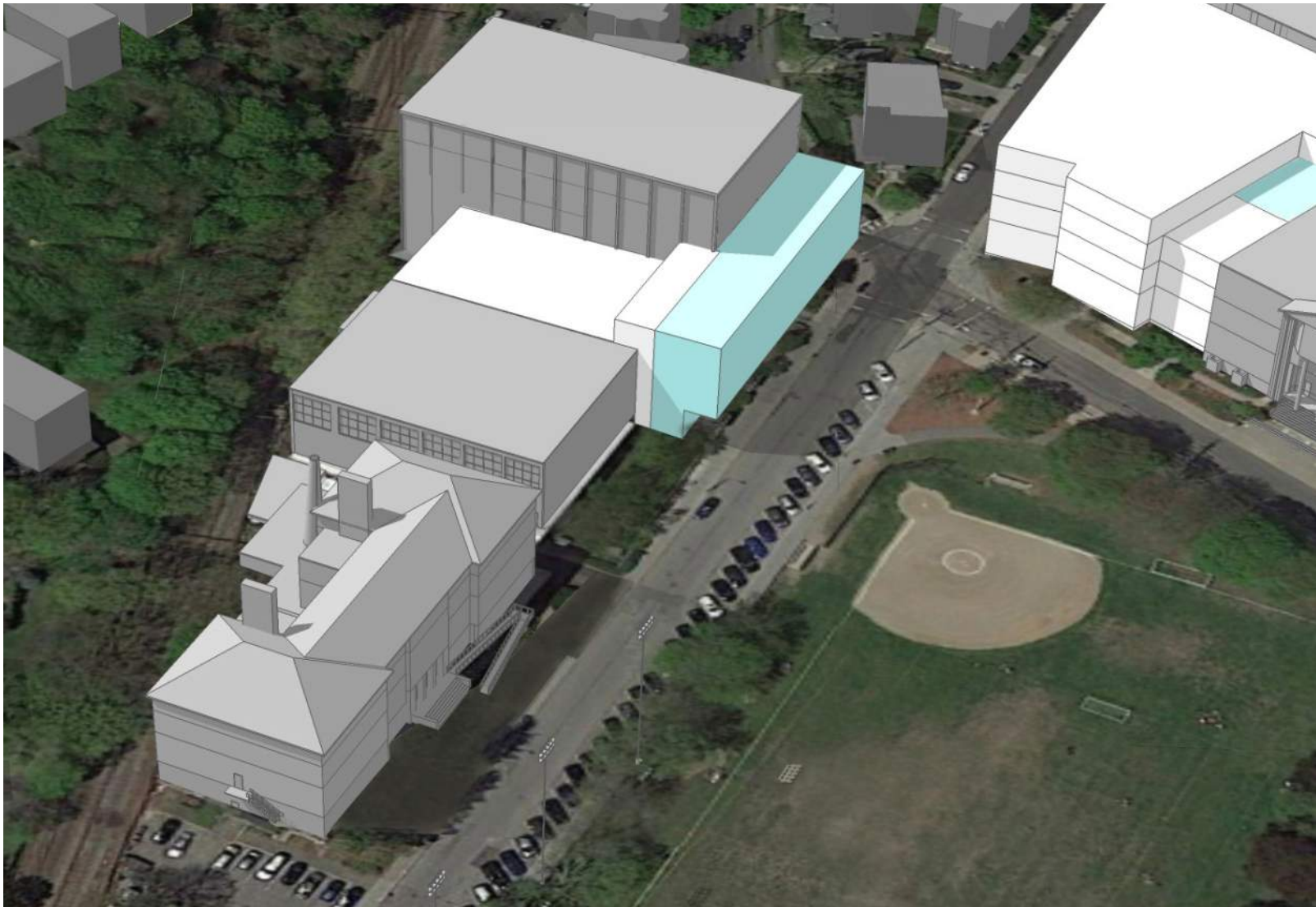
CIRCULATION HEALTH AND PHYSICAL EDUCATION

EXTENT OF DEMOLITION EXISTING - NO WORK RENOVATION

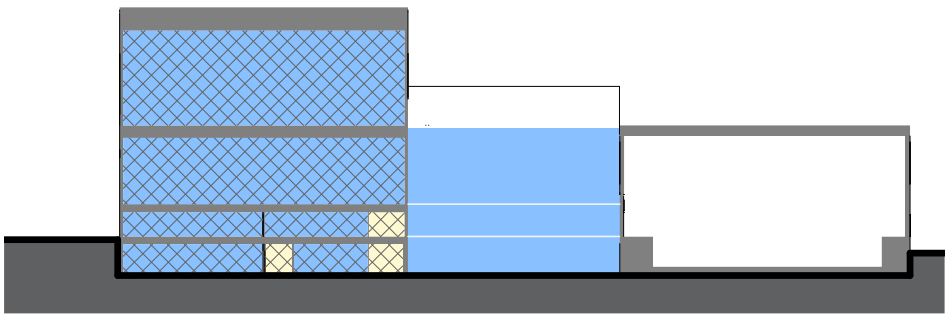
OPTION 2

MA State High School Standards

2. Tappan Gym - Moderate Expansion with Competition & Practice Gym



Tappan Gym Massing



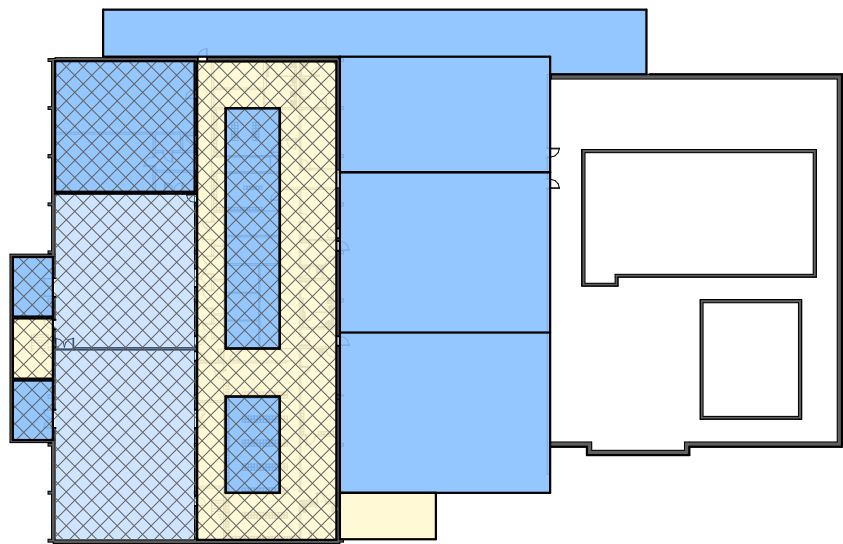
Tappan Gym Section

CIRCULATION HEALTH AND PHYSICAL EDUCATION EXTENT OF DEMOLITION EXISTING - NO WORK RENOVATION

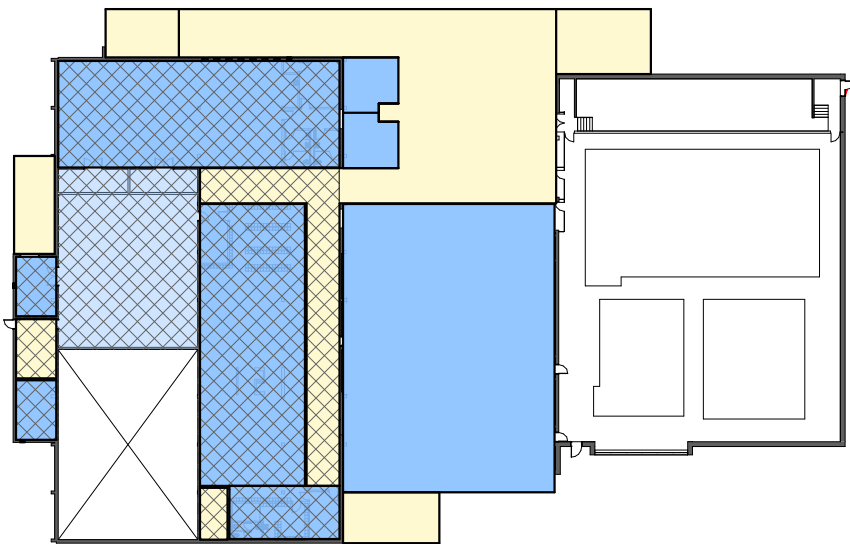
OPTION 2

MA State High School Standards

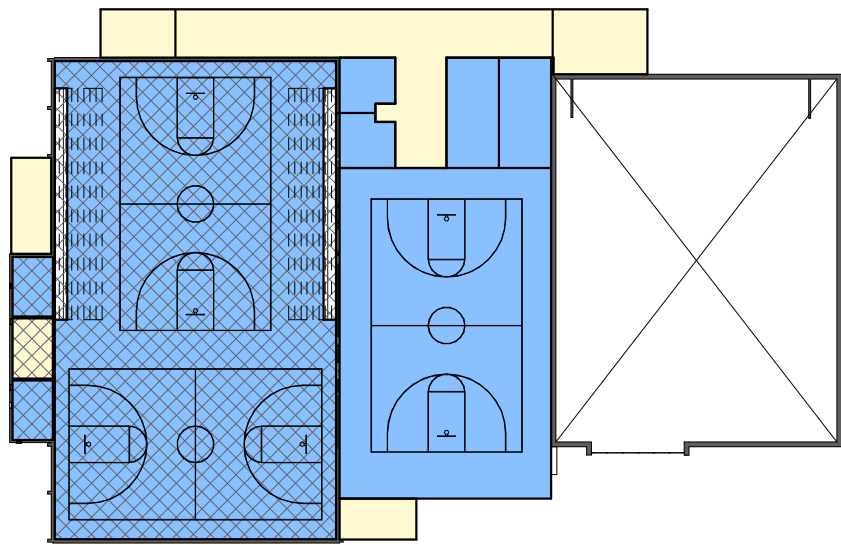
2. Tappan Gym - Moderate Expansion with Competition & Practice Gym



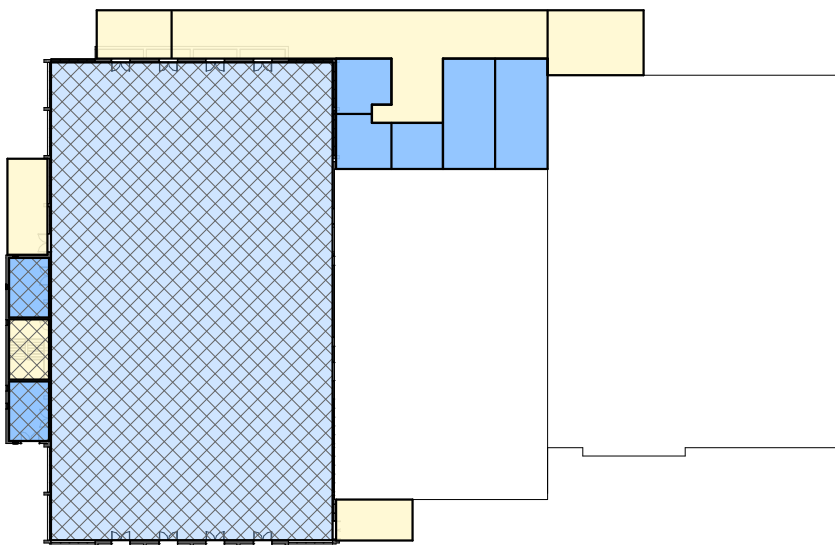
Basement



First Floor



Second Floor



Third Floor

CIRCULATION

HEALTH AND PHYSICAL EDUCATION

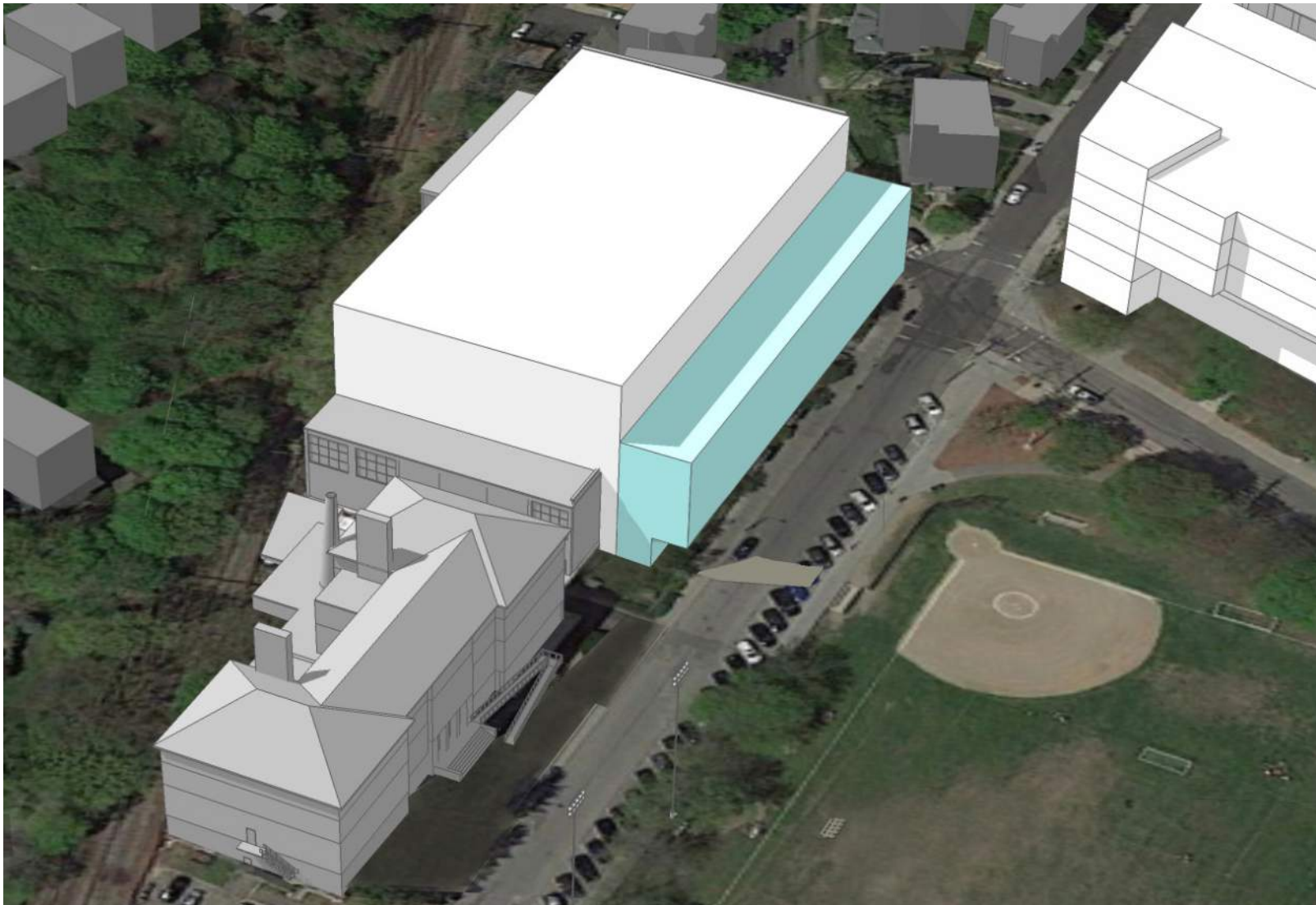
.....

EXTENT OF DEMOLITION

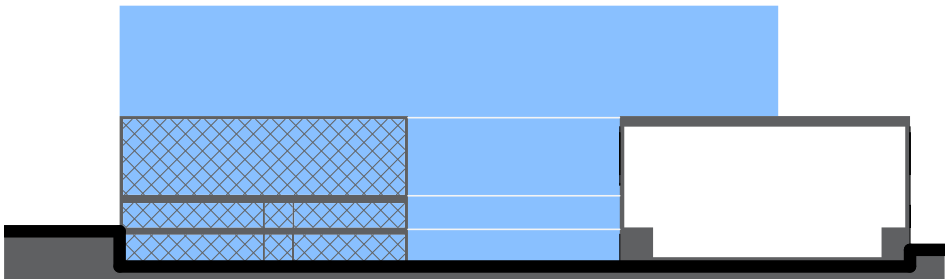
EXISTING - NO WORK

RENOVATION

OPTION 3
Brookline High School Education Plan
3. Tappan Gym - Education Plan with Field House



Tappan Gym Massing



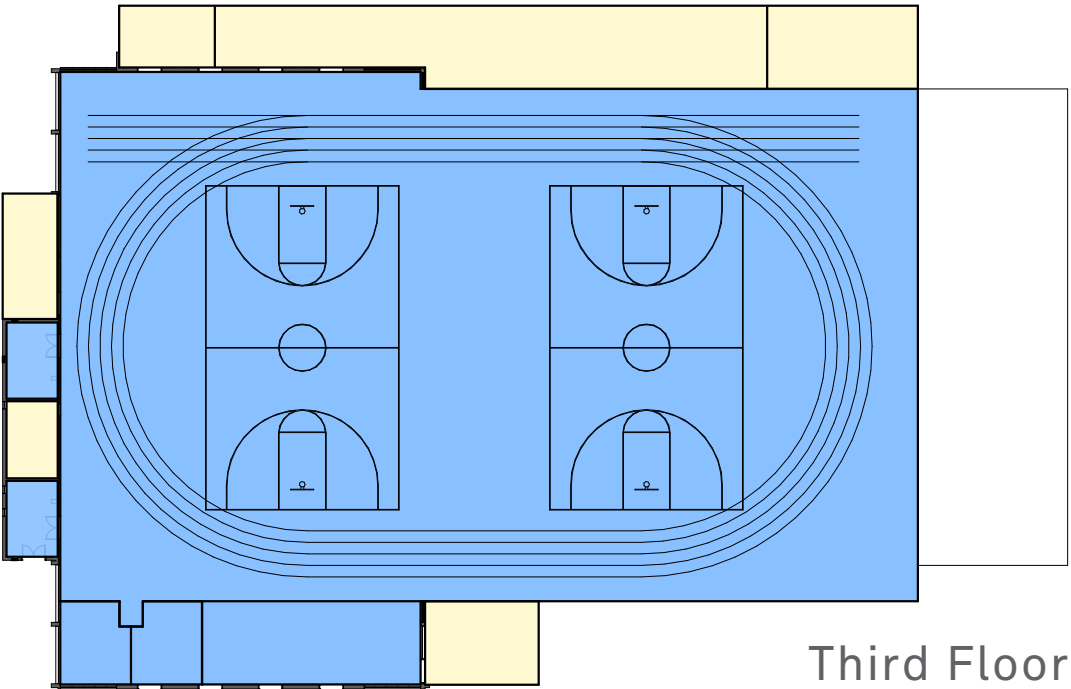
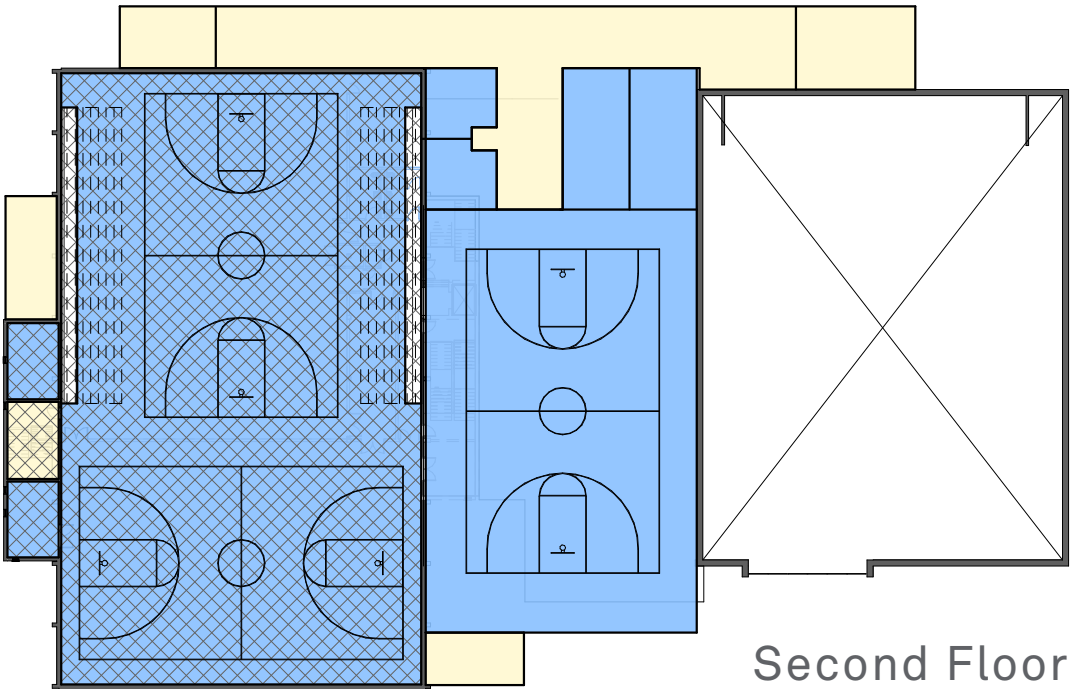
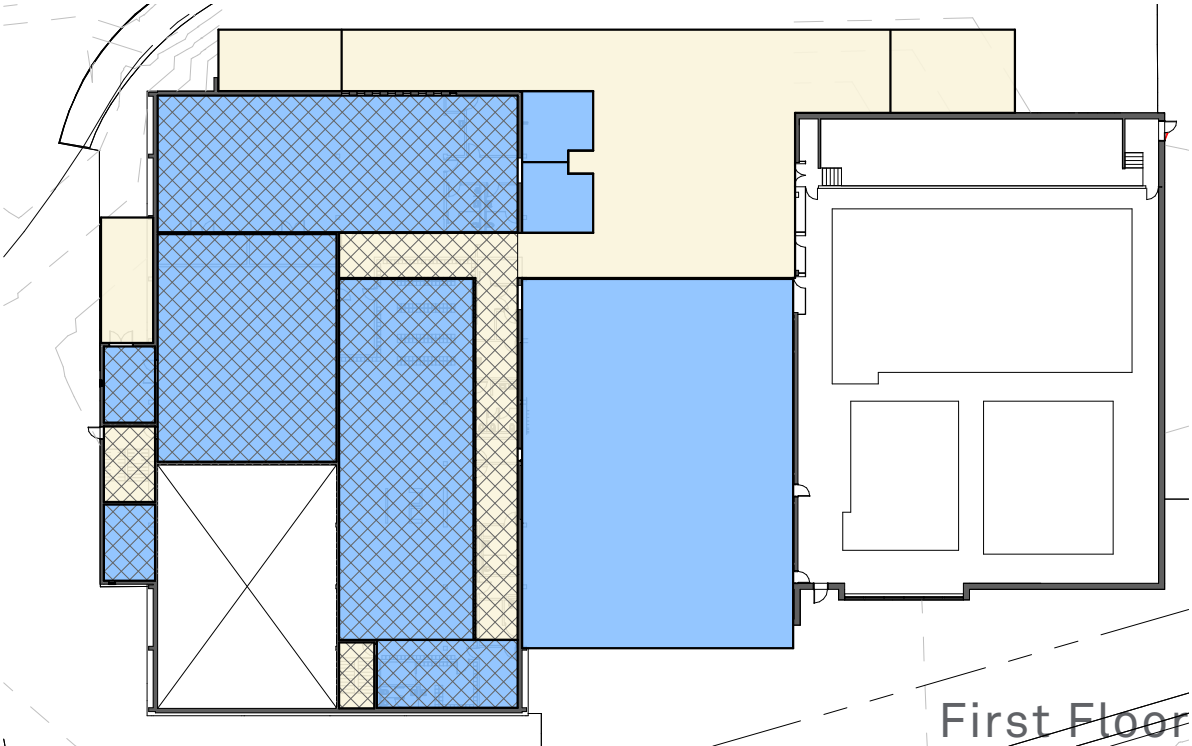
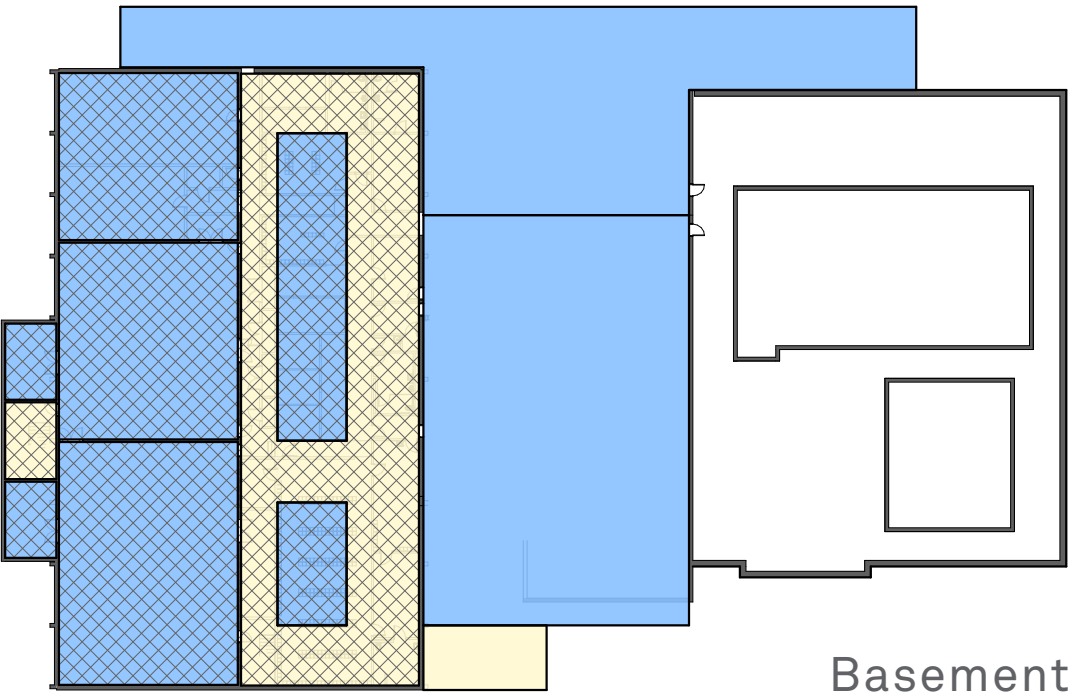
Tappan Gym Section

CIRCULATION HEALTH AND PHYSICAL EDUCATION EXTENT OF DEMOLITION EXISTING - NO WORK RENOVATION

OPTION 3

Brookline High School Education Plan

3. Tappan Gym - Education Plan with Field House



CIRCULATION

HEALTH AND PHYSICAL EDUCATION

..... EXTENT OF DEMOLITION

EXISTING - NO WORK

RENOVATION

Program Priorities

	Option 1	Option 2	Option 3	Option 4	Option 4	Option 4
Core Academics	Enrollment Accommodation	MA State High School Standards	Brookline BHS Education Plan	Stand-alone Cypress Building	Main Building Renovate Science	Main Building New STEM Wing
Classrooms						
Existing	62	38	40		65	56
New Classrooms	30 New	57 New	58 New	26 New	5 New	14 New
Science Classrooms						
New Science	24 New	24 New	24 New	6 New		18 New
Renovate Science					18	
Special Education						
Learning Centers	8	12	13	6 New	Existing to Remain	Existing to Remain
OT/PT	1 New	1 New	1 New	1 New		

Program Priorities

	Option 1	Option 2	Option 3	Option 4	Option 4	Option 4
	Enrollment Accommodation	MA State High School Standards	Brookline BHS Education Plan	Stand-alone Cypress Building	Main BHS Renovate Science	Main BHS New STEM Wing
Collaborative Learning						
Small Group Rooms		16 New	34 New	1 New		
Maker-Spaces	1 New	2 New	4 New	1 New	Existing to Remain	2 New
Collab.Learning Common	Existing to Remain	4 New	11 New	5 New	MLK Room to Remain	2 New
Library	Relocate Reno/Expand	New & Expanded	New & Expanded	1 New Satellite	Existing to Remain	Existing to Remain
Cafeteria	Reno/Expand Existing	Reno/Expand Existing	New & Expanded	1 New Satellite	Existing to Remain	Existing to Remain
Career Technology Ed.	New Culinary Arts at BHS	New Culinary Arts at BHS	New Culinary Arts at BHS	Culinary Arts at BHS	Existing to Remain	New Culinary Arts at BHS
OPTIONAL Project	Existing UAB to Remain	Renovations at UAB	Renovations & Expand at UAB	Existing UAB to Remain	Existing UAB to Remain	Existing UAB to Remain

Program Priorities

	Option 1	Option 2	Option 3	Option 4	Option 4	Option 4
	Enrollment Accommodation	MA State High School Standards	Brookline BHS Education Plan	Stand-alone Cypress Building	Main BHS Renovate Science	Main BHS New STEM Wing
Performing/ Visual Arts						
Music & Perform Space	New Black Box	New Black Box and White Box	New Black Box, White Box and 400 seat Theater	1 New	Existing to Remain	Existing to Remain
Ensemble & Music Practice	New Chorus	New Band and Chorus	New Expanded Band & Chorus	3 New		
Arts Studios	New at BHS	New at BHS	New at BHS	1 New	Existing to Remain	Existing to Remain
OPTIONAL Project	Existing UAB to Remain	Renovations at UAB	Renovations & Expand at UAB	and at UAB to Remain	Existing UAB to Remain	Existing UAB to Remain
Physical Ed. Facilities						
Schluntz Gym (Competition)	Existing to Remain	Replace at Tappan	Replace at Tappan	Existing to Remain	Existing to Remain	Existing to Remain
Health/Fitness	Existing to Remain	2. Tappan Moderate Renovation	3. Tappan Renovation Expansion with Field House & Competition Gym	2 New Health Classrooms	Existing to Remain	Existing to Remain
Tappan Facility	Existing to Remain	Expansion with Competition Gym		Existing to Remain	Existing to Remain	Existing to Remain
OPTIONAL Project	1. Renovate Tappan Gym			1. Renovate Tappan Gym		

Option 1, 2 & 3

Summary and Cost

Option 1



Existing to Remain at BHS
294,310 sf

New Construction at BHS
165,000 sf

Project Cost
\$181,572,560

Option 2



Existing to Remain at BHS
242,280 sf

New Construction at BHS
228,690 sf

Project Cost
\$212,261,728

Option 2 with Tappan



Existing to Remain at BHS
242,280 sf

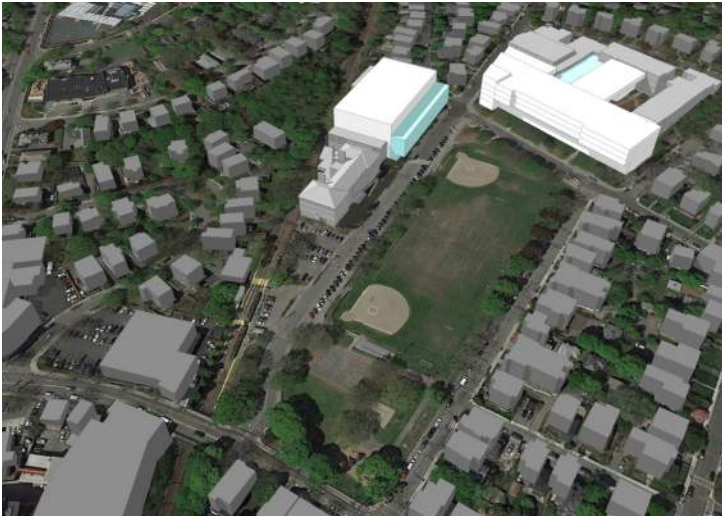
New Construction at BHS
228,690 sf

Renovation at Tappan Gym
60,330 sf

New Construction at Tappan Gym
49,300 sf

Project Cost
\$271,178,244

Option 3



Existing to Remain at BHS
181,050 sf

New Construction at BHS
298,130 sf

Renovation at Tappan Gym
60,130 sf

New Construction at Tappan Gym
64,050 sf

Project Cost
\$348,352,647

Option 4

Summary and Cost

Option 4A



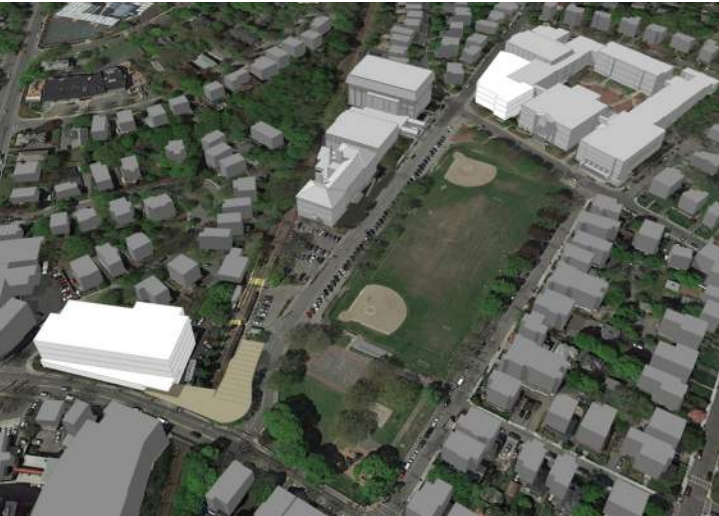
Renovate Existing BHS
Science
35,500 sf

New Construction at BHS
7,500 sf

New Construction
Cypress Building
109,810 sf

Project Cost
\$105,342,903

Option 4B



Renovate Existing BHS
Science into Classrooms
36,500 sf

New Construction at BHS
44,100 sf

New Construction
Cypress Building
109,810 sf

Project Cost
\$131,117,378

Option 4C



Renovate Existing BHS
Science
35,500 sf

New Construction at BHS
7,500 sf

New Construction Cypress
Building over the MBTA
112,700 sf

Project Cost
\$110,838,332

Option 4D



Renovate Existing BHS
Science into Classrooms
36,500 sf

New Construction at BHS
50,400 sf

New Construction Cypress
Building over the MBTA
112,700 sf

Project Cost
\$136,612,807

OPTIONAL PROJECTS

Tappan Gym Upgrades

1. Minimal Renovations



Project Cost
\$13,290,000

2. Moderate Expansion with Competition & Practice Gym



Project Cost
\$62,266,000

Note: Option 2. Tappan Gym Moderate Expansion is included in the program, planning for the BHS OPTION 2. State High School Standards. Costs have been summarized both with and without this Tappan Gym Upgrade

2.1 Min.Moderate Expansion with Practice Gym only



Project Cost
\$52,706,000

3. Education Plan with Field House



Project Cost
\$74,000,000

Note: Option 3. Tappan Education Plan with Field House is included in the program, planning, and costs for the BHS OPTION 3. Education Plan

OPTIONAL PROJECTS

Cypress Park Field & Playground

1. Field & Playground Renovation



Project Cost
\$13,335,500

2. Field & Playground Renovation with Greenough St. Re-alignment



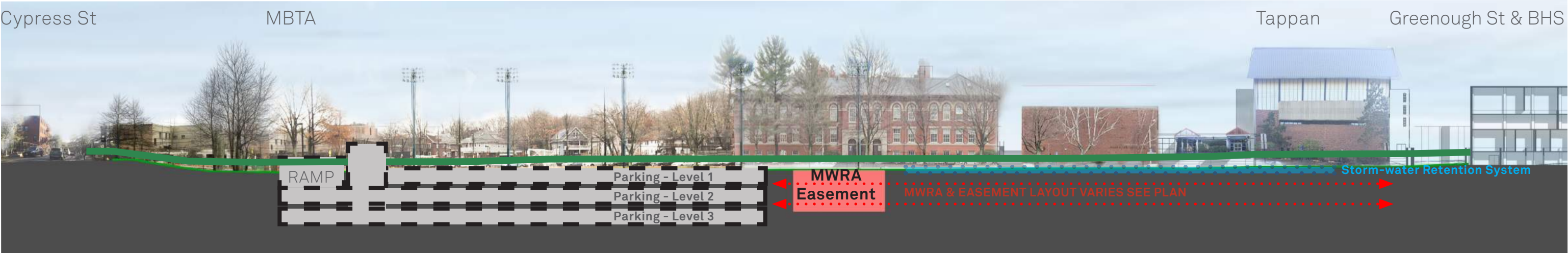
Project Cost
\$14,638,500

Underground Parking

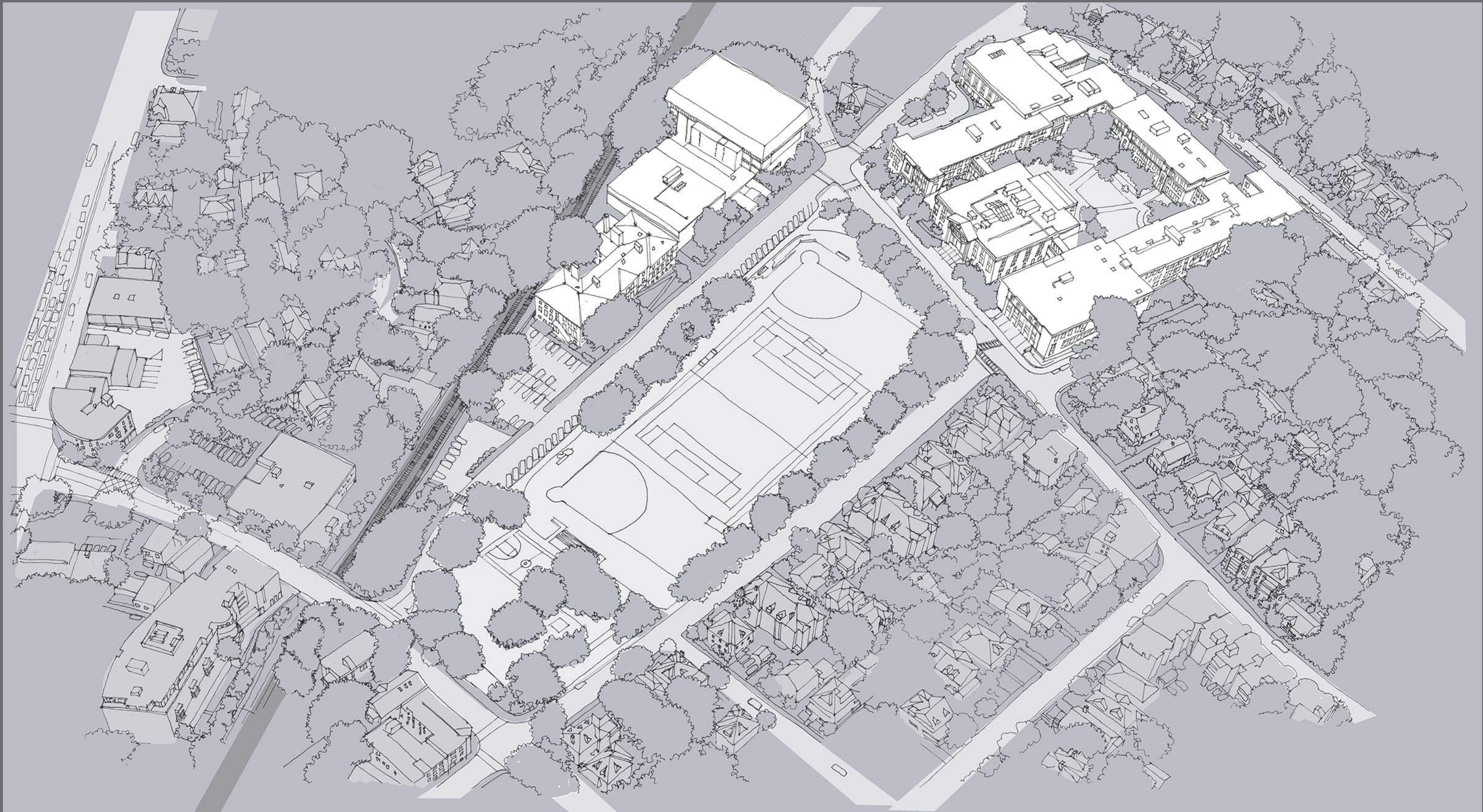
1. Typical Plan for Parking below Cypress Park



Project Cost
1-Level 100 cars = \$21,051,500
2-Levels 200 cars = \$33,769,600
3-Levels 300 cars = \$46,806,000



Site Section: Underground Parking Structure Below Cypress Park - Field @ +/- 100 Cars per Level



WARRANT ARTICLE 16

Submitted by: Board of Selectmen

To see if the Town will authorize and empower the Board of Selectmen to lease for a term not to exceed twenty (20) years rooftop space on the Runkle School building for the purpose of hosting a rooftop solar photovoltaic installation (solar panels and appurtenant equipment), subject to any authorizations, approvals and reviews, on such terms and conditions determined by the Board of Selectmen to be in the best interest of the Town. Or act on anything relative thereto.

EXPLANATION ARTICLE 16

Submitted by: Board of Selectmen

The Town has been in contract negotiations with developer BlueWave Capital, LLC, Boston, Massachusetts, to install solar photovoltaic arrays on several municipal buildings, including the Runkle School.

As of late 2016, BlueWave Capital partnered with a new installer, IGS Solar LLC, which requires an easement or roof lease agreement to install the panels. Once a lease agreement is in place, the Town can resume negotiations with BlueWave to execute the contract. At this time, the Board is seeking permission to grant a lease agreement for rooftop space on the Runkle School, which would be the first site available for installation by summer 2017.

BACKGROUND FROM PLANNING DEPT.

Here is the information that has been made available to the Board of Selectmen for your reference. The contract is not final and therefore has not been submitted to the Board of Selectmen.

Hearings

SCAC hearing: June 13, 2016

BOS hearing: July 12, 2016

- **Agenda** <http://apps.brooklinema.gov/sirepub/mtgviewer.aspx?meetid=262&doctype=AGENDA>
- **Minutes** <http://apps.brooklinema.gov/sirepub/mtgviewer.aspx?meetid=262&doctype=MINUTES>
- Presentation to BOS: Blue Wave to BOS (attached)

School Committee: July 26, 2016

Vote to approve installation of solar arrays on Runkle School:

- **Agenda** http://www.brooklinema.gov/AgendaCenter/ViewFile/Agenda/_07262016-4012

- Minutes http://www.brooklinema.gov/AgendaCenter/ViewFile/Minutes/_07262016-4012

Library Trustees: July 12, I believe (though not pertinent to WA 16)

Other relevant information

1 – BOS Notices:

Mailed on June 28 for July 12 BOS Hearing on 4 sites (to abutters within 300 foot radius of each site)

2 – BOS Website resources

<http://www.brooklinema.gov/1326/Solar-Panels-Municipal-Sites>

3 – Status Contracts: Runkle installation is only site viable at this time. Contract with Runkle is not final.

4- Contract's Technical Reviewer: Cadmus Group

5 – Town Counsel: John Buchheit



Runkle School Solar Initiative
Metropolitan Area Planning Council Regional Solar Initiative

April 27, 2017

BROOKLINE SOLAR INITIATIVE UPDATE AGENDA

1. Update on Runkle School Project

- Project Design, Capacity, and Yield
- Engineering, Procurement and Construction Plans
- Environmental Benefits

2. Update on Massachusetts' Solar Incentives

- SREC II C Program
- SMART Program

3. Next Steps & Timing

- NMCPA and Lease Review
- Complete Due Diligence
- Construction & Interconnection



Regional Solar Initiative Work to Date in Brookline:

- **Site Assessments:** Assessed roof age, condition, and availability of suitable ground, and/or parking lots. Projects moving forward in this initial stage include the Runkle School.
- **LOI Execution:** BWC & Brookline executed a Letter of Intent for Short and Long Term projects.
- **Rate Calculations¹:** BlueWave and IGS Solar updated the NMCPA/PPA based on updated project design, anticipated solar generation, and long-term O&M needs, and incentive program.
- **Finalize System Designs¹:** BlueWave and Advanced Solar Products have finalized system designs needed to execute with interconnection applications with Eversource.

Next Steps:

- **Finalize Net Metering Credit Purchase Agreement and Lease:** BlueWave will work with the Town to finalize the agreements for the system.
- **Obtain approvals:** Our team will complete structural review of the rooftop, submit applications for building and electrical permits, and seek final approval from the Board of Selectmen.
- **Construction & Operations:** Advanced Solar will commence construction once all necessary permits and approvals are secured, likely in July, 2017. The system is anticipated to begin operating during the Fall, 2017.

¹Final system sizes will be refined based on customer feedback, compliance with incentive programs, and are subject to additional engineering and design verification.

RUNKLE SCHOOL SYSTEM UPDATE



System Summary

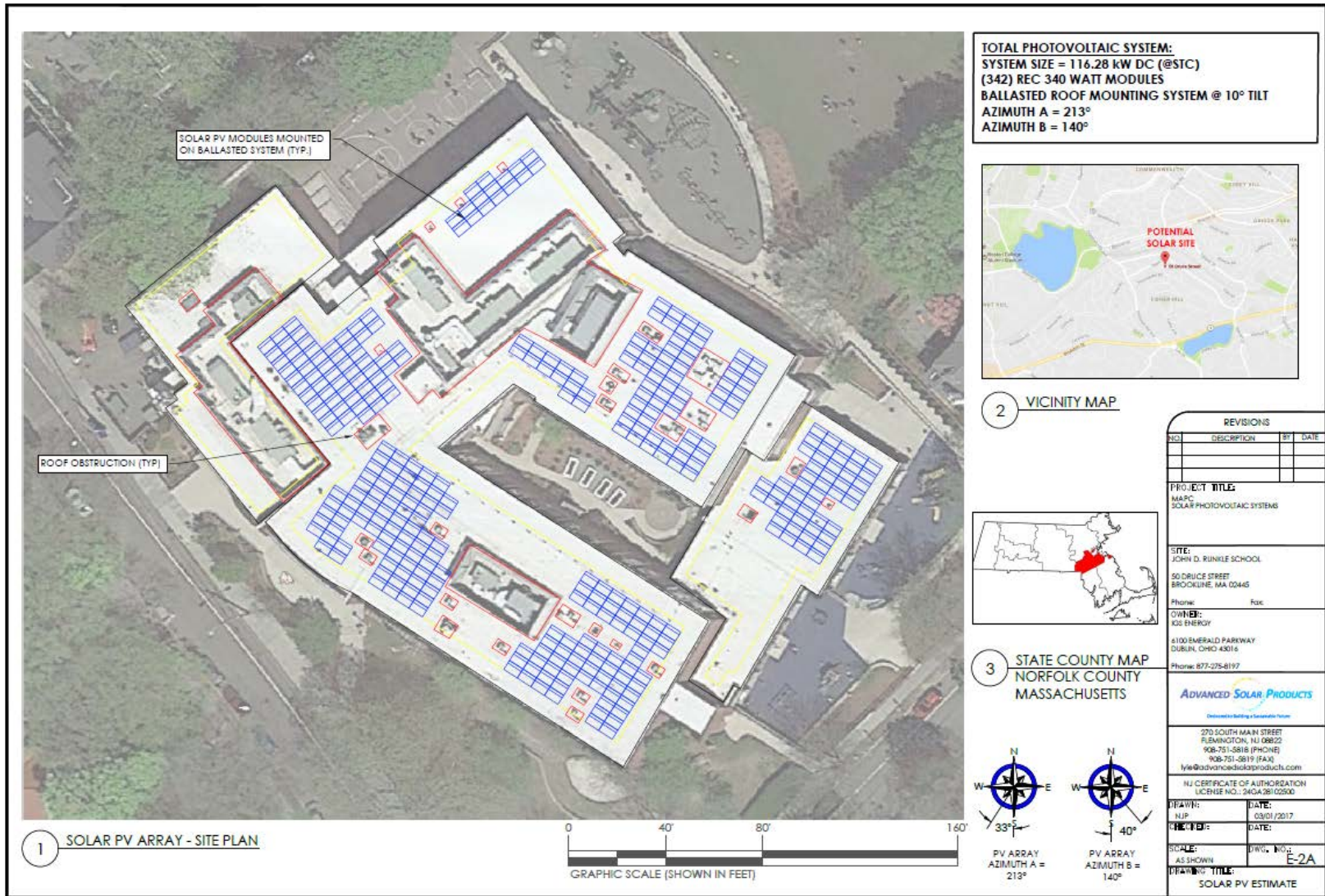
Site	Address	Type	Capacity KW DC ^A	Modules #	Estimated Yield KWH/KW/Year ^A	Year 1 KWH Production ^A
Brookline Runkle School	50 Druce Street	Rooftop	116.28	342	1,260	146,512.8

- **Key Updates:**

- BWC, IGS Solar, and Advanced Solar updated assumptions for the site, specifying high-quality Tier-1 materials of use, increasing the potential capacity of the system, and identifying scenarios to improve economics.
- Updated NMCPA reflects estimated Engineering, Procurement, and Construction costs from Advanced Solar. No utility upgrades are anticipated at this time, per the executed Interconnection Services Agreement from Eversource.
- Lower costs combined with increased system size system improved offer to the Town.

^A Estimated based on best-known parameters, realized savings achieved are contingent on final design and engineering.

RUNKLE SCHOOL SYSTEM UPDATE

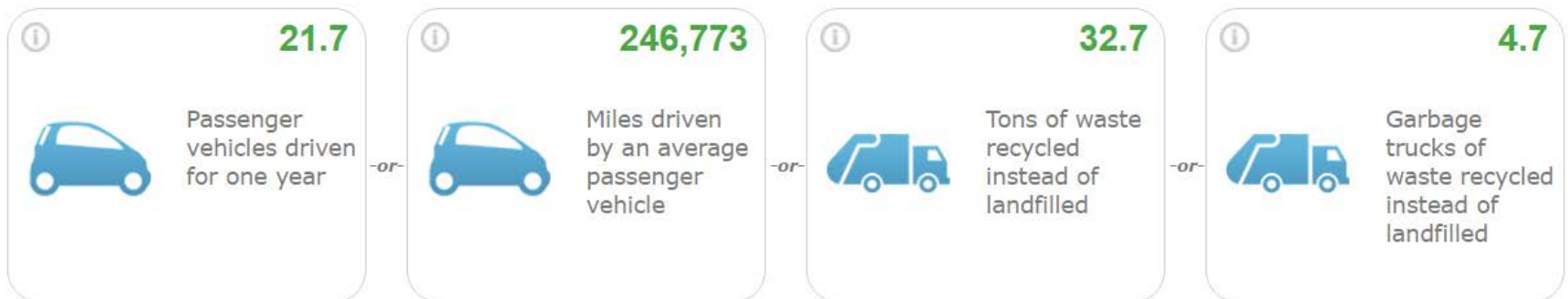


RUNKLE SCHOOL SYSTEM UPDATE

Runkle School Environmental Benefits Summary

- Energy Generation of **154,635** kilowatt hours annually → **103** Metric Tons of Carbon Dioxide Equivalents
- Meets **22.5%** of energy use at the Runkle School (657,976 kWh in 2015)

Green House Gas emissions from:



CO2 Emissions from:



MASSACHUSETTS SOLAR INCENTIVES UPDATE

Policy	Overview
Federal Investment Tax Credit	• Provides a 30% tax credit for solar systems on commercial solar projects, and can be monetized through the third-party ownership structure. • The ITC is based on the amount of investment in solar property, and is equal to 30% of the basis that is invested in eligible property which commences construction through 2019. • The ITC will step down to 26% in 2020 and 22% in 2021, and 10% after 2023.
Solar Carve-out Program	• Solar Carve-out requires Massachusetts' utilities to purchase SRECs from qualified solar energy generators. In 2014, the SREC II program established market sectors, providing preferential incentives for carports (100%), and rooftop (90%) projects. • MA DOER is currently accepting requests for extension to any facility that has not already secured qualification for SRECs, provided the facility will be mechanically complete in advance of the SMART programs' start date. Projects qualified currently under the SREC II program are eligible for SRECs at a discount, 70% for carports and 60% for rooftop projects.
Upcoming SMART Program	• Developed as a replacement to both the Solar Carve Out and Net Metering programs, DOER's SMART program will provide a production based incentive to qualified projects • Value of incentive is based on the project's capacity, with adders for preferential project types, including: rooftops, brownfields, landfills, and carport projects; projects which serve low income, community solar, or public off-takers; and those including batteries. • Developed by MA DOER and implemented by MA Department of Public Utilities, the SMART program should be operational by March, 2018.

NEXT STEPS

- **Finalize System Design, EPC Costs** May, 2017
BWC and Advanced Solar will finalize the system design, and provide an estimate of engineering, procurement, and construction costs which will be used to provide a updated terms.
- **Contract and Lease Review** June, 2017
BWC will provide a fully vetted/pre-negotiated contract reviewed by Cadmus, BWC, and the Metropolitan Area Planning Council for Brookline's review.
- **Population Of Contract Pricing And Terms** June, 2017
Contracts will be completed and submitted for final review and execution by Brookline.
- **Due Diligence and Permitting** June, 2017
Complete structural and electrical review, submit applications for building and electrical permits, and move ahead with final approval from Board of Selectmen.
- **System Construction** Summer, 2017
BlueWave and Advanced Solar will begin construction once the final agreements are executed, during Summer, 2017. Construction will last approximately 6 weeks.
- **System Operation** Fall, 2017
IGS Solar will begin to operate the system once the local utility issues permission to operate.

BlueWave Capital, LLC

137 Newbury Street
Boston, MA 02116
Tel: 617.209.3122
www.bluewave-capital.com

Sandra Brown
Director
Tel: 404.695.0555
sbrown@bluewave-capital.com

PETITIONED ARTICLE FOR SPRING 2017 TOWN MEETING

PETITIONERS: Werner Lohe, Alan Christ, Kathleen Scanlon

ARTICLE 21:

A RESOLUTION REGARDING A NET ZERO ENERGY NINTH ELEMENTARY SCHOOL AND THE EXPANSION OF BROOKLINE HIGH SCHOOL

To see if the Town will vote to adopt the following resolution:

Whereas our town, the nation, and the world are increasingly aware of the need to address climate change and of the importance of better protection of the environment in general, and

Whereas Net Zero Energy LEED Platinum schools create an environment that supports student learning and health through improvements in daylighting, indoor air quality, thermal comfort, acoustics, and classroom design, all of which have an impact on a child's ability to learn and a teacher's ability to teach, while saving energy, resources, and money, and

Whereas Net Zero Energy LEED Platinum schools increase energy efficiency, thereby reducing greenhouse gas emissions, cost less to operate, utilize durable materials, reduce water and energy use, and provide other benefits; while providing an educational experience that transcends the classroom by creating opportunities for curriculum innovation and hands-on, project-based learning in which the building itself becomes an interactive teaching tool, and

Whereas decisions made now about the design of the Ninth Elementary School at Baldwin and the expansion of Brookline High School will determine each school's environmental footprint, particularly greenhouse gas emissions, for decades to come, and

Whereas the technical ability to create energy-efficient, high performing buildings has increased significantly by incorporating systems thinking into design processes, and

Whereas construction of new schools in Massachusetts and around the nation during the past five years has shown the feasibility and desirability of Net Zero Energy schools, that is, schools in which the amount of energy used on an annual basis is equal to the amount of renewable energy created on the site, and

Whereas an international standard known as LEED (Leadership in Energy and Environmental Design) allows for a building's environmental and energy performance to be accurately measured and provides a benchmark to assist in designing a Net Zero Energy building,

Whereas the most accurate measure of energy efficiency for a building is EUI (Energy Use Intensity), calculated by dividing total energy consumed annually by the gross floor area of the building,

Now therefore be it Resolved that in order for the Ninth Elementary School at Baldwin to be a Net Zero Energy school, it shall be designed to obtain LEED v4 certification at the Platinum rating level, and in addition, specifically, to achieve 16 of the possible 16 points available in the Optimize Energy Performance category and to achieve an EUI of 25 kBtu/sq.ft.,

And, be it further Resolved that while overall net zero energy consumption is unlikely to be achieved for Brookline High School and even the degree to which the as-of-yet-undefined, expanded portion of the school can approach Net Zero Energy is currently uncertain, nevertheless, Net Zero Energy principles shall be applied, to the extent feasible, during all design phases of Brookline High School,

or act upon anything relative thereto.

EXPLANATION

Summary

Brookline was one of the first communities in Massachusetts to address climate change, adopting its first Climate Action Plan in 2002. In 2012, it accepted Massachusetts' target for greenhouse gas emissions—the reduction in emissions to 80% below 1990 levels by 2050. See “2012 Climate Action Plan,” p. 9 (<http://www.brooklinema.gov/DocumentCenter/View/2402>). In part in response to this statement of our community's environmental values, recent school construction (i.e., the Runkle School and the Devotion School) has achieved high standards of energy efficiency. Similarly, in the Educational Plan for the Ninth Elementary School at Baldwin, the School Committee has stated its strong commitment to a state-of-the-art school.¹ Nevertheless, overall town-wide progress toward reduced emissions has been slow, not yet approaching the rate needed to reach our goal. See, generally, “Selectmen's Climate Action Committee Report to Town Meeting, Spring 2015,” p. 2 (<http://www.brooklinema.gov/DocumentCenter/View/8158>).

New construction of any sort inevitably leads to a slight increases in emissions. Therefore, communities throughout Massachusetts and New England have begun to address that reality by designing “Net Zero Energy” (NZE) schools, that is, schools that minimize on-site energy use as much as possible, and offset that energy use with renewable energy generated on site, with the goal of equalizing, on an annual basis, energy consumed and renewable energy generated on site. (Examples of such schools are the Martin Luther King School, Cambridge, MA, the King Open School, Cambridge, MA, and the Pell Elementary School,² Newport, RI.) Schools such as this not only address climate change, but typically save money by reducing energy costs. See below. Brookline now has an opportunity to design its own NZE school.

About ten years ago, Brookline improved its design process for municipal buildings by including consideration of “environmental and sustainability goals and objectives, includ[ing] design and construction practices that explicitly consider Green technologies.” Bylaws, § 7.3.2(a). Because of the Town's sound design practices and the Massachusetts School Building Authority's (MSBA) standards, the Runkle and Devotion projects have good energy performance, though they are not NZE. The Ninth Elementary School at Baldwin, however, is not subject to MSBA requirements, and no specific standards for energy performance have been set. This resolution provides direction

1. “ “Building a new school in the early 21st century when our community and society are more conscious than ever of the delicate balance between environmental sustainability and ongoing development provides an opportunity to have the physical plant itself play a significant role in the culture, educational approach and daily lives of students and teachers. Whether it's through monitoring waste water, understanding the science behind passive and active solar power, or studying conservation measures built into the new building, the physical plant can be used to help students learn about science, sustainability, and taking care of the environment. For example, signs and working exhibitions created by students could identify design elements that demonstrate architectural, structural, mechanical, and green building strategies. Student tour guides could be trained to introduce visitors to the building's features. Back-of the-house spaces could be used as instructional spaces for students and staff, and could be used by town building and maintenance staff for hands-on training. ***Brookline's new elementary school could stand as a physical demonstration of environmental stewardship and innovation, providing a local case study for sustainable school construction.***” 9th Elementary School Educational Program (11/29/16 draft), p. 13 (emphasis added). <http://www.brookline.k12.ma.us/cms/lib8/MA01907509/Centricity/Domain/722/Draft%209th%20School%20Ed%20Plan%20-%2011.29.2016.pdf>

2. “The Pell School is... considered a net-zero-ready building. With the addition of a 1,100-kW photovoltaic system in the future, the school will be an actual net-zero building.” <https://webspm.com/Articles/2014/08/01/Net-Zero-in-School-Design.aspx>. Binghamton, NY designed a NZE school, the MacArthur Elementary School, that is not only NZE, but also “fossil-fuel-free.” Building Energy, vol. 36, no. 1 (2017), p. 12,

to the design architects, under the supervision of the School Committee, the Building Commission, and the Board of Selectmen, by establishing Net Zero Energy as the community's goal for its new school. This goal encourages energy efficient building design, places a major focus on energy, and encourages building users to reduce their energy needs without compromising building programs or mission.

Mechanism for Assuring a NZE Design

The most widely used and accepted rating system in the United States and in the world for green buildings is LEED v4 (Leadership in Energy and Environmental Design, version 4). See <http://www.usgbc.org/LEED>. Building performance is measured by awarding up to a total of 110 checklist points in eight categories and 57 sub-categories. The single sub-category in which the most points are available—with 16 points—is Energy and Atmosphere: Optimize Energy Performance. Consultation with architects and energy efficiency consultants familiar with schools, NZE principles, and LEED indicates that the realistic goal for achieving NZE for the Ninth Elementary School at Baldwin is 16 Optimize-Energy-Performance points with certification at the Platinum rating level with Energy Use Intensity (EUI) of 25 kBTU/sq.ft.

Cost

NZE schools, in addition to addressing climate change, typically save money by reducing energy costs. See <https://www.districtadministration.com/article/net-zero-schools-save-big-energy-costs>. Further, the Town and its architect are committed to “integrated design,” which is the most effective process for ensuring not only energy efficiency, but also cost savings. See <http://www.facilitiesnet.com/facilitiesmanagement/article/When-talk-is-cheaper-Integrated-design-and-better-buildings-Facilities-Management-Facilities-Management-Feature--2138>.

Integrated design relies on careful planning and goal setting early in the design process. Project costs and savings are evaluated during the schematic design phase of the project. Thus, the savings (or costs) that may be realized by designing an NZE school cannot be accurately assessed at present since schematic design has not yet been authorized by Town Meeting. But, consistent with best practices, an important provision in the Town's bylaw requires that “the consultant [(the architect) prepare] a cost estimate for the project (including life-cycle costs) [and] consider the investigation, cost-benefit analysis, and recommendation of appropriate options that address the environmental and sustainability goals and objectives....” Bylaw, § 3.7.2(b).

The bylaw does not specify the methodology to be used in the cost-benefit analysis. The Town has frequently used “simple payback period” methodology. But, because of the complexity of the design of a large project such as the Ninth Elementary School at Baldwin and the desire to achieve the goal of NZE, financial analysis of energy-related features such as geothermal HVAC, solar PV panels, triple-glazed windows, and the like should include a Discounted Cash Flow (DCF) analysis, considering Internal Rate of Return (IRR) or Net Present Value (NPV) or both, based upon the life of the features,³ and should also consider the Social Cost of Carbon.⁴

3. Funding for the Ninth Elementary School at Baldwin will be provided by the issuance of municipal bonds. Energy-related features whose nominal rate of return, as compared to baseline conventional features, are projected to exceed the Town's cost of capital in the municipal bond market should be chosen because they are the more fiscally responsible alternative.

4. See “Technical Support Document: Technical Update of the Social Cost of Carbon for Regulatory Impact Analysis Under Executive Order 12866” published by the Interagency Working Group on Social Cost of Carbon, United States Government, in May 2013, and revised in November 2013 and July 2015.

Warrant Article 21 **NET ZERO ENERGY SCHOOLS**

Petitioners:

Werner Lohe
Climate Action Brookline (founding member 2000)
TMM, Pct. 13

Alan Christ AIA, LEED Accredited Professional
Member of Green Caucus & EDAB
TMM, Pct. 4

Kathleen Scanlon, R.A., LEED Accredited Professional
Member of Green Caucus, Selectmen's Climate Action
Committee, and Mothers Out Front
TMM, Pct. 3

WA 21: Net Zero Ninth Elementary School



Climate Change

WA 21: Net Zero Ninth Elementary School



The Political Climate Has Changed

ACT LOCALLY



- **Runkle School**
From 410 to 560 Students,
105,000 SF,
\$29 Million Construction Cost,
\$276/SF



- **Devotion School**
1010 Students,
227,087 SF,
\$120 Million Construction Budget
\$528/SF

MSBA Limitations

(Mass. School Building Authority)

We can do even better.

Ninth Elementary School



Ninth Elementary School **Unambiguous Goals**

Building Commission Bylaw

“Environmental and sustainability goals... include design and construction practices that explicitly consider Green technologies,... energy efficiency....”

—Building Commissions Bylaw, § 3.7.2(a).

BUT NO SPECIFIC STANDARDS

Goals the architects can design to....

- LEED Platinum Certification
- 16 of 16 LEED Energy Performance Points
- EUI of 25 (Energy Use Intensity, in kBtu/ft²/year)

...a significant advance toward a Net Zero Energy school...

Need for Flexibility to stay within Budget Constraints

Goals the architects can design to...

- LEED Platinum Certification
- 16 of 16 LEED Energy Points
- EUI of 25

Minimums

- LEED **Silver** Certification
- **13** of 16 Energy Points
- EUI of **30**

Goals the architects can design to...

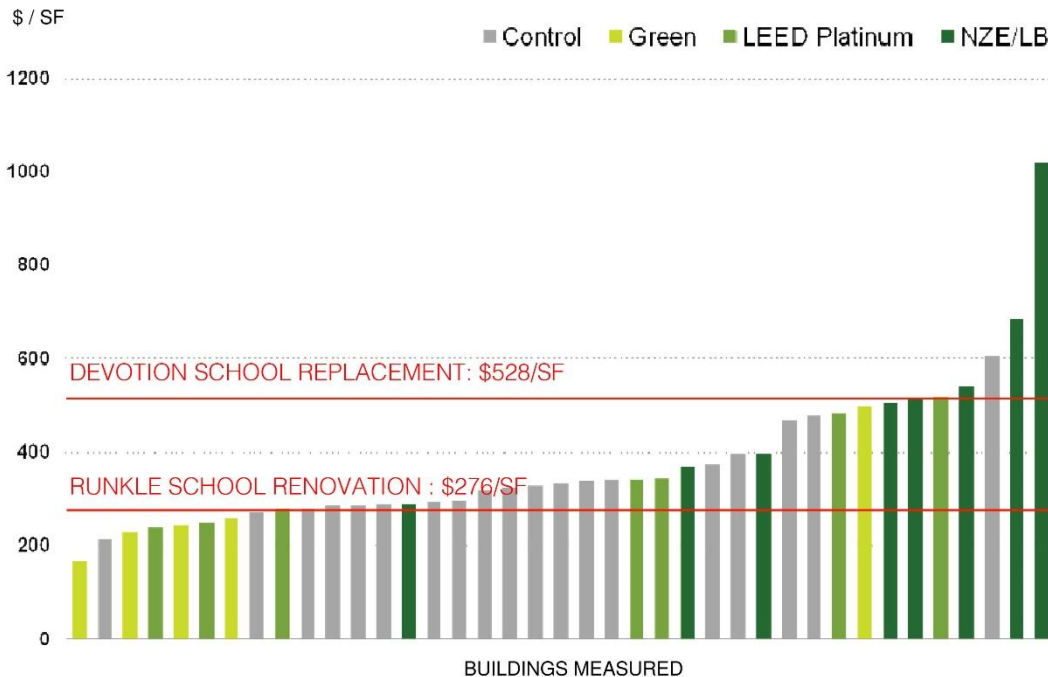
- *Do NOT change the Town design process*
- *Do NOT interfere with cost-benefit analysis*
- *Do NOT tie us down to a particular design*

Alan Christ AIA, LEED Accredited Professional
Member of Green Caucus & EDAB
TMM, Pct. 4

Cost of Green and NZE Schools Relative to Brookline Schools

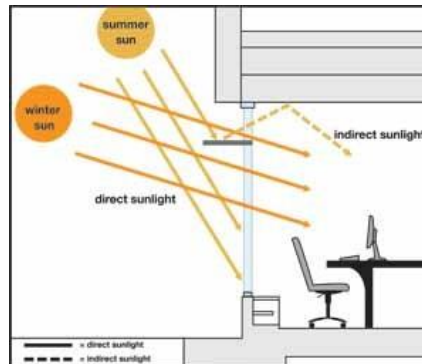
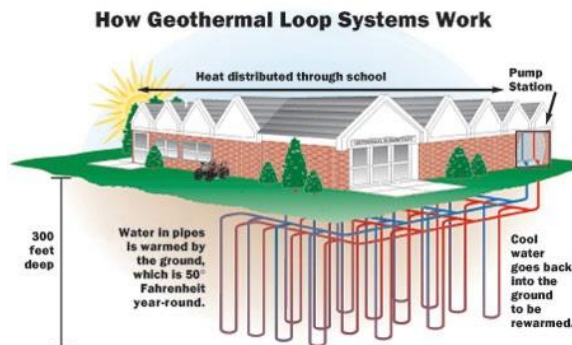
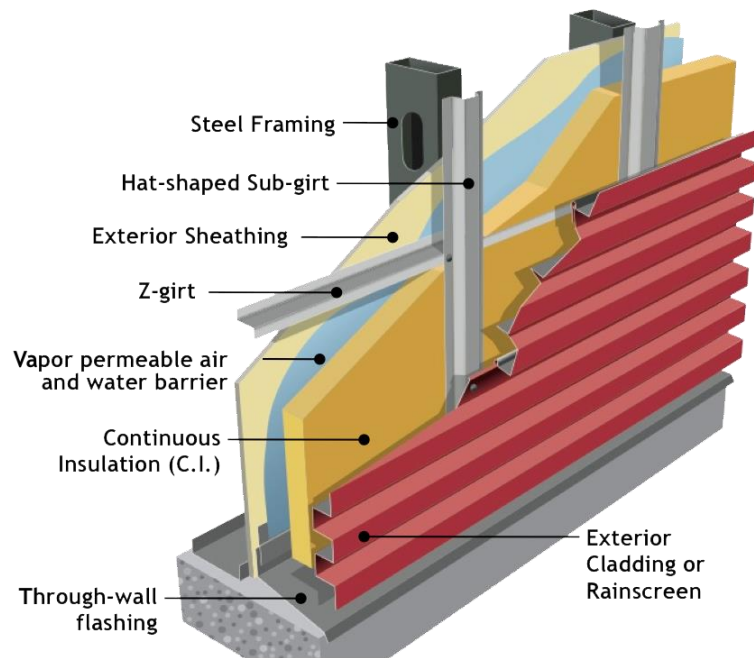
K-12

statistical analysis



There is substantial variation in the cost of Green and Net Zero Schools. This chart compares the cost of recent non-green “control” schools to green, LEED Platinum, and NZE schools, while using the recent Devotion and Runkle projects as reference points. The cost premium can vary depending upon the particular project and systems.

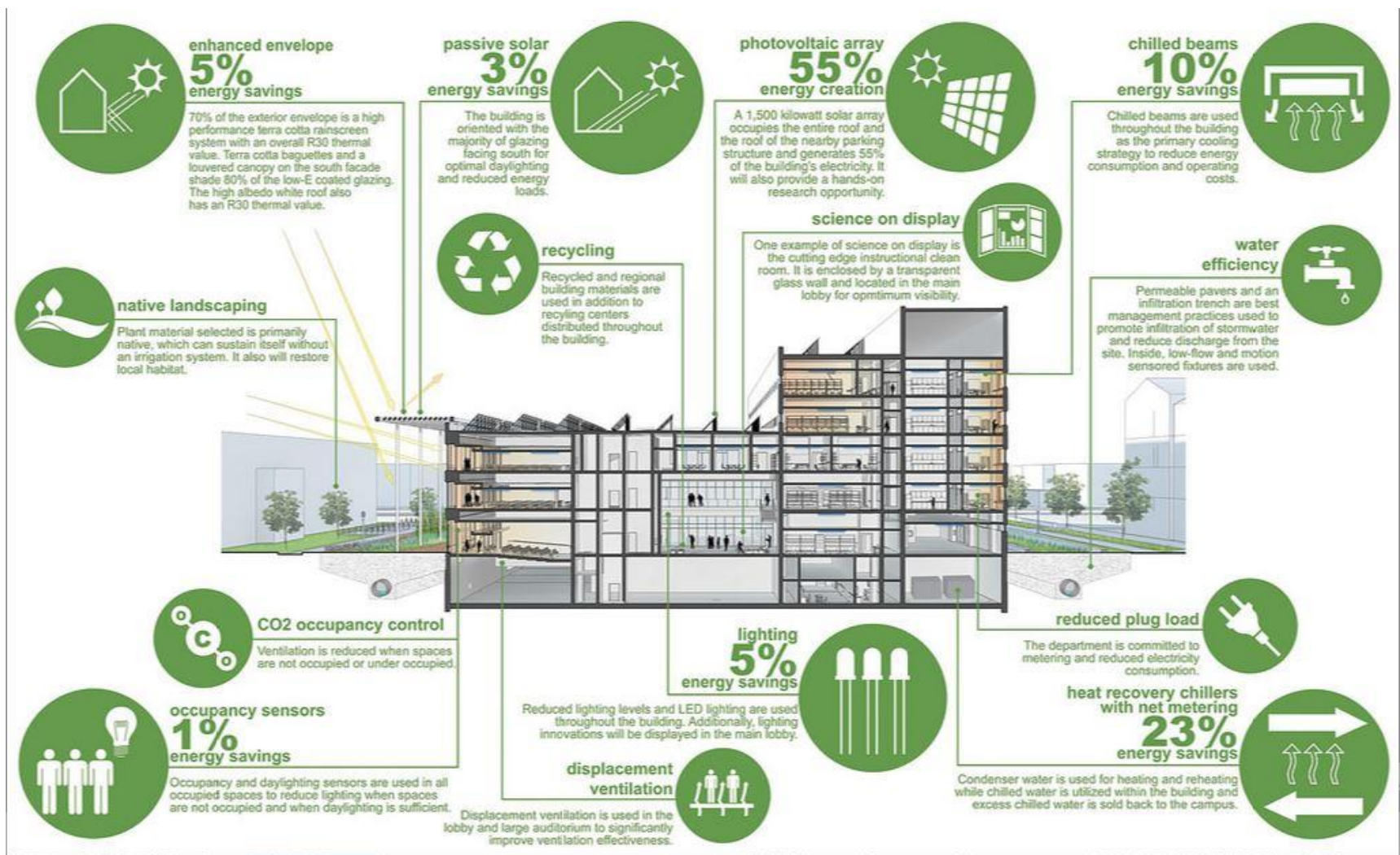
Net Zero Principles



Net-Zero Principles:

- Supertight, Heavily Insulated Envelope: R32 Walls, R50 Roof, R20 Slab
- Structural Thermal Breaks,
- Thermally Decoupled Sunshading Devices
- High Performance Windows with insulated, thermally broken frames
- Building Orientation: North and South Facing Window Openings
- Heating and Cooling are separated from ventilation
- 100% dedicated outdoor air unit for ventilation air
- Geothermal Heat Pump System
- PV or Solar Hot Water
- Energy recovery ventilators

Net Zero Principles: HVAC Systems



Components of a Net Zero Building: HVAC systems can include Geothermal Heat Pumps, Air Source Heat Pumps, Chilled Beams, or Dehumidification Displacement Systems

Claiborne Pell Elementary School, Newport, RI



Pell School, Newport, RI (2013):
105,565 sq. feet, \$28 Million
Const. Cost (\$265/SF) , EUI=35,
\$116,000 Yearly Energy Savings

- Dehumidification displacement system rather than a conventional air conditioning system (25% savings)
- Exterior Envelope R-value exceeds code by 35%
- Daylight harvesting with automated light fixture dimming system
- Light shelf at south-facing windows
- Minimally energy intensive north- and south-facing classrooms
- LED Lighting throughout

Claiborne Pell Elementary School, Newport, RI



MacArthur School



MacArthur School, Binghamton, NY (2015):

129,000 sq. feet, \$79.5 Million Construction Cost (\$616/SF), 10 EUI, LEED Platinum Certification Pending

- Rain gardens, bio-swales, and indigenous vegetation
- Northern exposure for classrooms
- Southern exposure for PV
- Use of energy budgets and energy modeling
- Ground source heat pumps
- exposed radiant concrete floors
- On-demand ventilation
- robust controls for lighting and energy
- Extensive commissioning



MacArthur School



MLK School



MLK School, Cambridge, MA (2015):
170,000 sq. feet, 33 EUI, \$80 Million (\$420/SF Including \$100/SF for Hazardous Materials Removal), LEED Platinum

- EUI 60% less than typical educational buildings in New England
- Ideal solar orientation
- Pervasive natural light in almost every space
- High-performance roof and wall assemblies.
- Rooftop photovoltaic panels
- Geothermal wells
- Automatic dimmer system
- Gray water storage and reuse



MLK School



The Opportunity for Brookline

Kathleen Scanlon, R.A., LEED AP accredited professional
Selectmen's Climate Action Committee & Green Caucus member
Brookline Mothers Out Front, founding member, 2014
TMM, Pct. 3

The Imperative of Acting Now in Our Communities



Our goal: *A significant advance*
toward a net zero energy ninth
school

Why a Net Zero School for Brookline?

- Greener buildings = healthier buildings
- Dramatically lower total life cycle cost
- High performance buildings = competitive advantage and differentiator
- More than just a new school



Why a Net Zero School for Brookline?



Source: http://cor.europa.eu/en/documentation/brochures/Documents/EU_cities_and_regions_leading_the_way_against_climate_change/COP21.pdf

Why a Net Zero School for Brookline?

- Significant impact on our emissions.
- Reinforces the Town's commitment to addressing Climate Change.
- Essential element to begin meeting 80% emission reduction goals by 2050.
- Brookline's legacy as a regional and national leader.



9th School Building Program

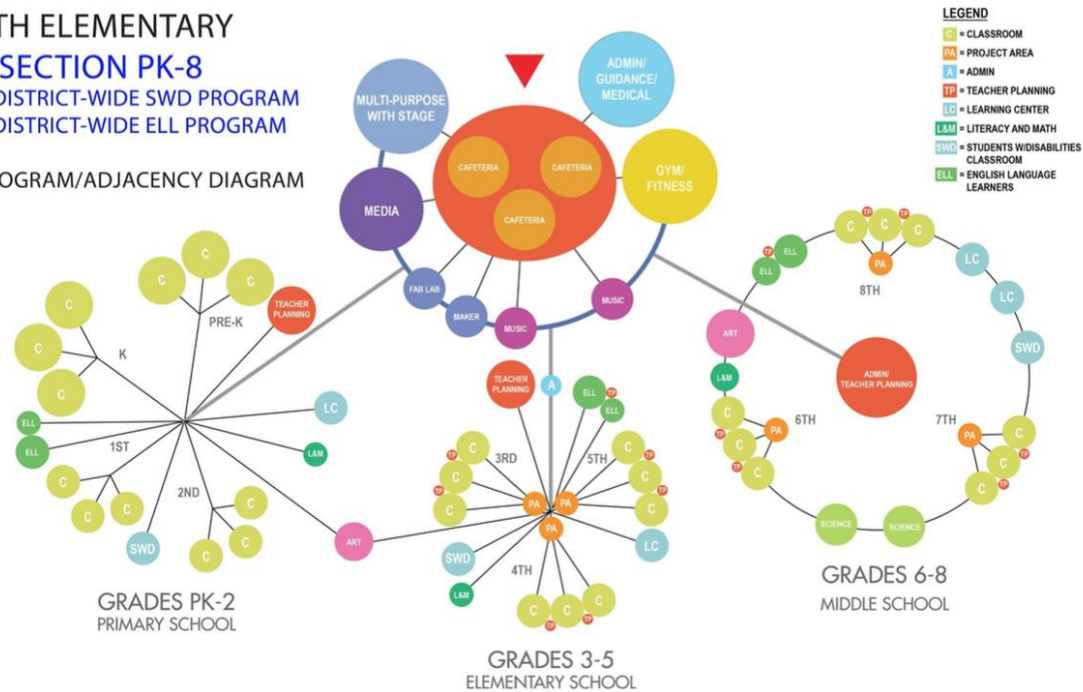
9TH ELEMENTARY

3 SECTION PK-8

+ DISTRICT-WIDE SWD PROGRAM

+ DISTRICT-WIDE ELL PROGRAM

PROGRAM/ADJACENCY DIAGRAM



 Jonathan Levi Architects

MARCH 6, 2017
BROOKLINE 9TH ELEMENTARY SCHOOL
FFASIRIITY STUDY

“Building a new school...when our community and society are more conscious than ever of the delicate balance between environmental sustainability and ongoing development provides an opportunity to have the physical plant itself play a significant role in the culture, educational approach and daily lives of students and teachers.”



Energy Dashboard

“...the physical plant can be used to help students learn about science, sustainability, and taking care of the environment. ... Brookline’s new elementary school could stand as a physical demonstration of environmental stewardship and innovation, providing a local case study for sustainable school construction.”

— 9th Elementary School Educational Program (11/29/16 draft), p. 13

Communities Pursuing Net Zero

- Cambridge, MA
- Somerville, MA
- Lexington, MA
- Austin, TX
- Seattle, WA
- Fort Collins, CO
- Boulder County, CO
- Rochester, MN
- Montpelier, VT
- Palo Alto, CA
- Binghamton, NY
- Staten Island, NY
- Newport, RI



Brookline High School Class of 2036



PETITIONED ARTICLE 21 - SPRING 2017 TOWN MEETING

Proposed Amendments voted 4-0 by Schools Subcommittee of the Advisory Committee 4/18/17

A RESOLUTION REGARDING A NET ZERO ENERGY NINTH ELEMENTARY SCHOOL
AND THE EXPANSION OF BROOKLINE HIGH SCHOOL

To see if the Town will vote to adopt the following resolution:

Whereas our town, the nation, and the world are increasingly aware of the need to address climate change and of the importance of better protection of the environment in general, and

Whereas an international standard known as LEED (Leadership in Energy and Environmental Design of the United States Green Building Council) allows for a building's environmental and energy performance to be accurately measured and provides a benchmark to assist in designing a net zero energy building, and

Whereas net zero energy LEED Platinum schools create an environment that supports student learning and health through improvements in daylighting, indoor air quality, thermal comfort, acoustics, and classroom design, all of which have an impact on a child's ability to learn and a teacher's ability to teach, while saving energy, resources, and money, and

Whereas net zero energy LEED Platinum schools increase energy efficiency, thereby reducing greenhouse gas emissions, cost less to operate, utilize durable materials, reduce water and energy use, and provide other benefits; while providing an educational experience that transcends the classroom by creating opportunities for curriculum innovation and hands-on, project-based learning in which the building itself becomes an interactive teaching tool, and

Whereas decisions made now about the design of the Ninth Elementary School ~~at Baldwin~~ and the expansion of Brookline High School will determine each school's environmental footprint, particularly greenhouse gas emissions, for decades to come, and

Whereas the technical ability to create energy-efficient, high performing buildings has increased significantly by incorporating systems thinking into design processes, and

Whereas construction of new schools in Massachusetts and around the nation during the past five years has shown the feasibility and desirability of net zero energy schools, that is, schools in which the amount of energy **delivered** on an annual basis is **less than or** equal to the amount of renewable energy **exported from** the site, and

~~*Whereas* an international standard known as LEED (Leadership in Energy and Environmental Design of the United States Green Building Council) allows for a building's environmental and energy performance to be accurately measured and provides a benchmark to assist in designing a net zero energy building, and~~

Whereas the most accurate measure of energy efficiency for a building is EUI (Energy Use Intensity), calculated by dividing total energy consumed annually by the gross floor area of the building,

Now therefore be it Resolved that in order for the Ninth Elementary School ~~at Baldwin~~ to be a **significant advance toward** a net zero energy school **and consistent with the projected energy efficiency results at the new Devotion School, the Ninth Elementary School at Baldwin** shall be ~~designed~~ **seek to obtain a minimum of LEED v4 certification at the Gold Silver rating level, with a goal of achieving the Platinum rating level; and in addition it shall be designed to prioritize achieving points in the Optimize Energy Performance category and shall seek to achieve a minimum of 14 13 of the possible 16 points available in that category, with the goal of achieving 16 of the possible 16 points available in that category; and, finally, it shall be designed seek to achieve a minimum EUI of 30 kBtu/sq.ft./yr., with the goal of achieving an EUI of 25 kBtu/sq.ft./yr.,**

And, be it further Resolved that while overall net zero energy is unlikely to be achieved for Brookline High School and even the degree to which the as-of-yet-undefined, expanded portion of the school can approach net zero energy **design** is currently uncertain, nevertheless, net zero energy principles shall be **appropriately** applied, to the extent feasible, during all design phases of Brookline High School,

or act upon anything relative thereto.

Advisory Committee

School Subcommittee

Warrant Article 21, a Resolution Regarding a Net Zero Energy Ninth Elementary School and the Expansion of Brookline High School

Report and Recommendations

Attendance:

The Schools Subcommittee held a public hearing on Warrant Article 21 on Tuesday, April 18th from 6:00 to 7:30 p.m. in Brookline Town Hall, employee lounge, third floor. In attendance were subcommittee members Mariah Nobrega, Shaari Mittel, Bobbie Knable, and David Lescohier; petitioners Werner Lohe (TMM), Alan Christ (TMM), and Kathleen Scanlon (TMM); School Committee member and former Building Commission member David Pollak (TMM), Deputy Superintendent of Brookline Schools MaryEllen Dunn, former school committee member Rebecca Stone (TMM), Ray Masak from the Building Department, Claire Stampfer (TMM), Virginia La Plante (TMM), Tom Kilday, Regina Frawley (TMM), Paul Harris (TMM), Richard Nangle (TMM), and Bonnie Rotenberg.

Voted:

The subcommittee voted 4-0-0 to recommend WA 21 as amended.

Background:

This Warrant Article sets out energy efficiency goals and minimum standards for the Ninth School construction and the expansion of Brookline High School that will be a significant advance toward “net zero energy” schools for Brookline, using LEED, EUI and NetZero energy standards.

LEED stands for “Leadership in Energy Efficiency Design”. This standard has been developed by the United States Green Building Council. LEED has evolved as building science technology has advanced. The current version is 4. Buildings can be qualified at four levels: certified, silver, gold or platinum. The level depends on the number of points earned by the building design in each category. A minimum of 40 points are required for the lowest level, ‘certified’. With additional points, the building earns silver, gold, or platinum levels up to a possible 110 points. The categories are:

- Location and Transportation
- Sustainable Sites
- Water Efficiency
- Energy and Atmosphere
- Materials and Resources
- Indoor Environmental Quality
- Innovation, and Regional Priority

The Energy and Atmosphere category offers the most points, and the Optimize Energy Performance subcategory offers the most points within the Energy and Atmosphere category, a maximum of 16 points.

LEED uses a modeling method to determine the LEED Optimize Performance score. It works somewhat like Turbo Tax software or an accreditation manual. As the designer answers building characteristics and energy optimization questions, the modelling software or the accreditation worksheet forms assign points as appropriate.

EUI stands for Energy Use Intensity. It is commonly expressed in thousand British Thermal Units (kBtu) per square foot per year. A square foot is readily understood. BTU and kBtu are less common. kBtu measures energy just like kilowatt hours (kWh). A kilowatt hour is the amount of energy used by a 100 watt bulb every 10 hours. kBtu measures the same energy as kWh. kBtu can be converted into kWh and vice-versa. One kBtu is the same as 0.29 kWh.

Net zero energy schools are schools for which the energy delivered to the building for heating, ventilation, cooling, and plug-in loads, all the energy the building uses, is less than the amount of offsetting energy produced (exported) by the building. Net zero energy buildings typically employ a solar PV array as a means of providing an offsetting energy source. There are a wide range of understandings and definitions for how to calculate the actual net zero energy budget of a building. The term net zero energy building does not yet have a clearly established, universally accepted meaning even though the US Department of Energy, through the National Institute of Building Sciences has published “A Common Definition for Zero Energy Buildings” in September 2015. Therefore, comparing the performance of buildings that claim to be net zero energy buildings is currently not straightforward. In view of this uncertainty, the Warrant Article, unlike LEED and EUI, does not identify an explicit minimum or goal for net zero energy performance. Regardless, the Warrant Article seeks to set in motion use of net zero energy building principles, thereby aiming to decrease the carbon footprint of our schools as much as feasible.

Brookline’s most recent Climate Action Plan, updated in 2012, accepted Massachusetts’ Green Communities Act target for greenhouse gas emissions, i.e. a reduction in emissions to 80% below 1990 levels by 2050. However, the overall town-wide progress toward reduced emissions has been slow, not yet approaching the rate needed to reach this goal.

While all efforts to achieve reductions in greenhouse gas emissions are important, the emissions under the direct control of the Town are a small (3%) part of the whole and offer only limited scope for achieving the overall reduction goal. However the actions of the Town – leading by example, demonstrating what is possible, and acting on the Town’s stewardship of the environment values – will likely have an impact beyond the specific reductions or avoided increases associated with the construction of the Ninth School and the expansion of Brookline High School. In addition, there are clearly extremely significant but less measurable benefits, such as these school buildings serving as teaching assets, particularly for the science, social studies, and environmental aspects of the curriculum. Today’s students are likely to be very aware and concerned about future climate change since, in view of their age, they are the most vulnerable to the expected consequences. Aiming for constructing net zero energy schools is consistent with Brookline’s educational and environmental

stewardship values. The PSB Education Plan for the Ninth School says: “Building a new school...when our community and society are more conscious than ever of the delicate balance between environmental sustainability and ongoing development provides an opportunity to have the physical plant itself play a significant role in the culture, educational approach, and daily lives of students and teachers.”

Discussion:

A lengthy and wide ranging discussion among the committee, the petitioners, and the numerous members of the public touched on a number of points.

Within this discussion, it was agreed that this Warrant Article is not:

- About the siting of the Ninth School or the expansion of Brookline High School decisions;
- Altering in any way in the design processes required by Town Bylaw Article 3.7, the role of the Board of Selectmen, the Building Commission, or the School Committee;
- Changing applicable building codes or other requirements;
- Affecting established methods for cost-benefit analysis;
- Affecting funding decisions ultimately requiring Town Meeting approval.

It was widely recognized that in order to most cost-effectively implement these LEED/EUI/NetZero principles, high performance environmental and sustainability goals must be incorporated into the planning at the beginning of the design process, which is why this Warrant Article is timely. Ultimately, there was interest among many in the room about the suggestion that these principles should possibly be applied to all appropriate, future Town building construction projects, and agreement that this warrant article was a good beginning.

Other key points:

There was concern that, while the environmental impact reduction goals should definitely be supported, that the impact of the original wording of this Warrant Article could create the impression that we are imposing costs that future Town Meetings would be forced to accept. This was addressed through amendments to the article accepted by the petitioners to change instances of ‘be designed’ to ‘seek’ in the first resolved clause. This recommended change is intended to make it plain that the Warrant Article is not to be implemented in a manner that would create a cost burden for a future Town Meeting.

Questions were asked about how these standards compared to the new Devotion building, which, along with Runkle, are high performance buildings. How much additional performance would this WA require? Are the goals and minimums feasible, or would meeting them be a stretch? How certain can we be that the minimums are practical? Based on the advice of Ray Masak and David Pollak, the committee suggested some edits to the language of the Warrant Article, which the petitioners accepted, to align minimums specified in the Warrant Article to be consistent with the expected energy efficiency performance of the new Devotion School, and with what the Building Department feels confident can be achieved. Specifically the warrant article was amended to:

1. Change 'Gold' to 'Silver for LEED. While Devotion is very, very close to Gold, it is slightly short of the points needed to be 'Gold'. Therefore, to be cautious and guard against over promising, the committee recommends that the Warrant Article should set 'Silver' rather than 'Gold' as the minimum, while 'Platinum' is still the goal. (With respect to new Devotion, only one additional point would raise it to the Gold level.)
2. Change the recommended minimum energy optimization points from 14 to 13, while keeping a goal of 16.

Finally, the committee recommended:

1. Moving the 'whereas' clause that defines LEED to second place, before other whereas clauses that reference this term.
2. Deleting 'Baldwin' throughout so that the Ninth School is not specific to any site for the purposes of this Warrant Article.

VOTED:

THE SUBCOMMITTEE VOTED 4-0-0 TO RECOMMEND WA 21 AS AMENDED BELOW.

ARTICLE 21

To see if the Town will vote to adopt the following resolution:

Whereas our town, the nation, and the world are increasingly aware of the need to address climate change and of the importance of better protection of the environment in general, and

Whereas an international standard known as LEED (Leadership in Energy and Environmental Design of the United States Green Building Council) allows for a building's environmental and energy performance to be accurately measured and provides a benchmark to assist in designing a net zero energy building, and

Whereas net zero energy LEED Platinum schools create an environment that supports student learning and health through improvements in daylighting, indoor air quality, thermal comfort, acoustics, and classroom design, all of which have an impact on a child's ability to learn and a teacher's ability to teach, while saving energy, resources, and money, and

Whereas net zero energy LEED Platinum schools increase energy efficiency, thereby reducing greenhouse gas emissions, cost less to operate, utilize durable materials, reduce water and energy use, and provide other benefits; while providing an educational experience that transcends the classroom by creating opportunities for curriculum innovation and hands-on, project-based learning in which the building itself becomes an interactive teaching tool, and

Whereas decisions made now about the design of the Ninth Elementary School and the expansion of Brookline High School will determine each school's environmental footprint, particularly greenhouse gas emissions, for decades to come, and

Whereas the technical ability to create energy-efficient, high performing buildings has increased significantly by incorporating systems thinking into design processes, and

Whereas construction of new schools in Massachusetts and around the nation during the past five years has shown the feasibility and desirability of net zero energy schools, that is, schools in which the amount of energy delivered on an annual basis is less than or equal to the amount of renewable energy exported from the site, and

Whereas the most accurate measure of energy efficiency for a building is EUI (Energy Use Intensity), calculated by dividing total energy consumed annually by the gross floor area of the building,

Now therefore be it Resolved that in order for the Ninth Elementary School be a significant advance toward a net zero energy school and consistent with the projected energy efficiency results at the new Devotion School, the Ninth Elementary School shall **seek** to obtain a minimum of LEED v4 certification at the **Silver** rating level, with a goal of achieving the Platinum rating level; and in addition it shall prioritize achieving points in the Optimize Energy Performance category and shall **seek** to achieve a minimum of **13** of the possible 16 points available in that category, with the goal of achieving 16 of the possible 16 points available in that category; and, finally, it shall **seek** to achieve a minimum EUI of 30 kBTU/sq.ft./yr., with the goal of achieving an EUI of 25 kBTU/sq.ft./yr.,

And, be it further Resolved that while overall net zero energy is unlikely to be achieved for Brookline High School and even the degree to which the as-of-yet-undefined, expanded portion of the school can approach net zero energy design is currently uncertain, nevertheless, net zero energy principles shall be appropriately applied, to the extent feasible, during all design phases of Brookline High School,

or act upon anything relative thereto.